THE EVOLUTION OF ENGINEERING TEAMS

Why Engineering Never Seems To Build "The Right Thing"

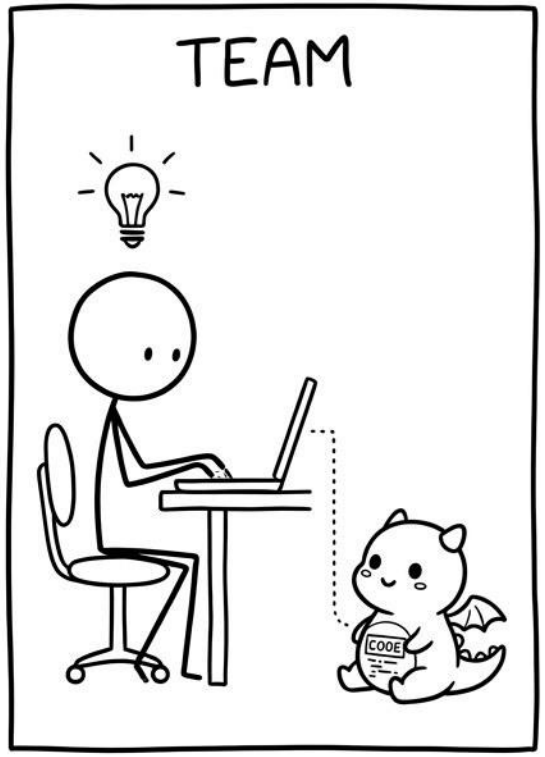
A Systems Explanation of Why It's So Damn Hard

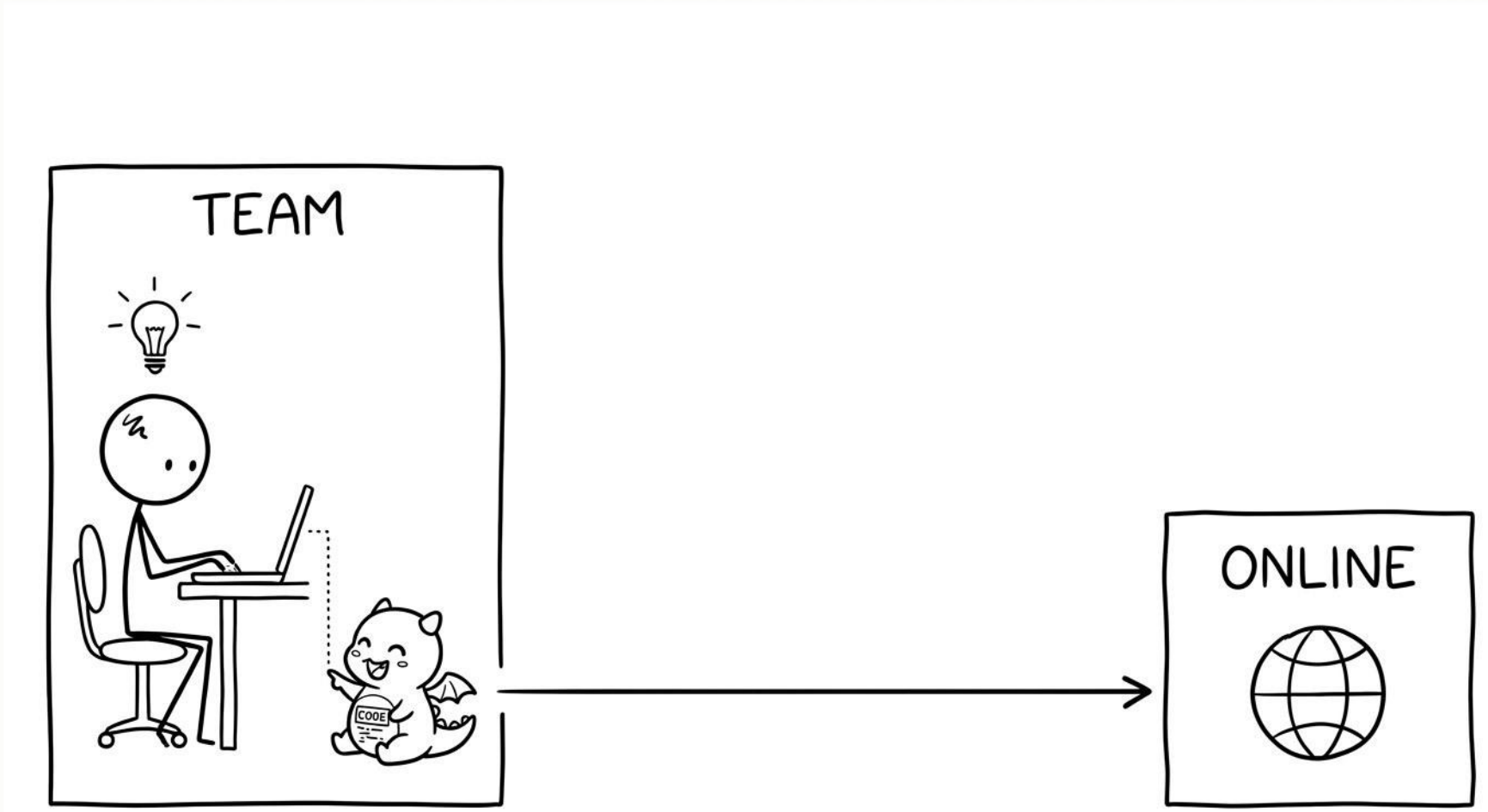
A field guide for non-engineers and new managers, in stick figures.

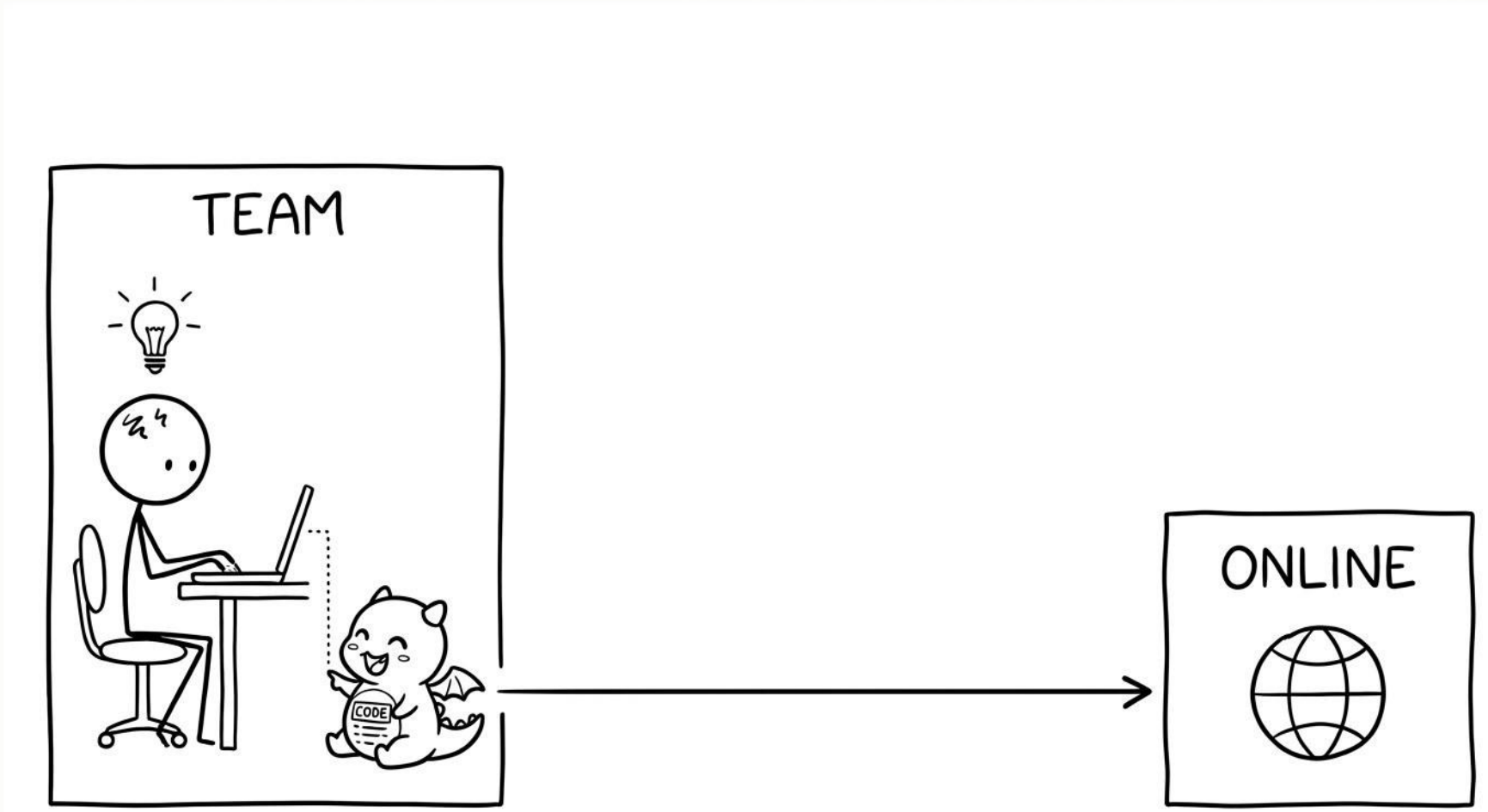
PART 1

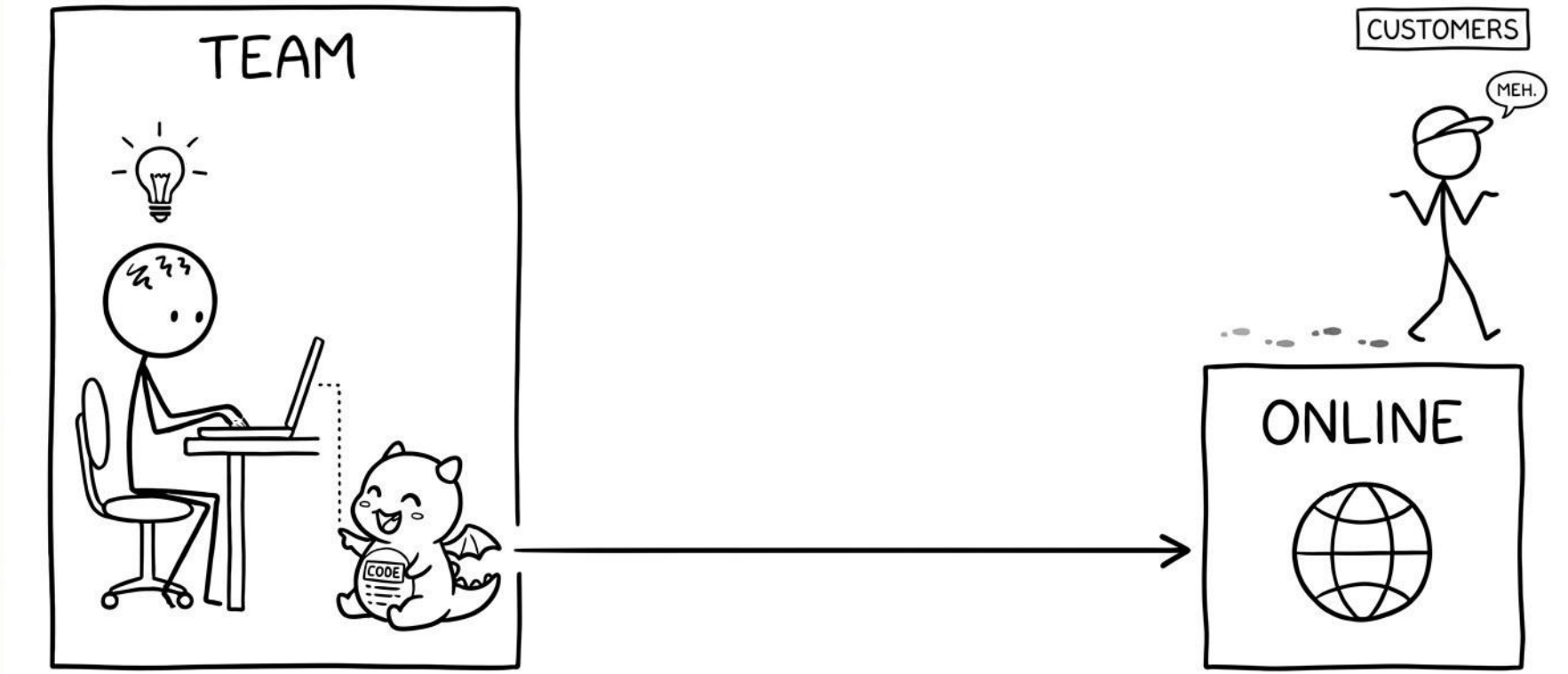
How an Engineering Org Grows

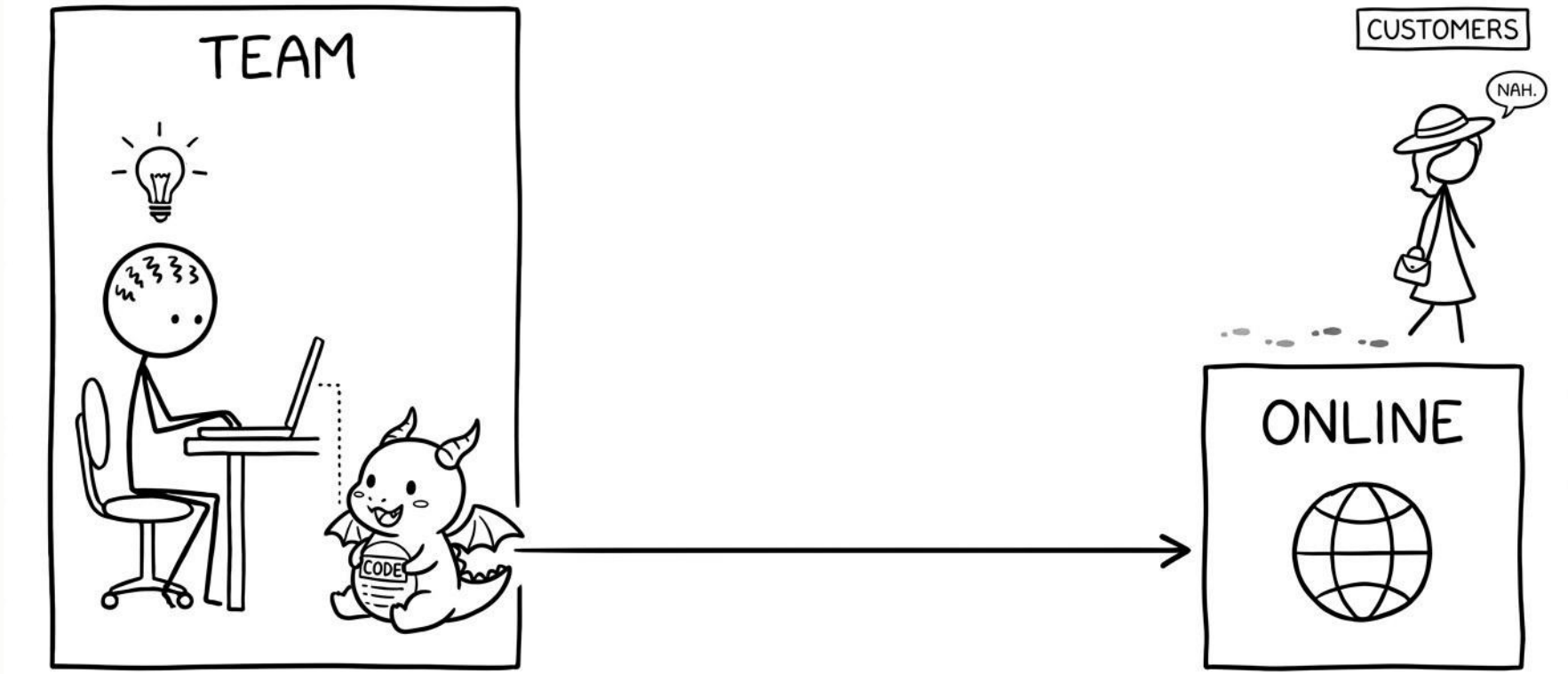
From one engineer to a whole org, one sensible step at a time

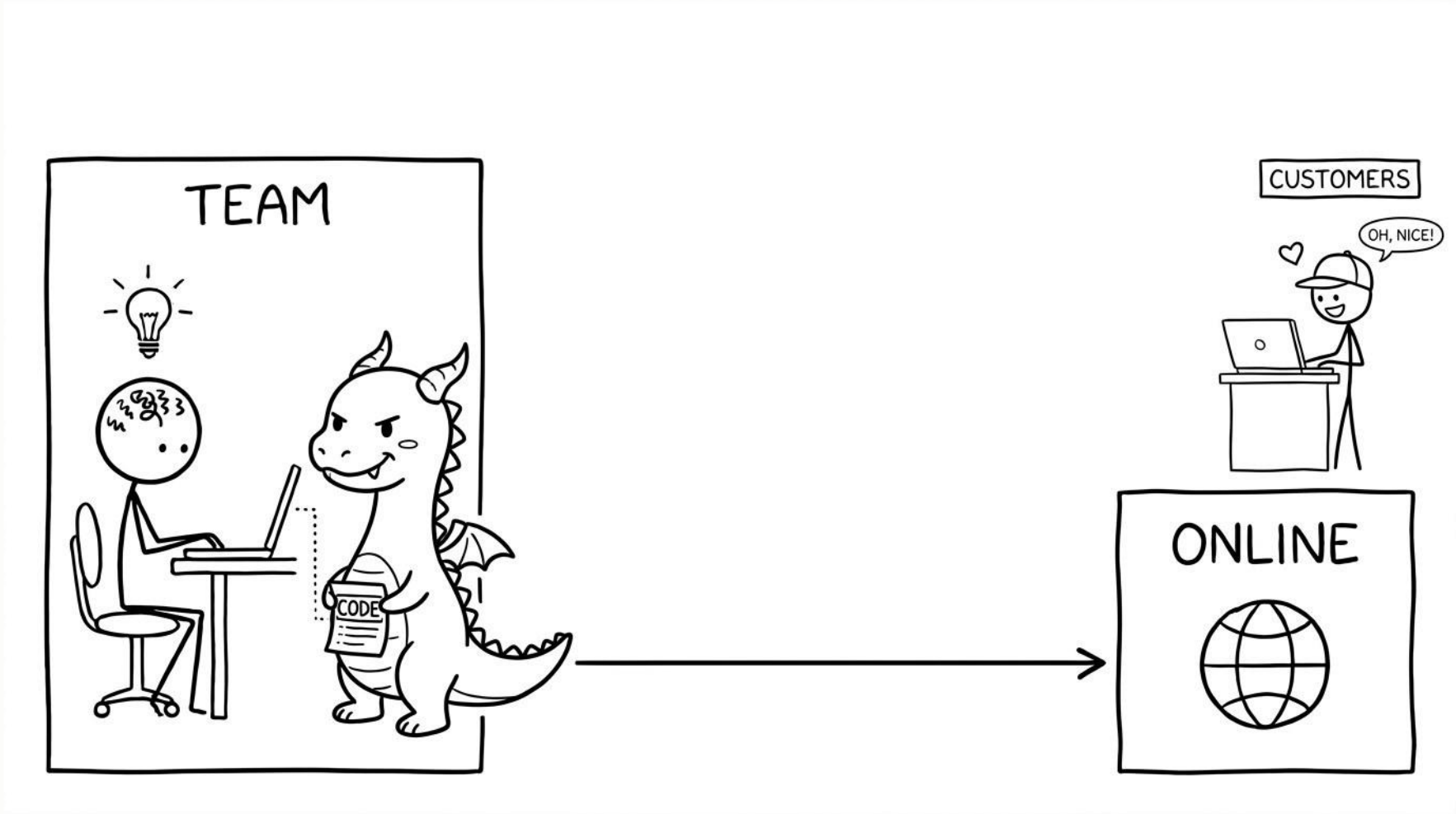

TEAM
COOE

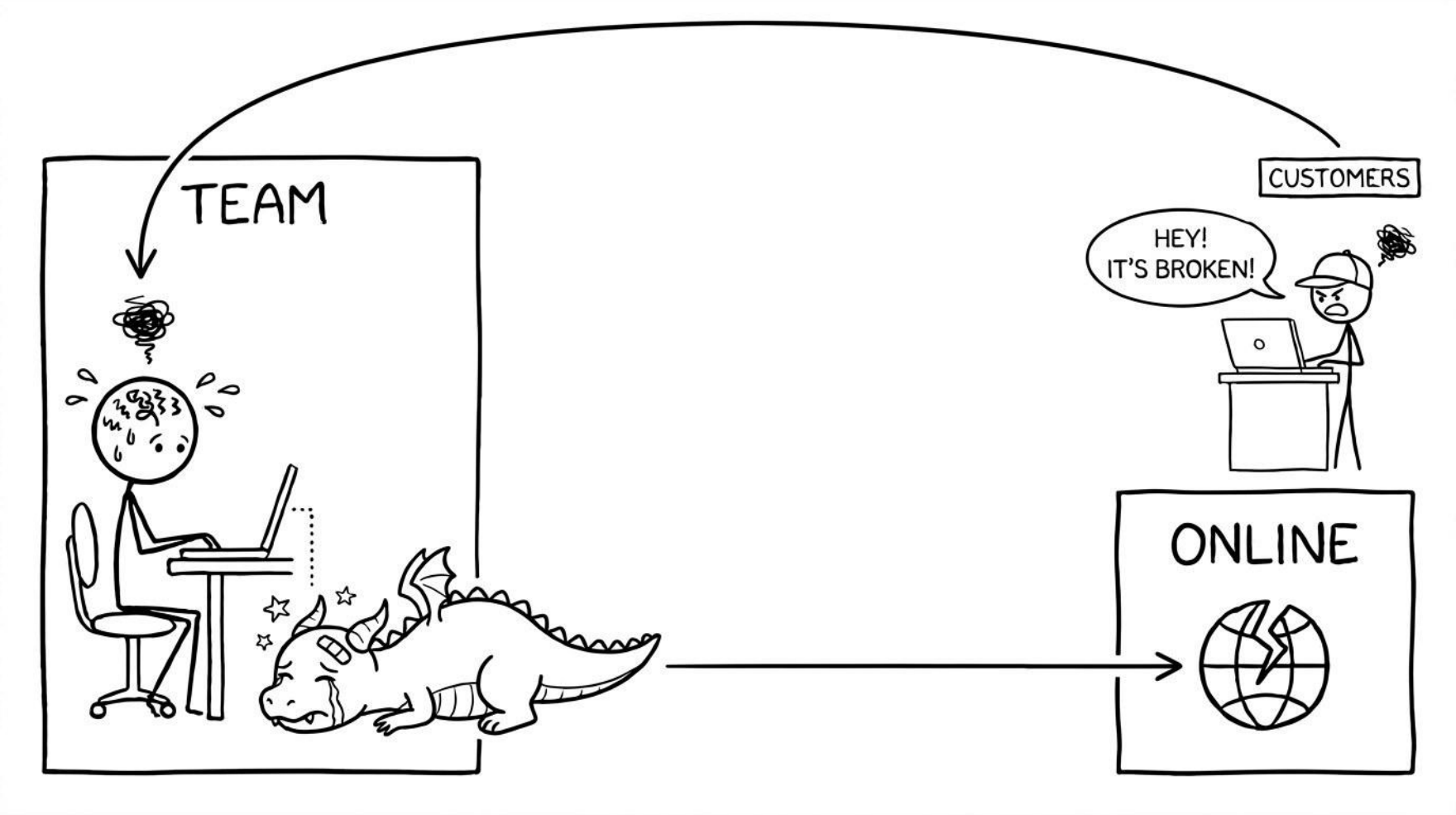

TEAM
CODE
ONLINE

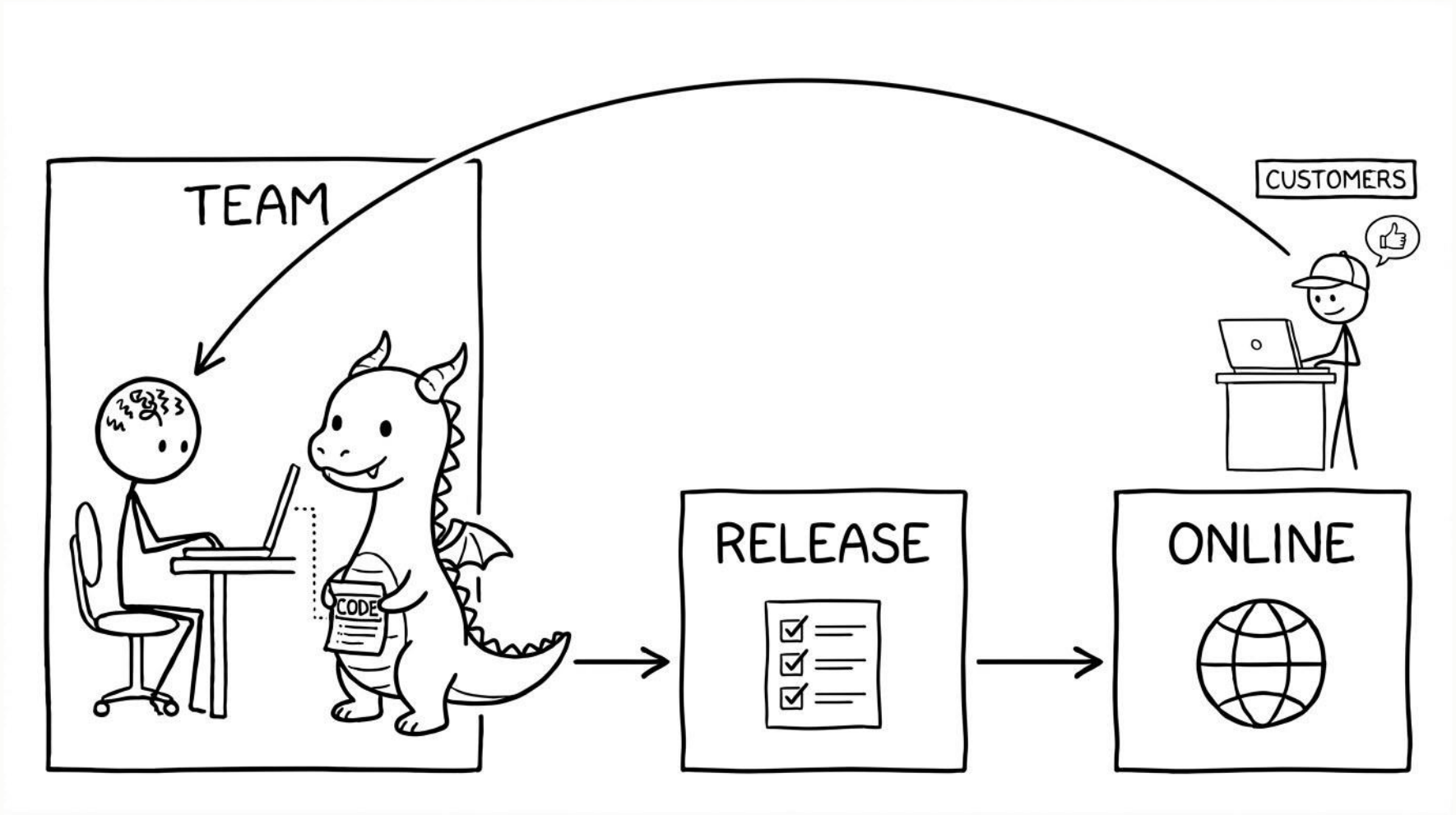

TEAM
CODE
ONLINE

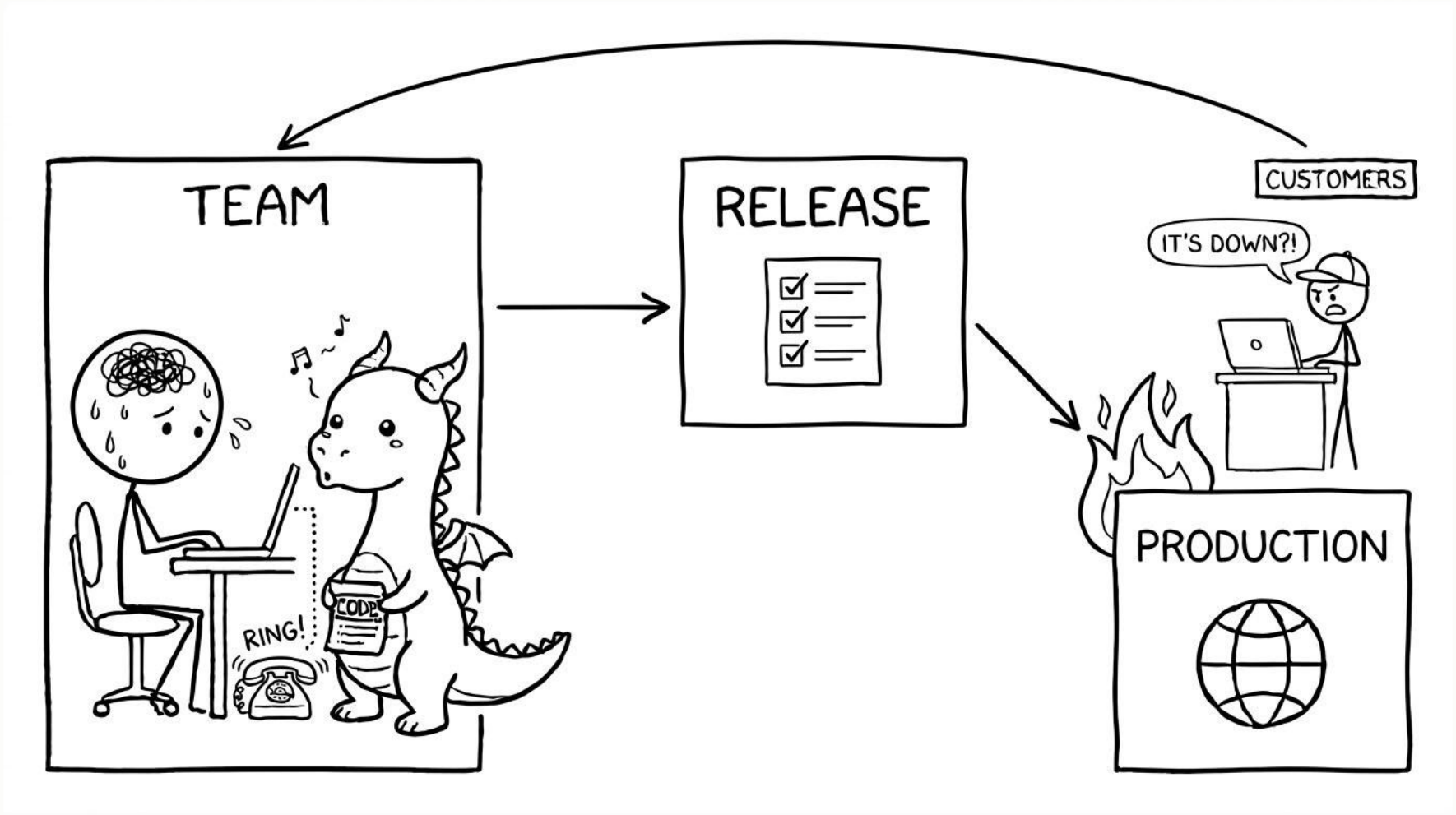

TEAM
CODE
CUSTOMERS
MEH.
ONLINE

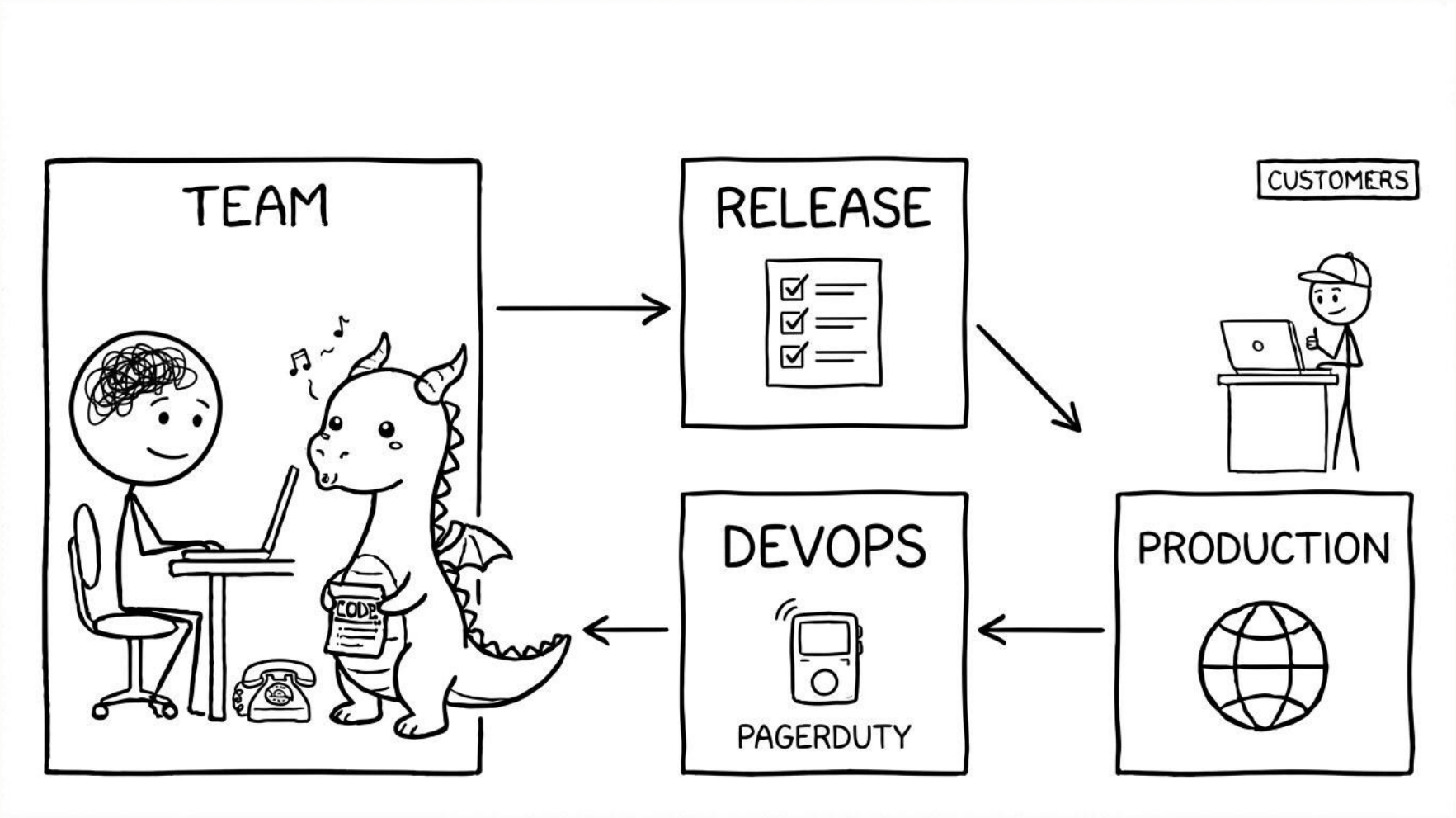

TEAM
CODE
CUSTOMERS
NAH.
ONLINE

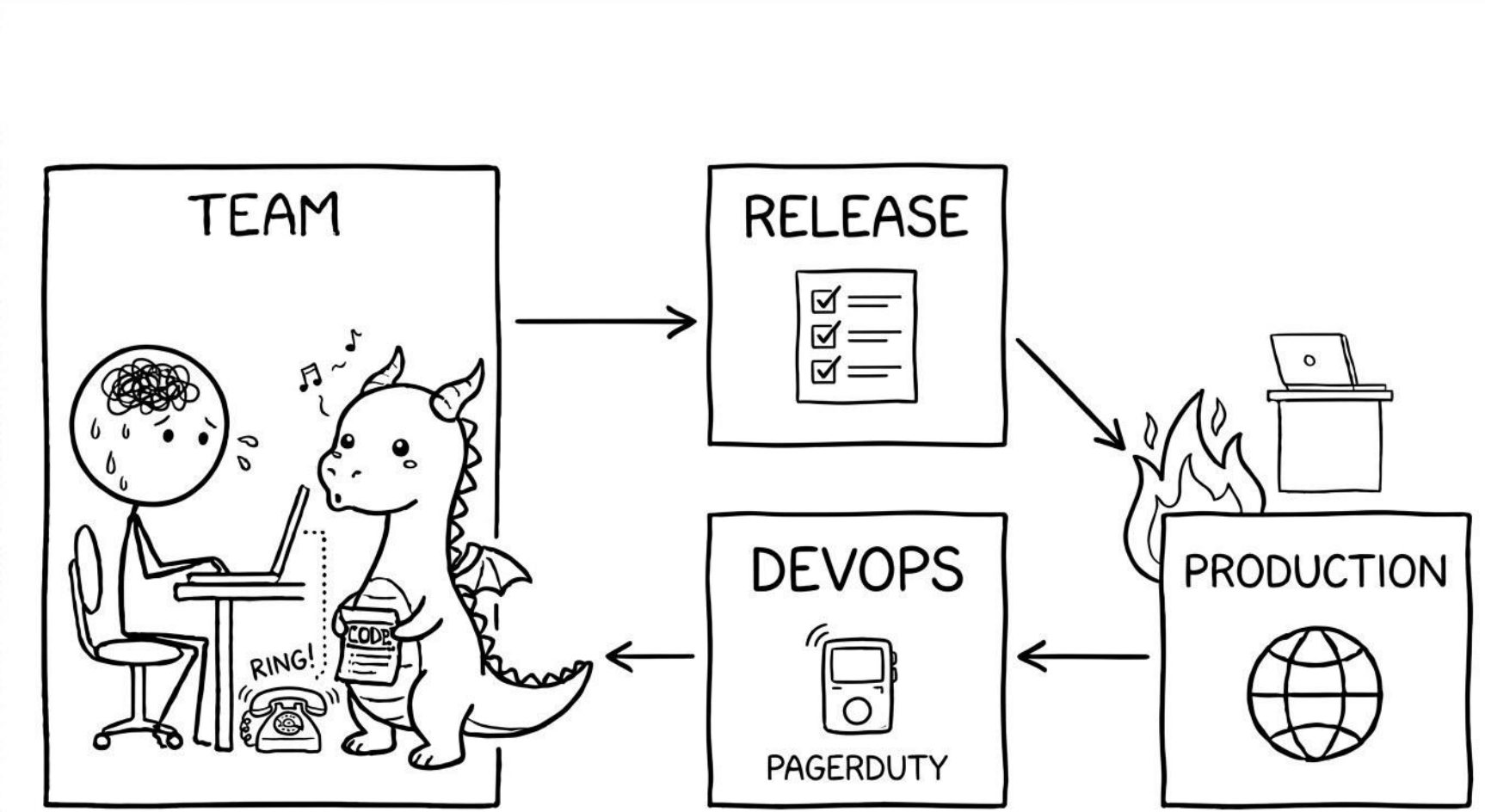

TEAM
CODE
CUSTOMERS
OH, NICE!
ONLINE

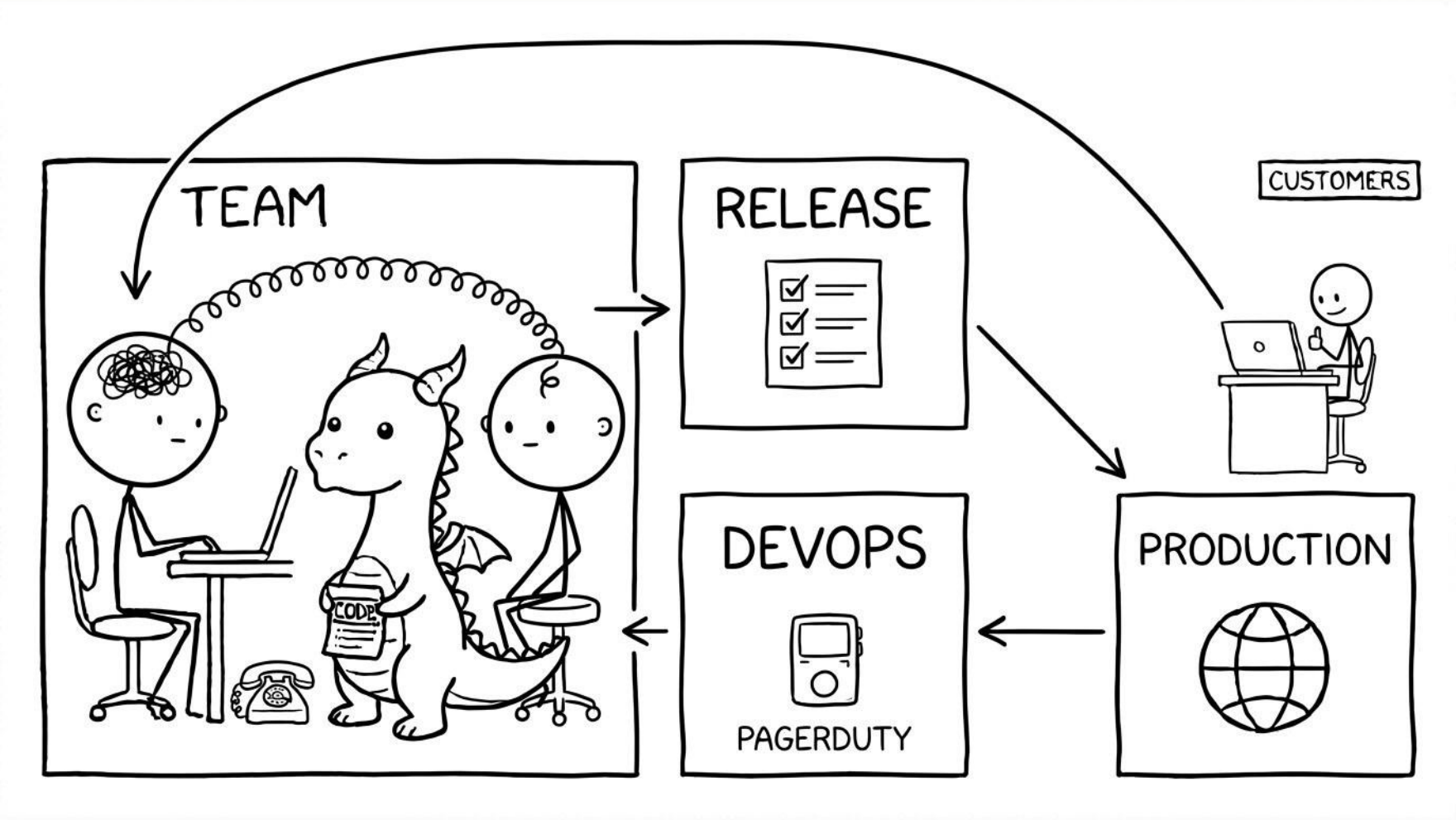

TEAM
CUSTOMERS
HEY!
IT'S BROKEN!
ONLINE

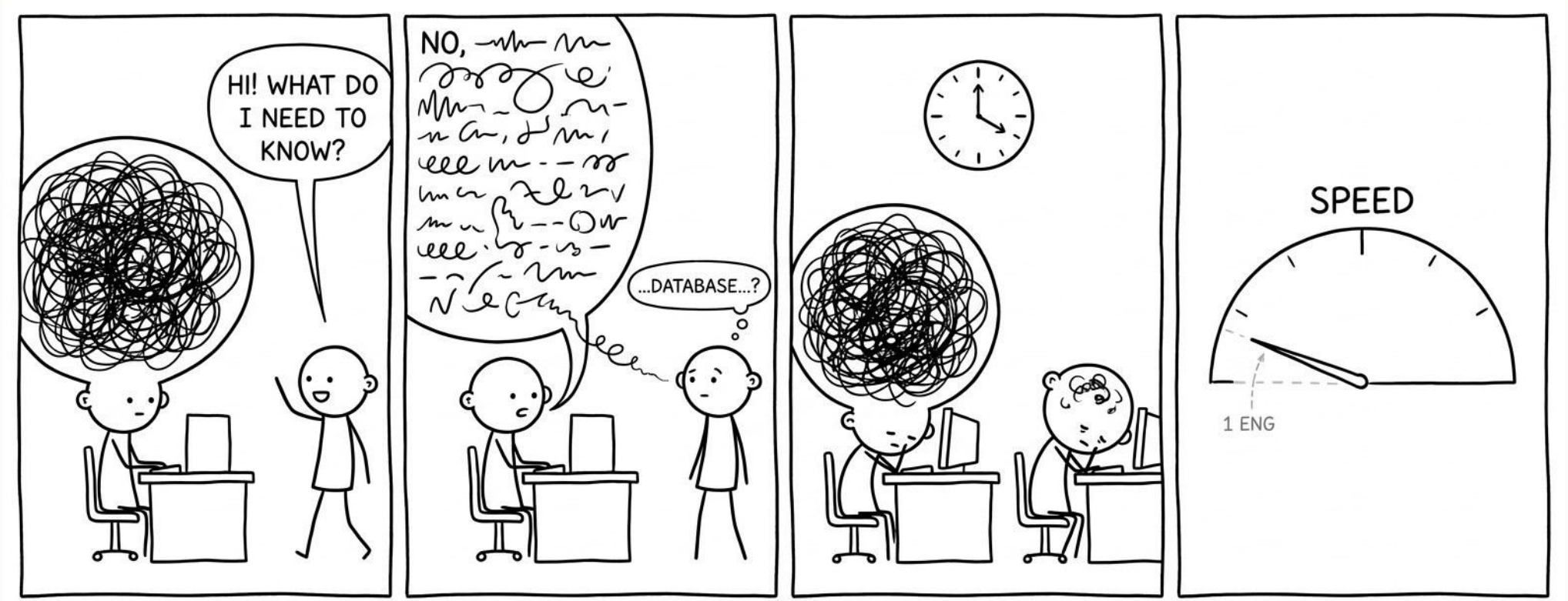

TEAM
CODE
RELEASE
ONLINE
CUSTOMERS


TEAM
RELEASE
CUSTOMERS
IT'S DOWN?!
PRODUCTION
RING!
CODE

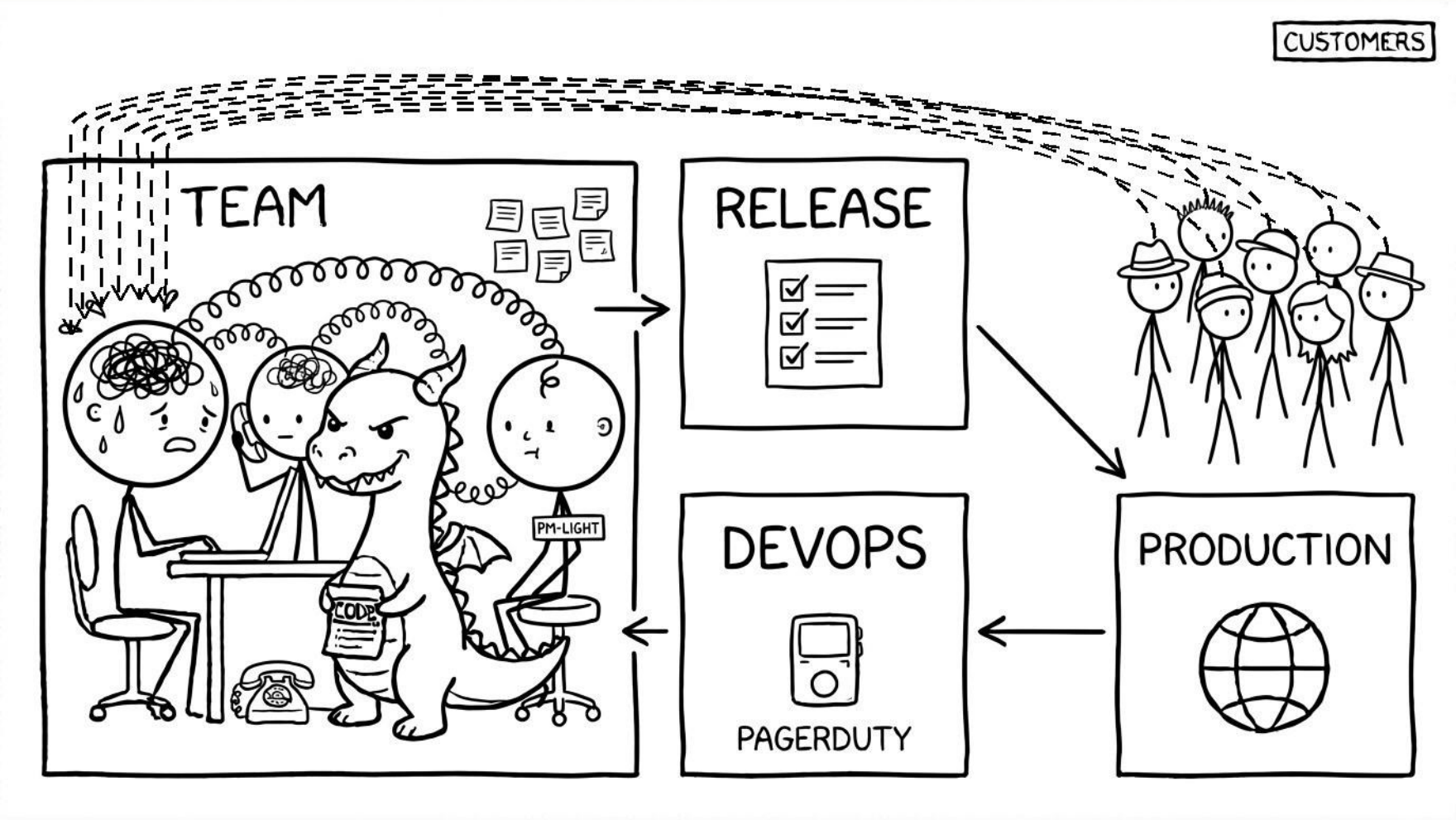

TEAM
CODE
RELEASE
CUSTOMERS
PRODUCTION
DEVOPS
PAGERDUTY

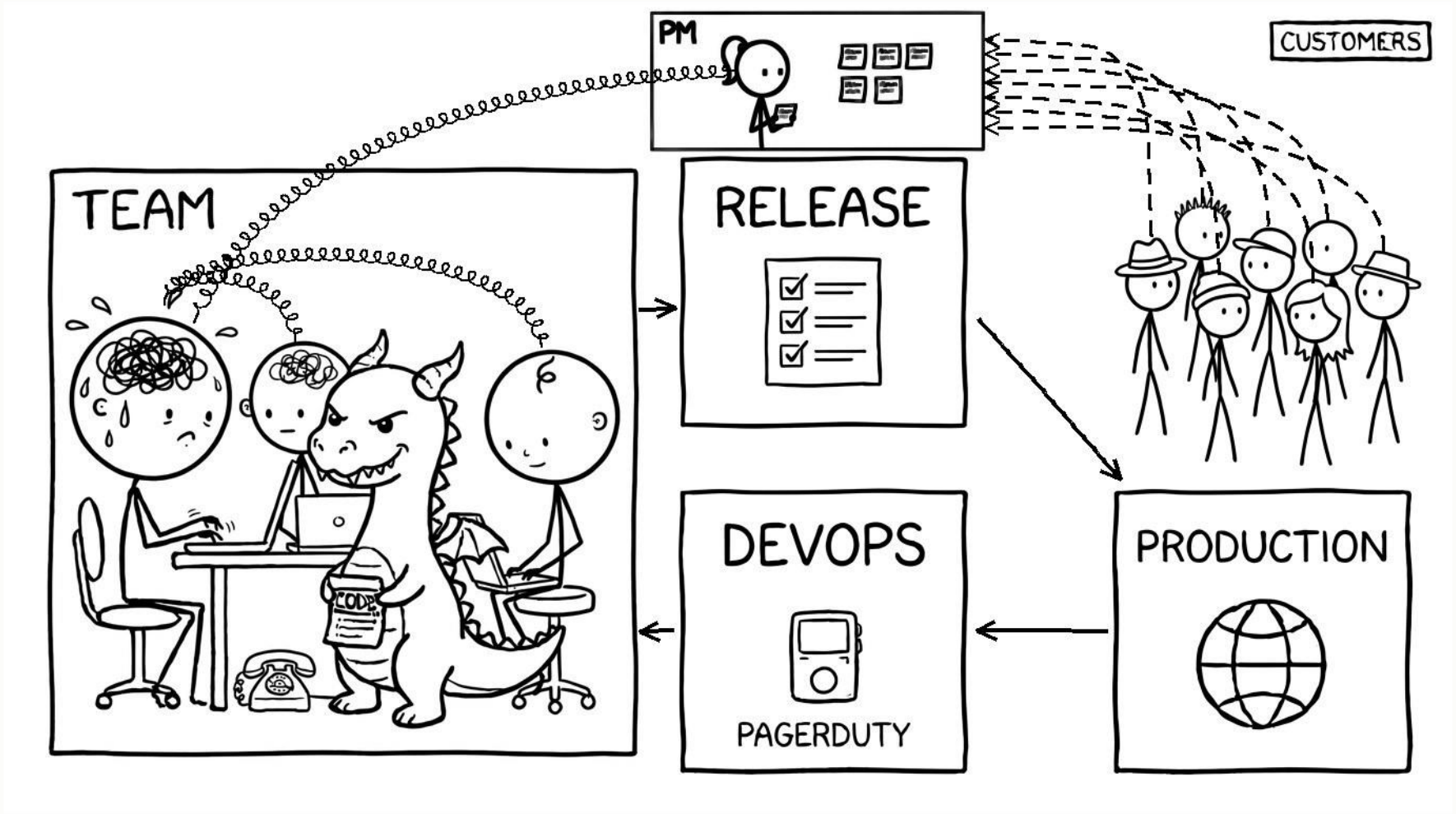

TEAM
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RING!
CODE

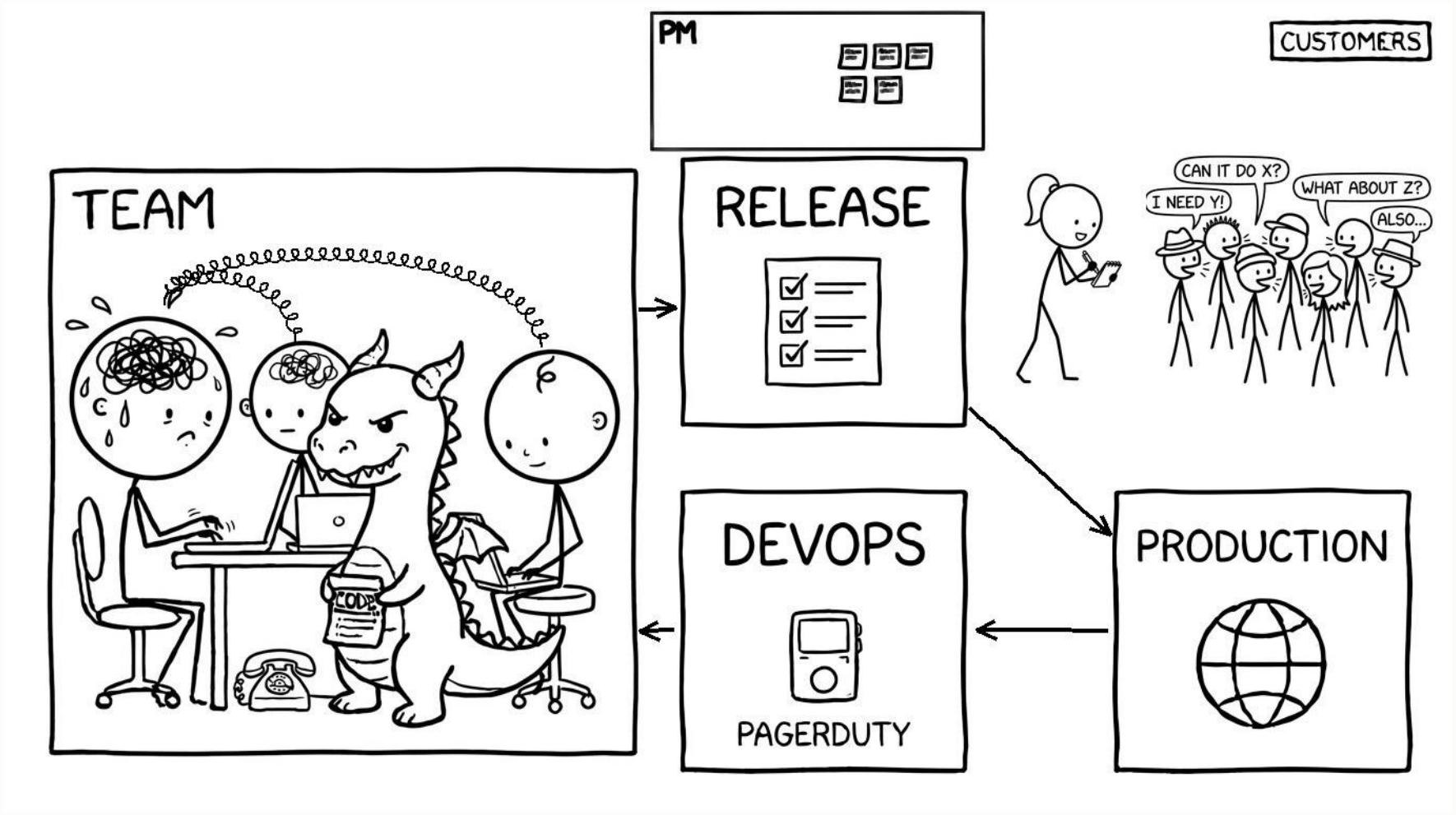

TEAM
CODE
RELEASE
CUSTOMERS
DEVOPS
PAGERDUTY
PRODUCTION

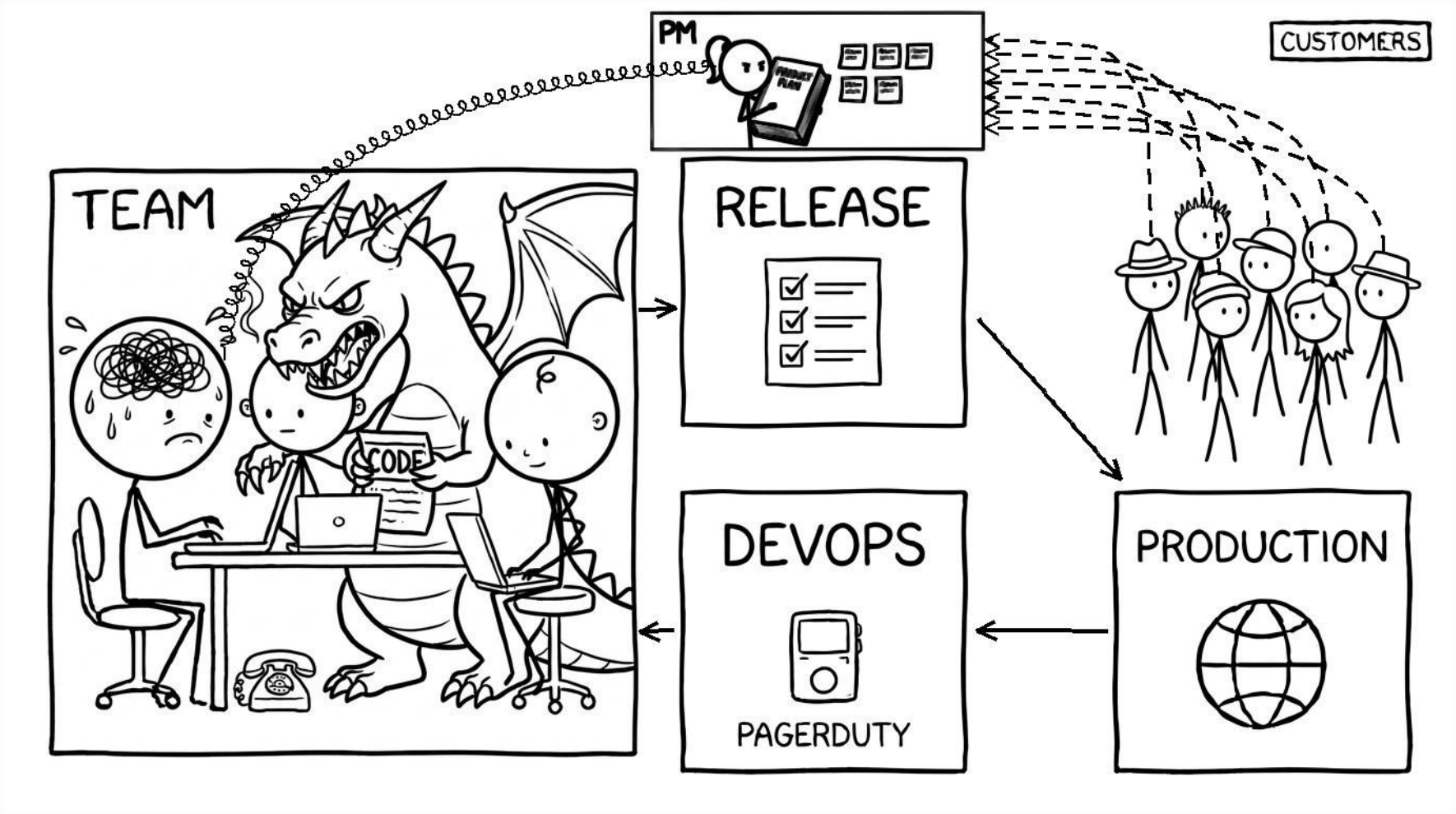


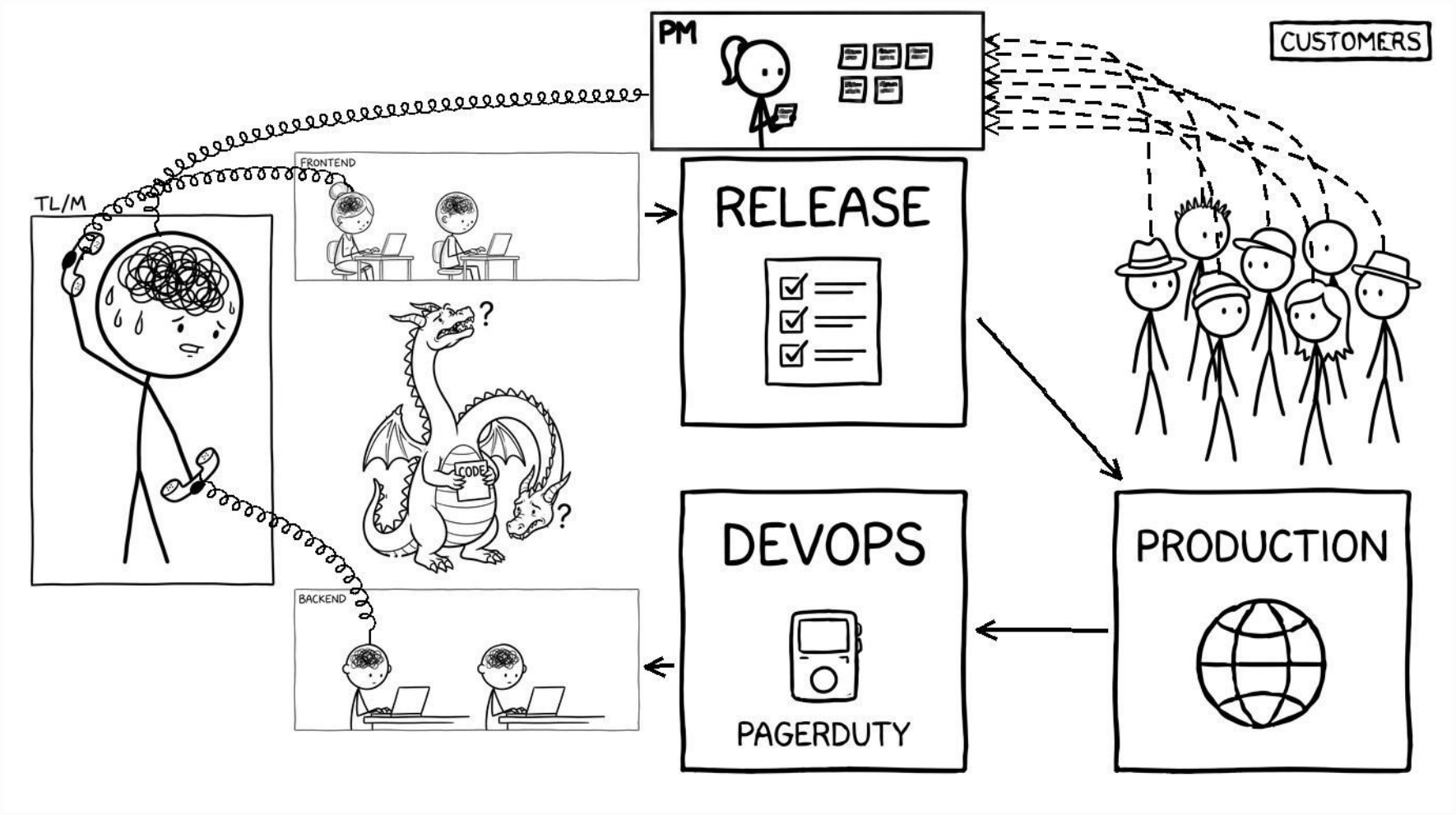
TWO ENGINEERS ≠ TWICE AS FAST.

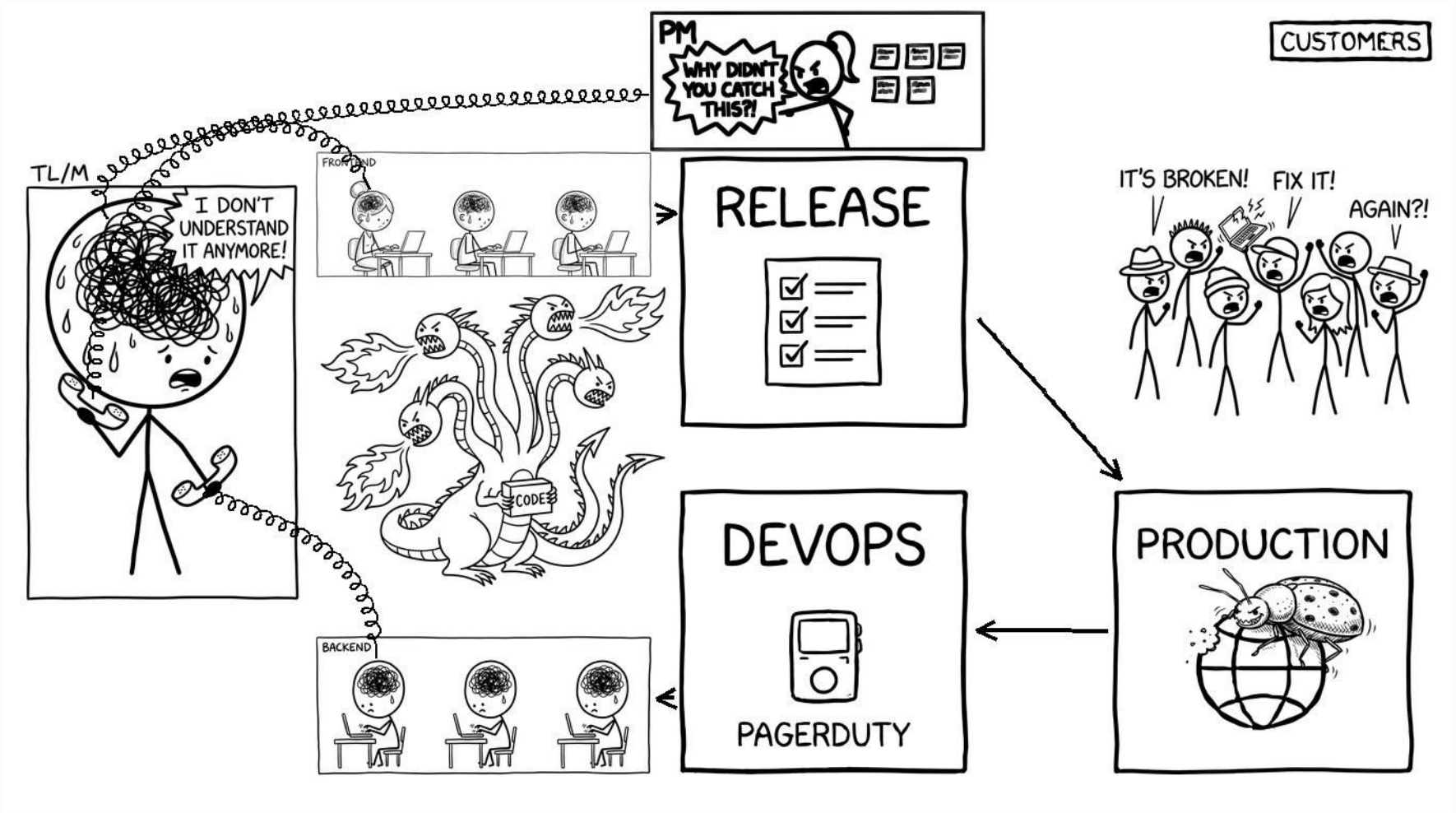

CUSTOMERS
TEAM
PM-LIGHT
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

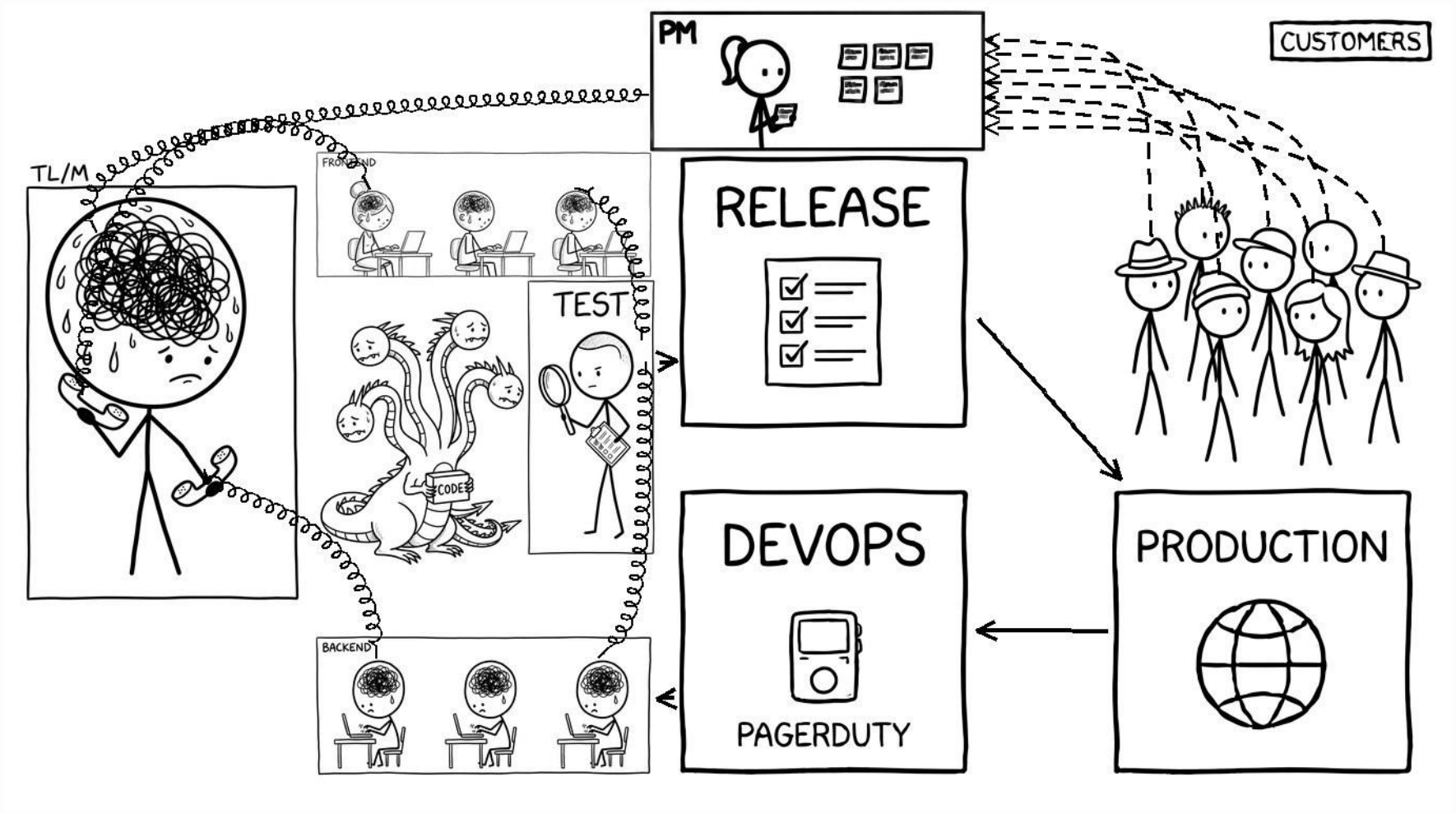

PM
CUSTOMERS
TEAM
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

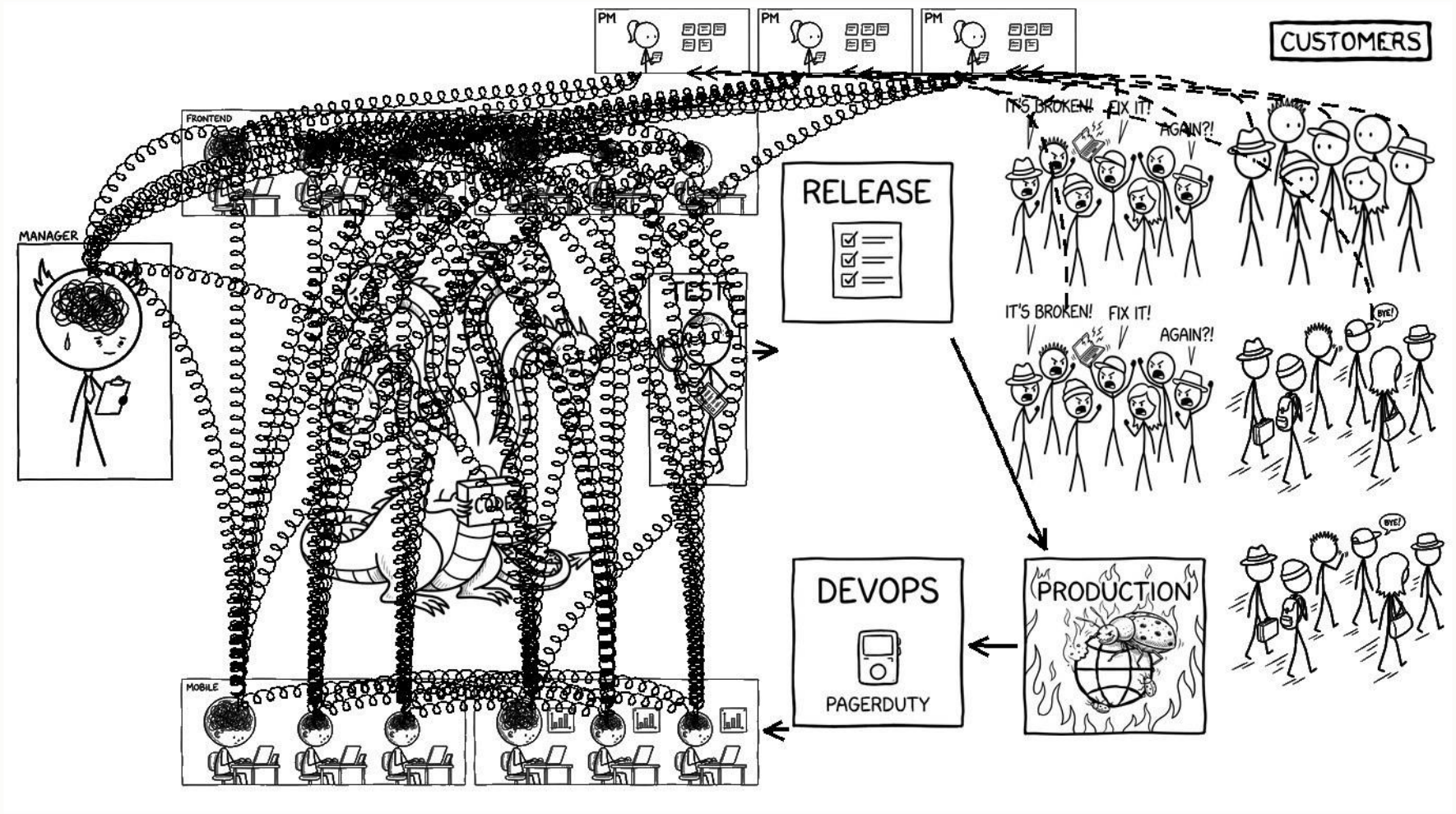

PM
CUSTOMERS
TEAM
RELEASE
CAN IT DO X?
I NEED Y!
WHAT ABOUT Z?
ALSO...
CODE
DEVOPS
PAGERDUTY
PRODUCTION

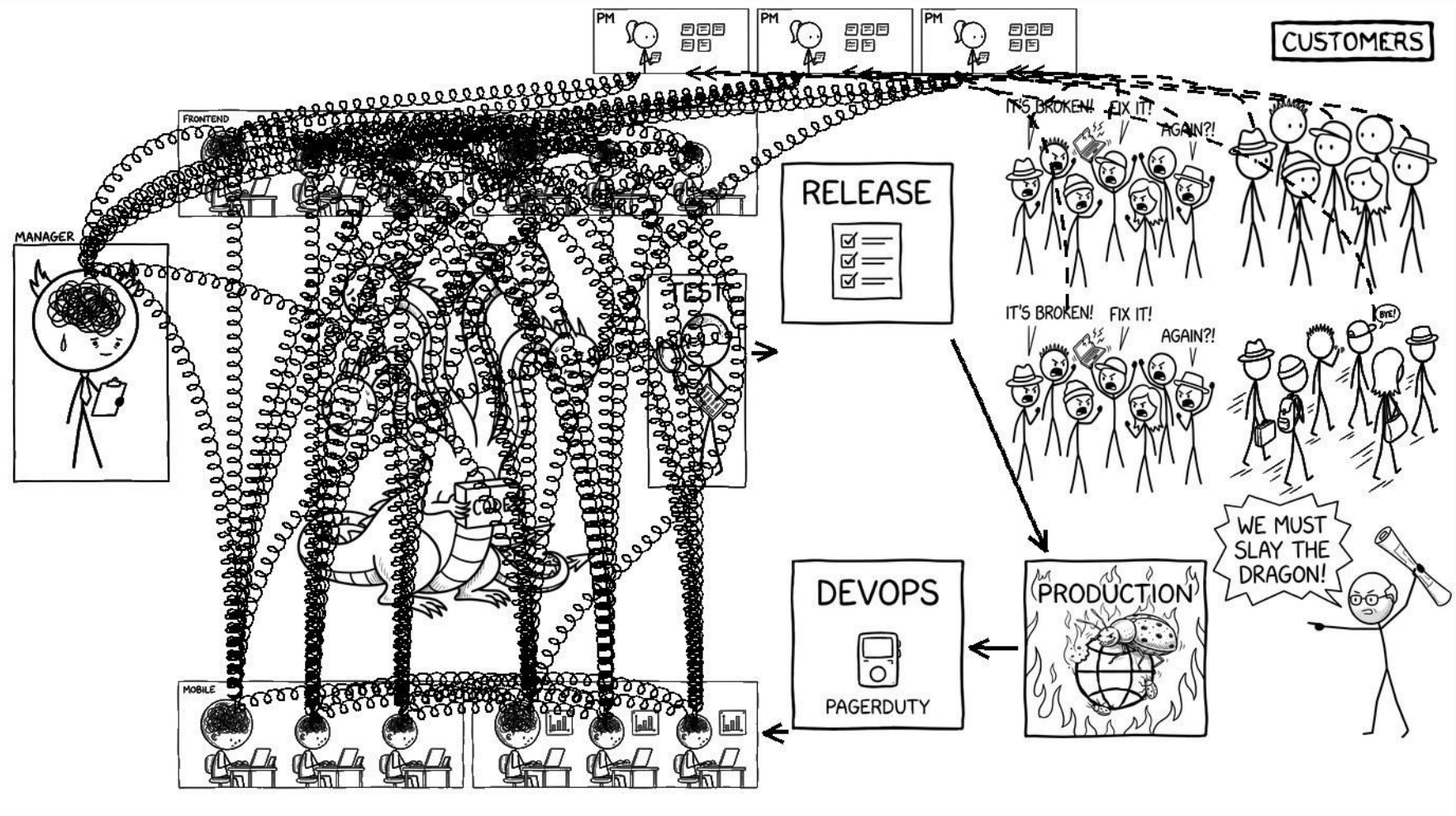

PM
PRODUCT PLAN
CUSTOMERS
TEAM
CODE
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

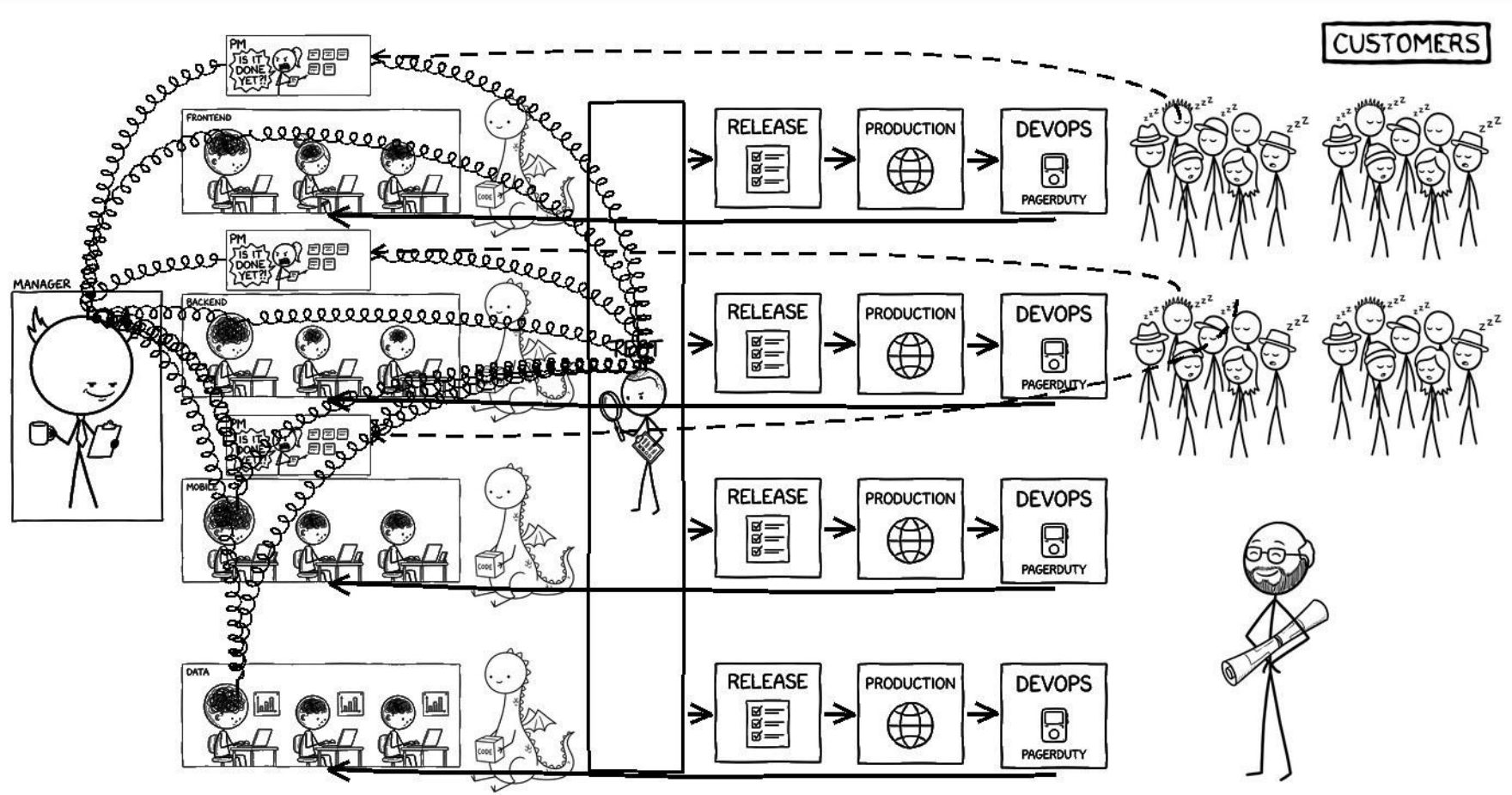

PM
CUSTOMERS
TL/M
FRONTEND
RELEASE
CODE
BACKEND
DEVOPS
PAGERDUTY
PRODUCTION

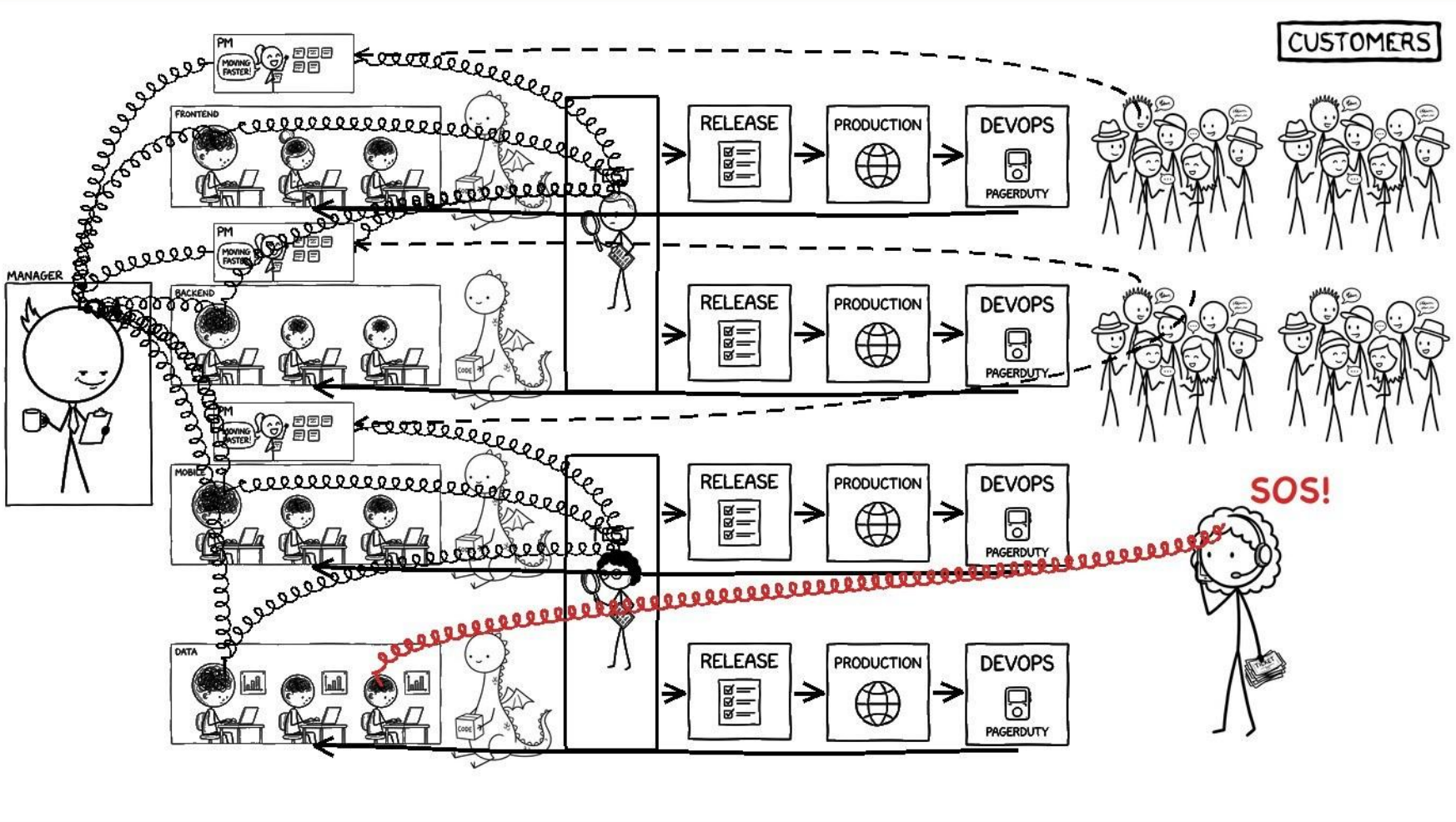

PM
WHY DIDN'T YOU CATCH THIS?!
CUSTOMERS
TL/M
I DON'T UNDERSTAND IT ANYMORE!
FRONTEND
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
CODE
DEVOPS
PAGERDUTY
PRODUCTION
BACKEND

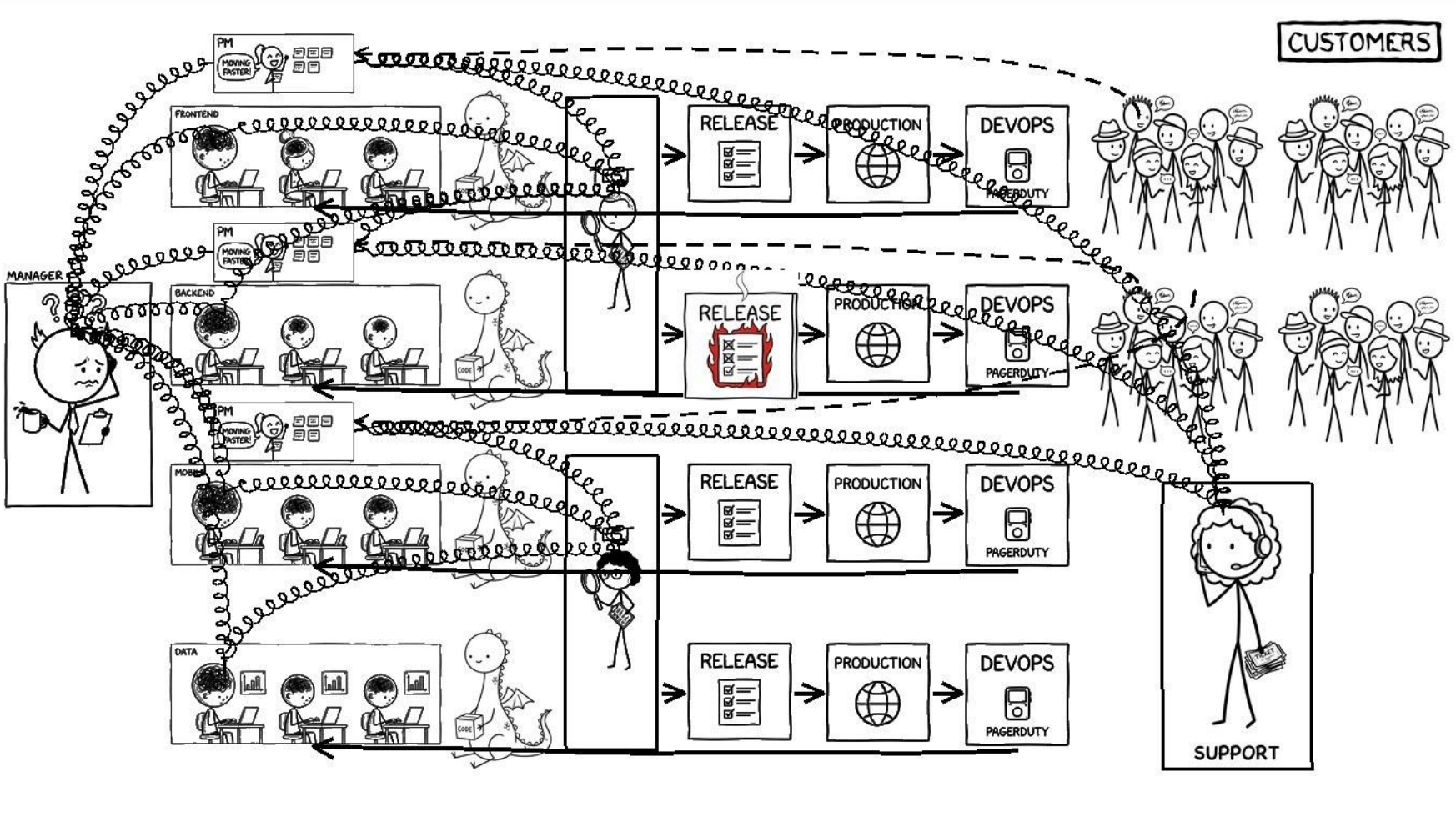

PM
CUSTOMERS
TL/M
FRONTEND
TEST
RELEASE
CODE
DEVOPS
PRODUCTION
BACKEND
PAGERDUTY

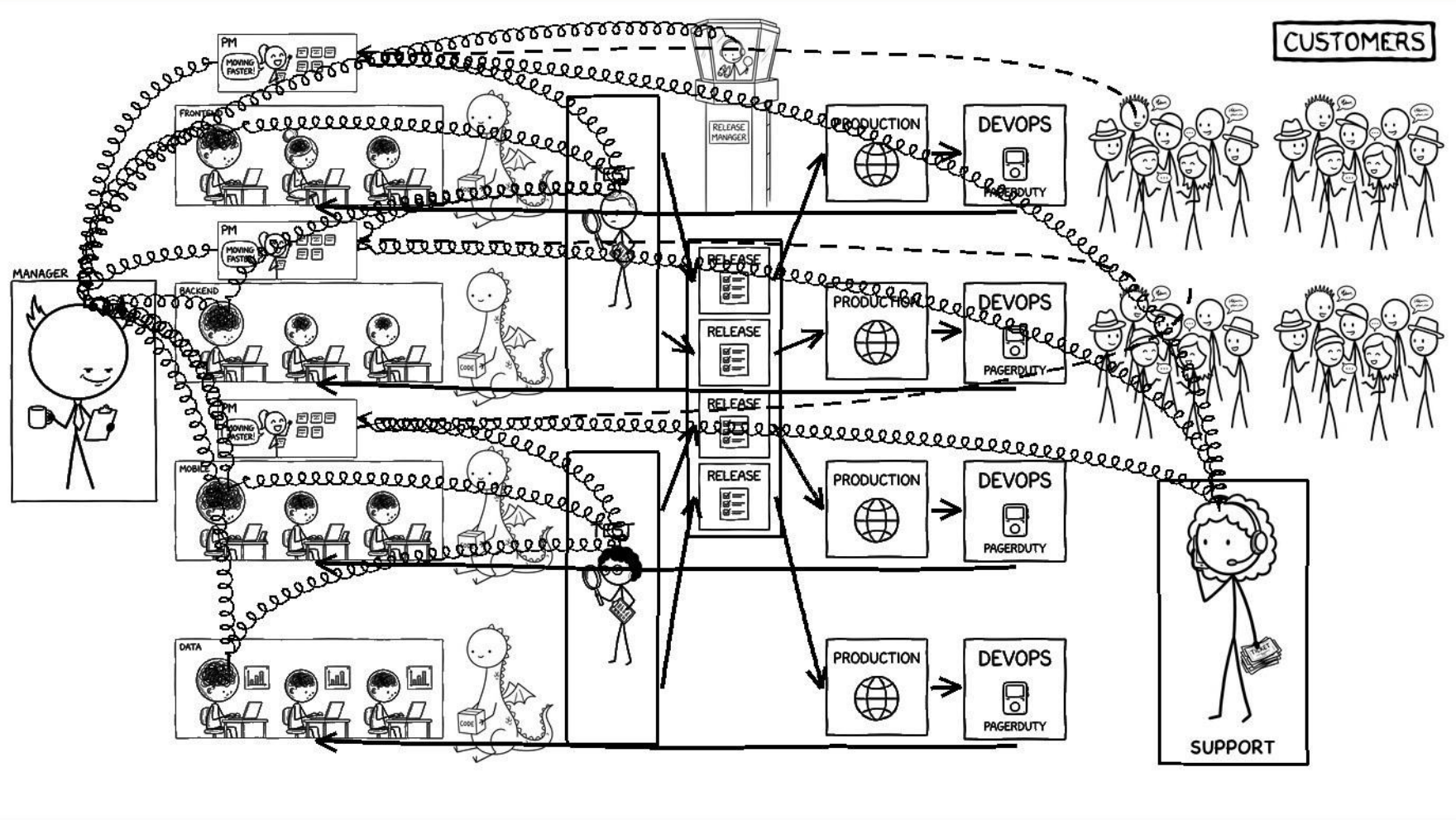

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
BYE!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

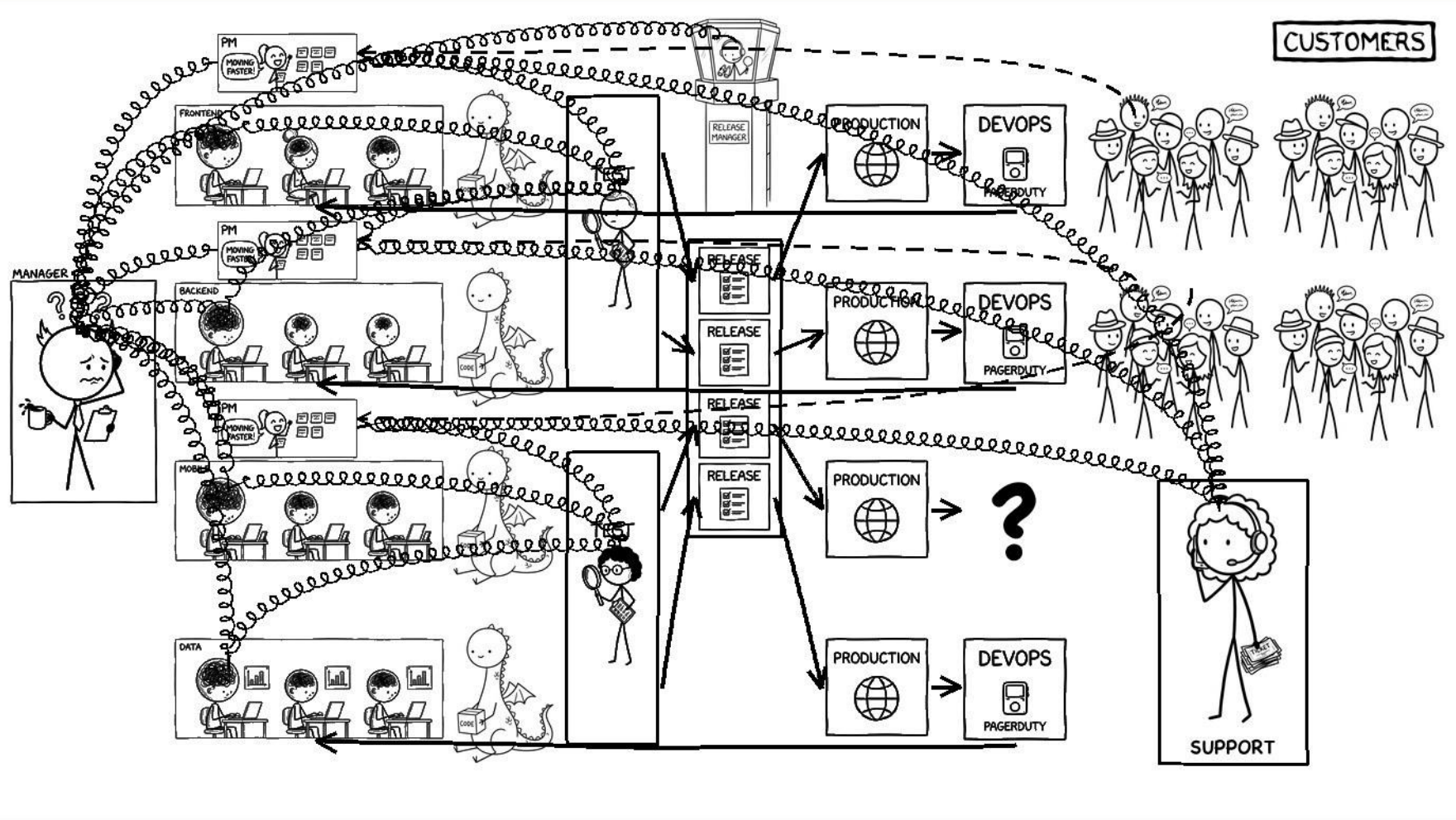

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
WE MUST SLAY THE DRAGON!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

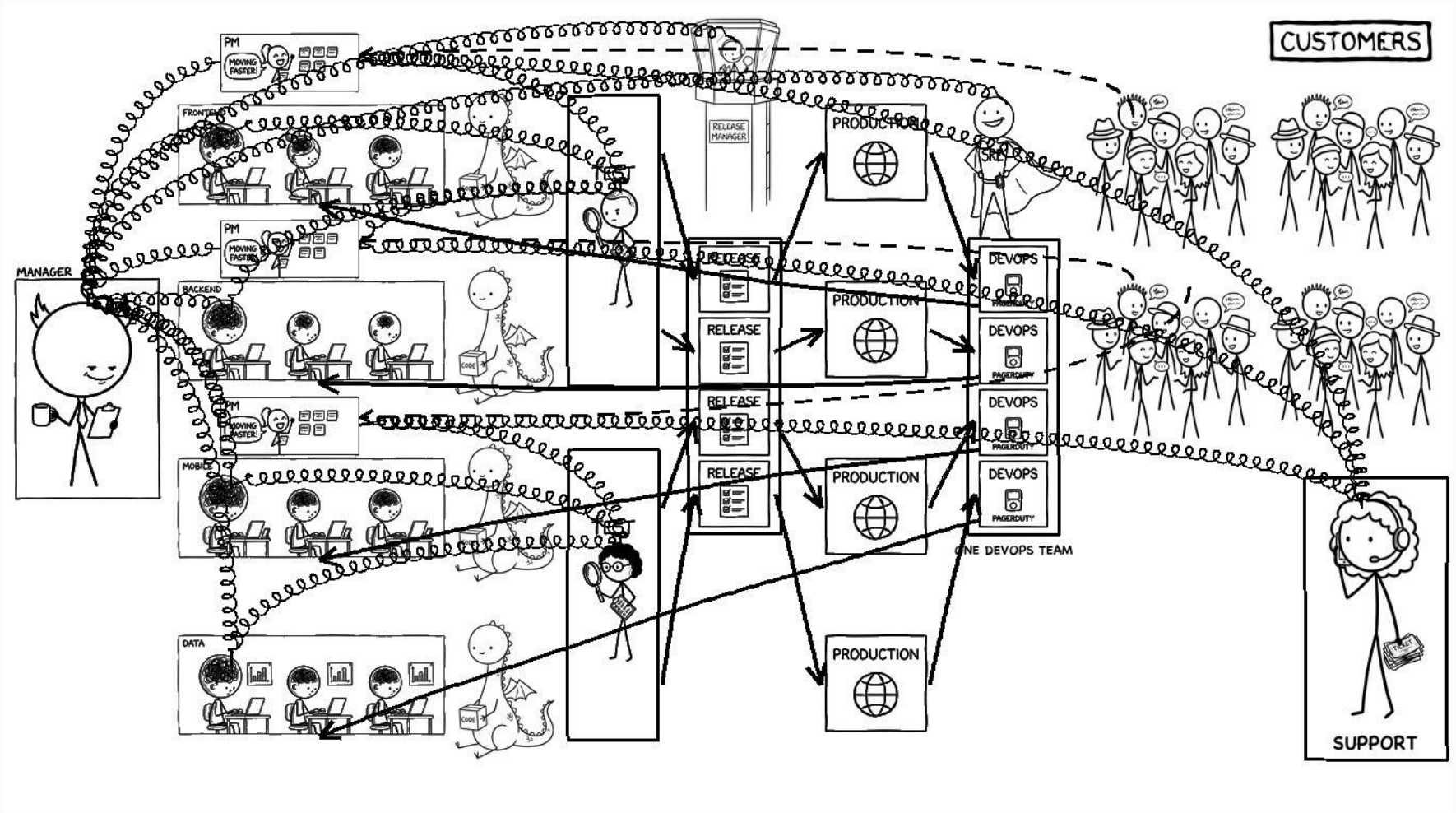

CUSTOMERS
PM
IS IT DONE YET?!
FRONTEND
BACKEND
MOBILE
DATA
MANAGER
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY

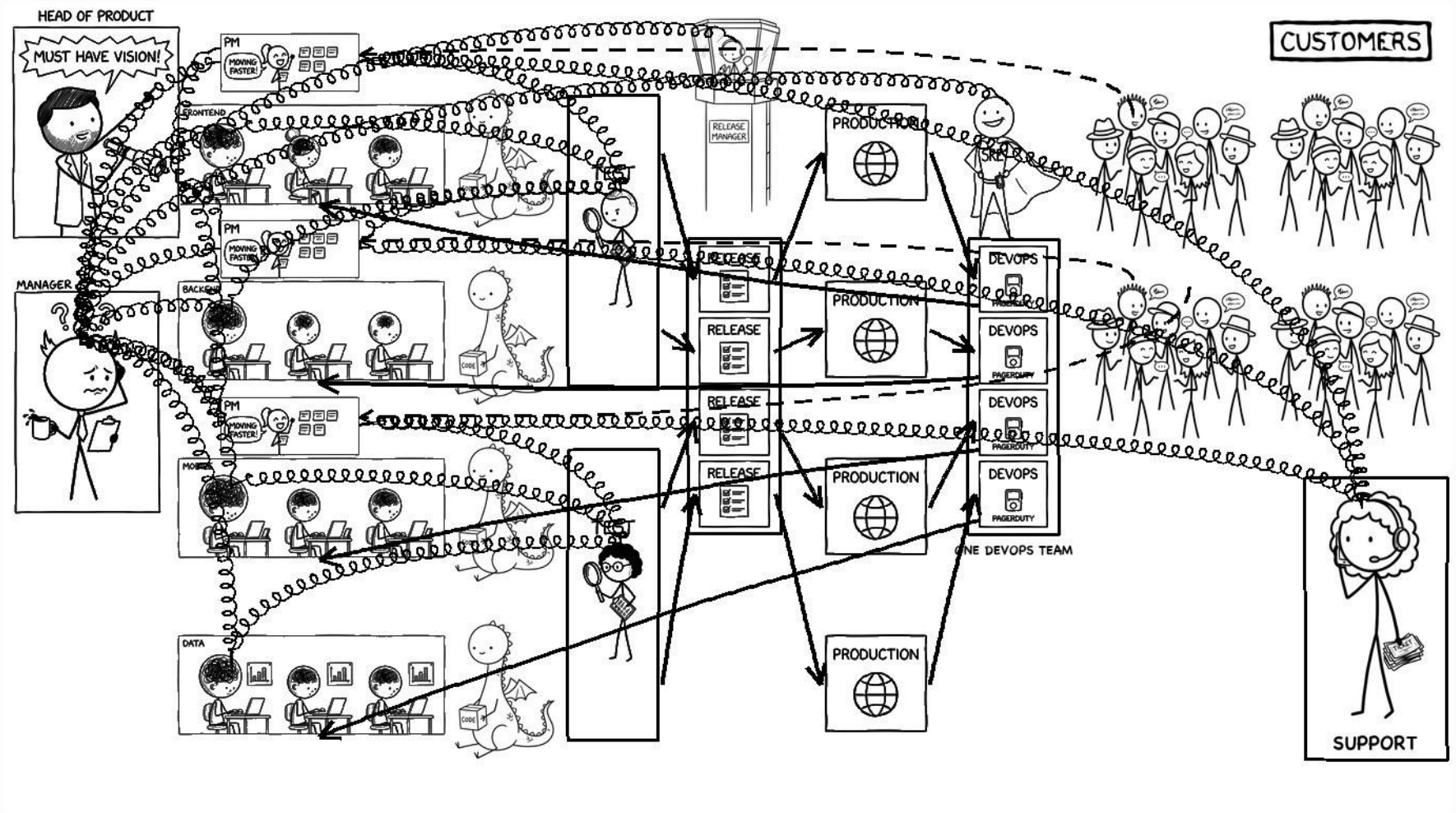

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
CODE
PM
MOVING FASTER!
MOBILE
DATA
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
SOS!


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
MANAGER
BACKEND
CODE
MOBILE
DATA
SUPPORT
TICKET

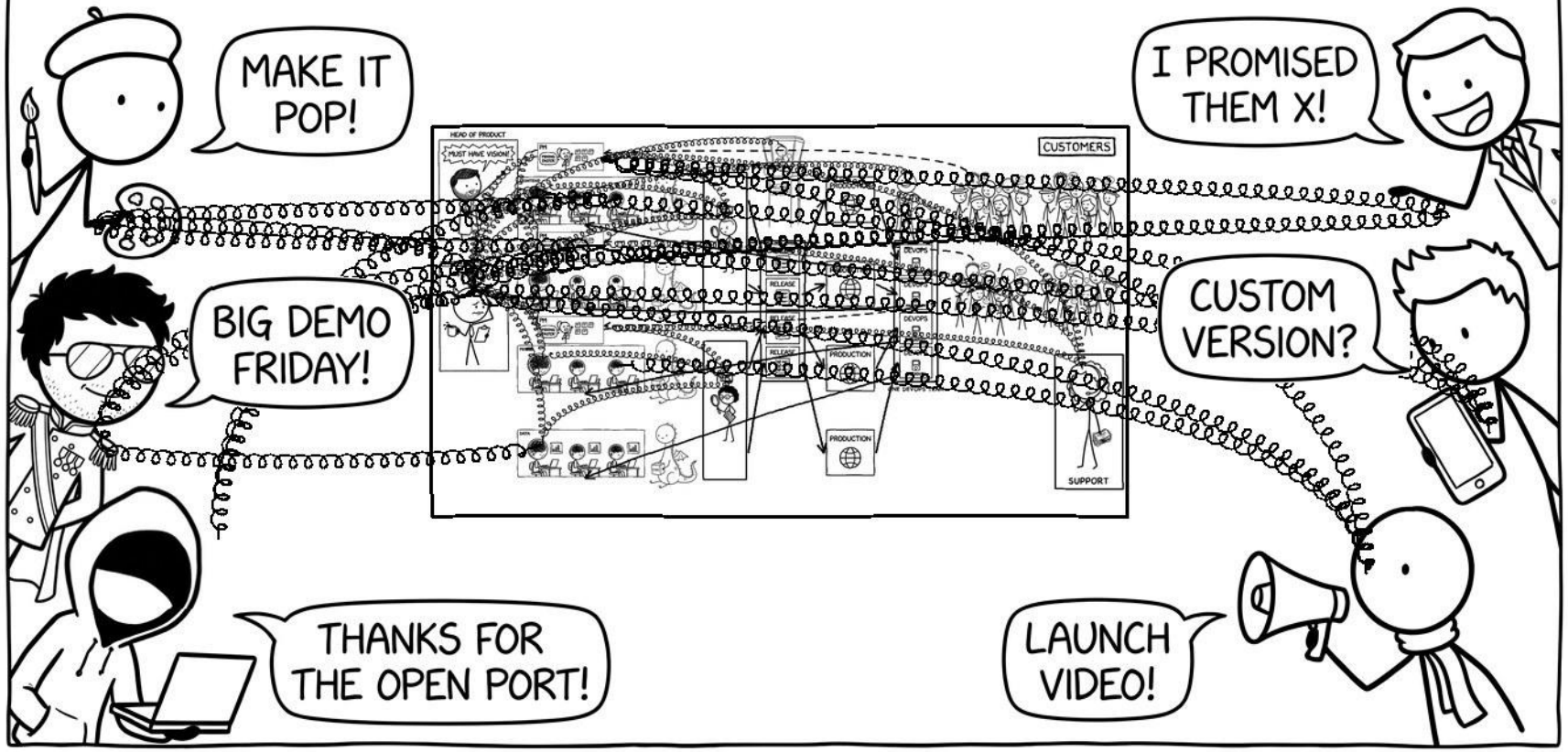

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
?
PRODUCTION
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
SRE
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
SUPPORT


HEAD OF PRODUCT
MUST HAVE VISION!
PM
MOVING FASTER!
FRONTEND
BACKEND
MOBILE
DATA
MANAGER
RELEASE MANAGER
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
CUSTOMERS
SUPPORT

OH, AND DON'T FORGET...

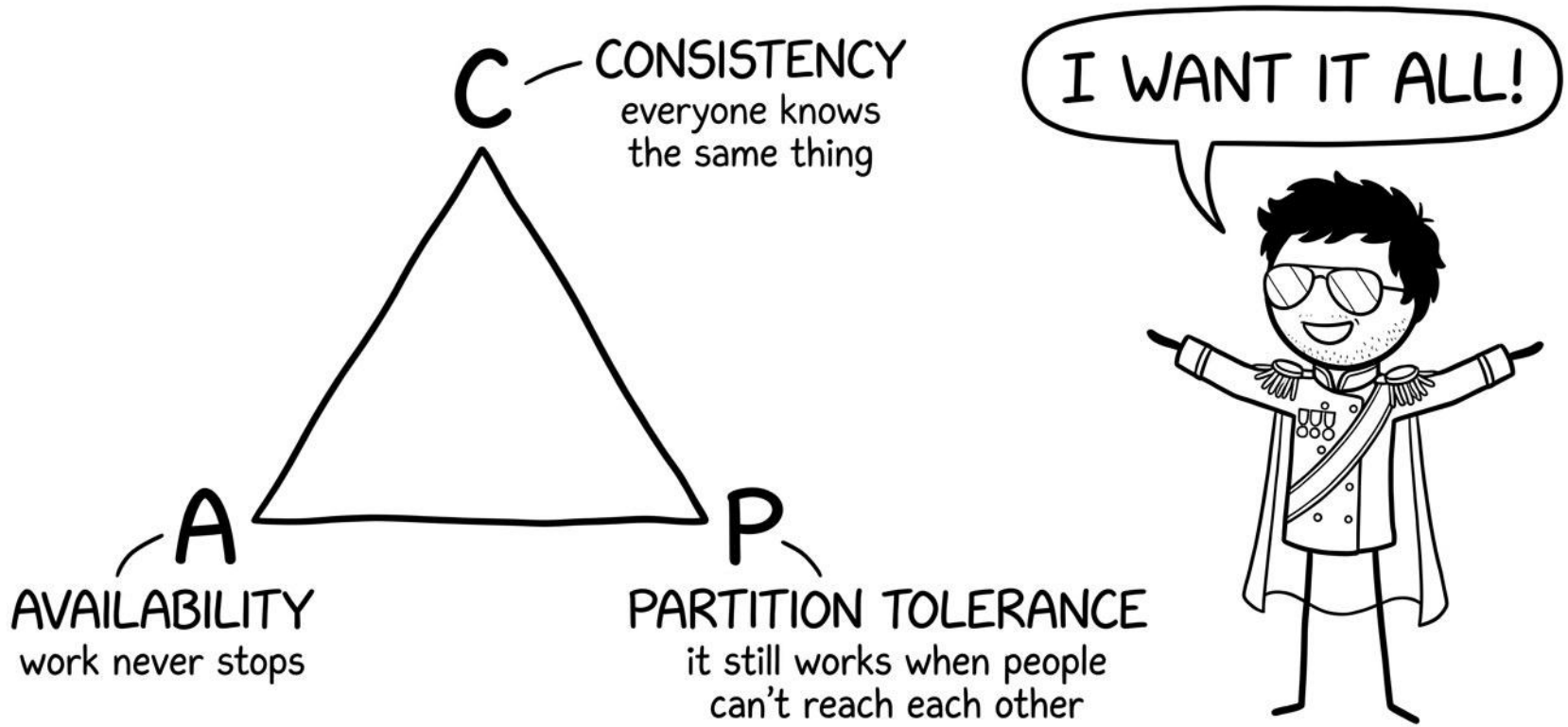


PART 2

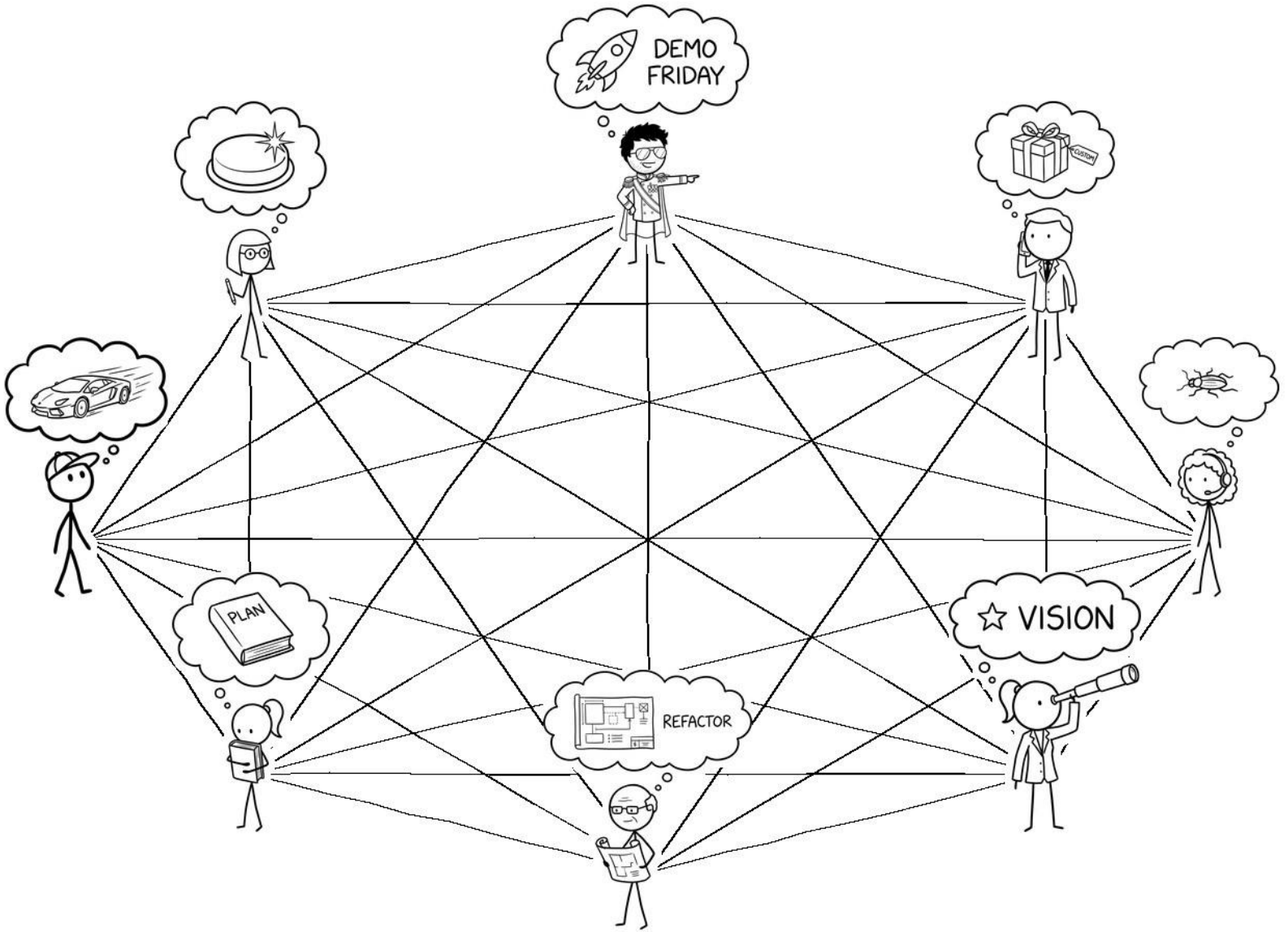
Why It's So Hard

Everyone's signal is different, and that can't be avoided

THE CAP THEOREM



CONSISTENCY: everyone knows the same thing



Every line is a conversation that has to happen.

2 people = 1 line

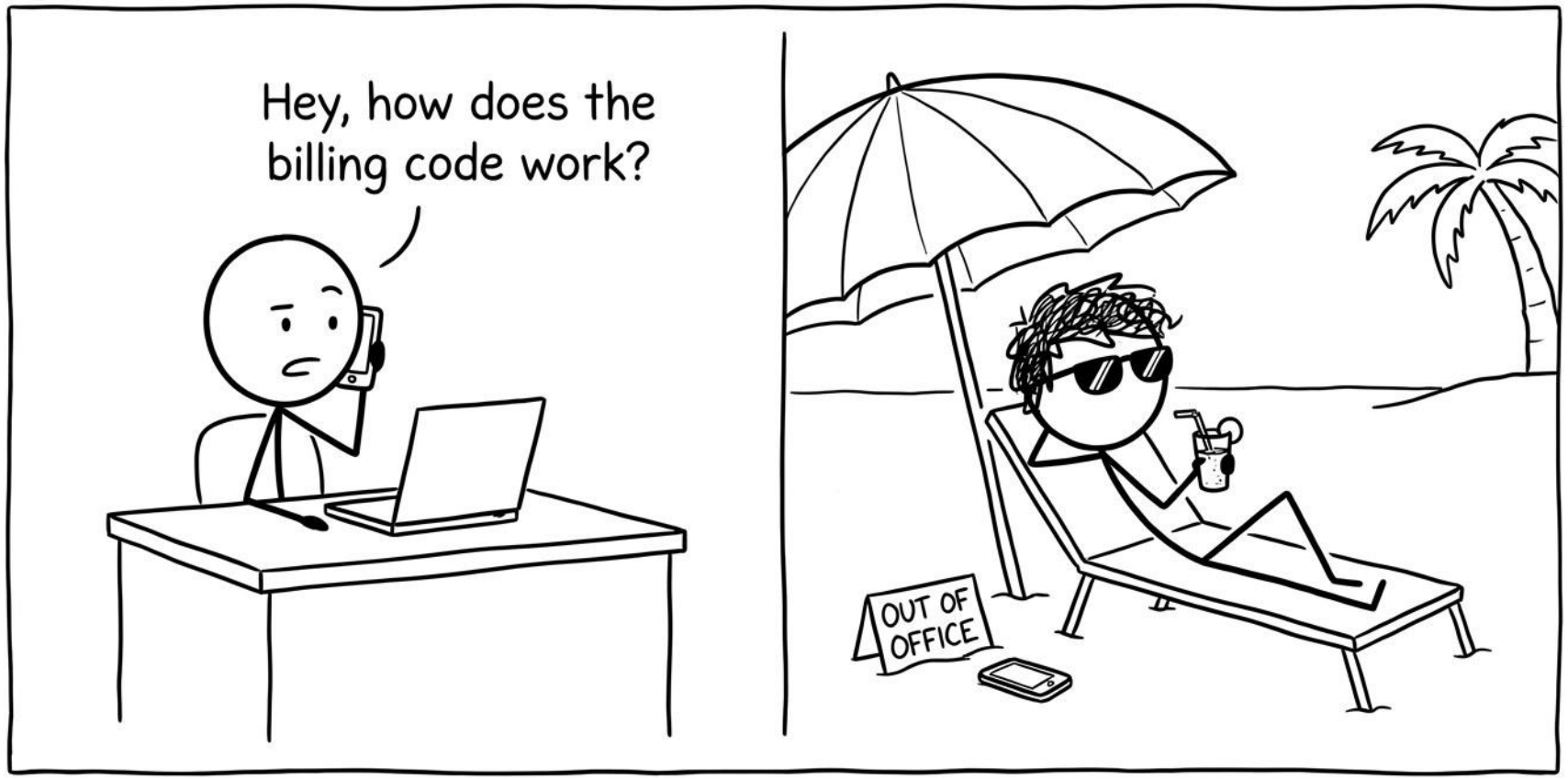
4 people = 6 lines

8 people = 28 lines

50 people = 1,225 lines

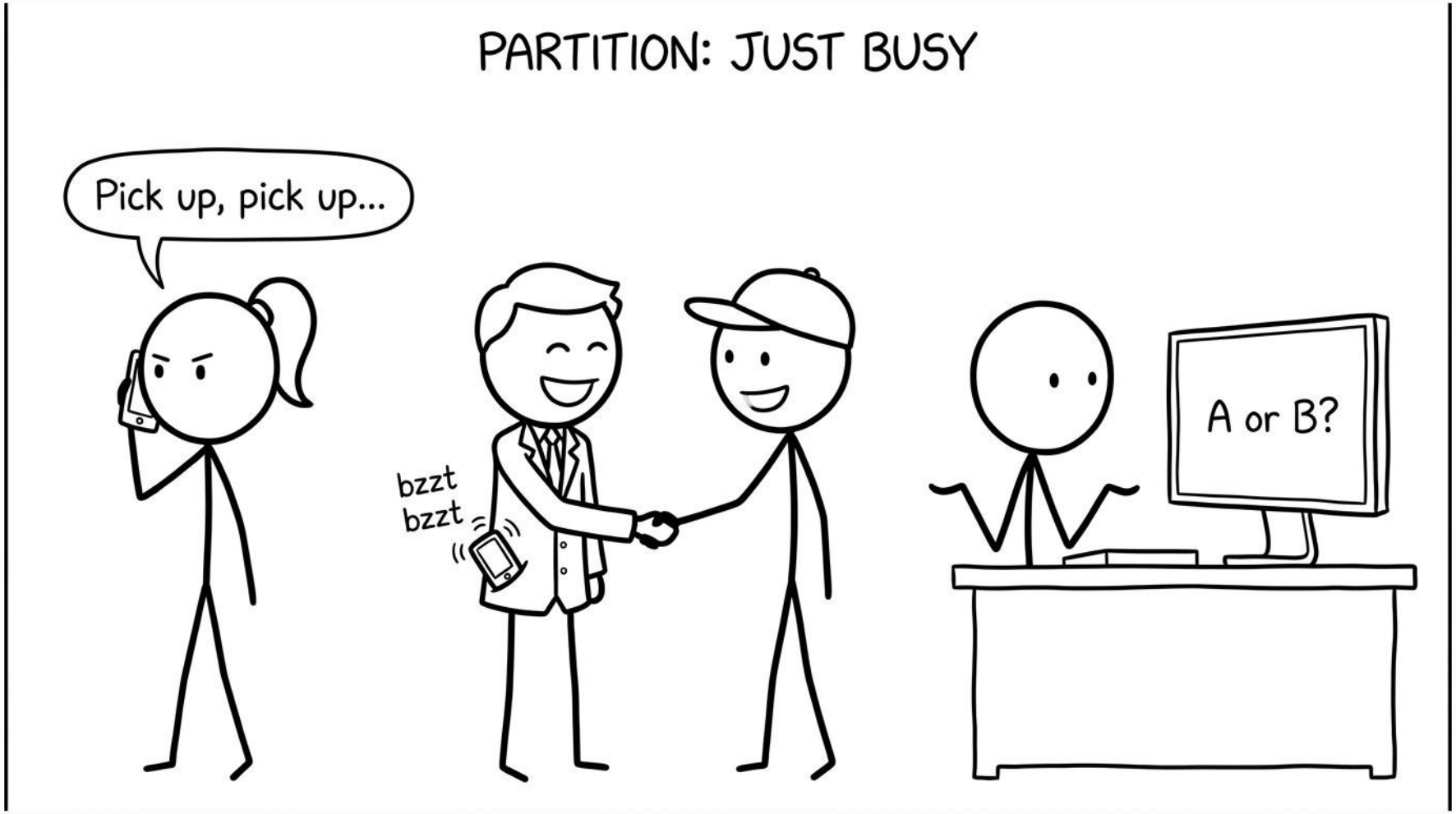
...and they all want something different.

PARTITION: ON VACATION

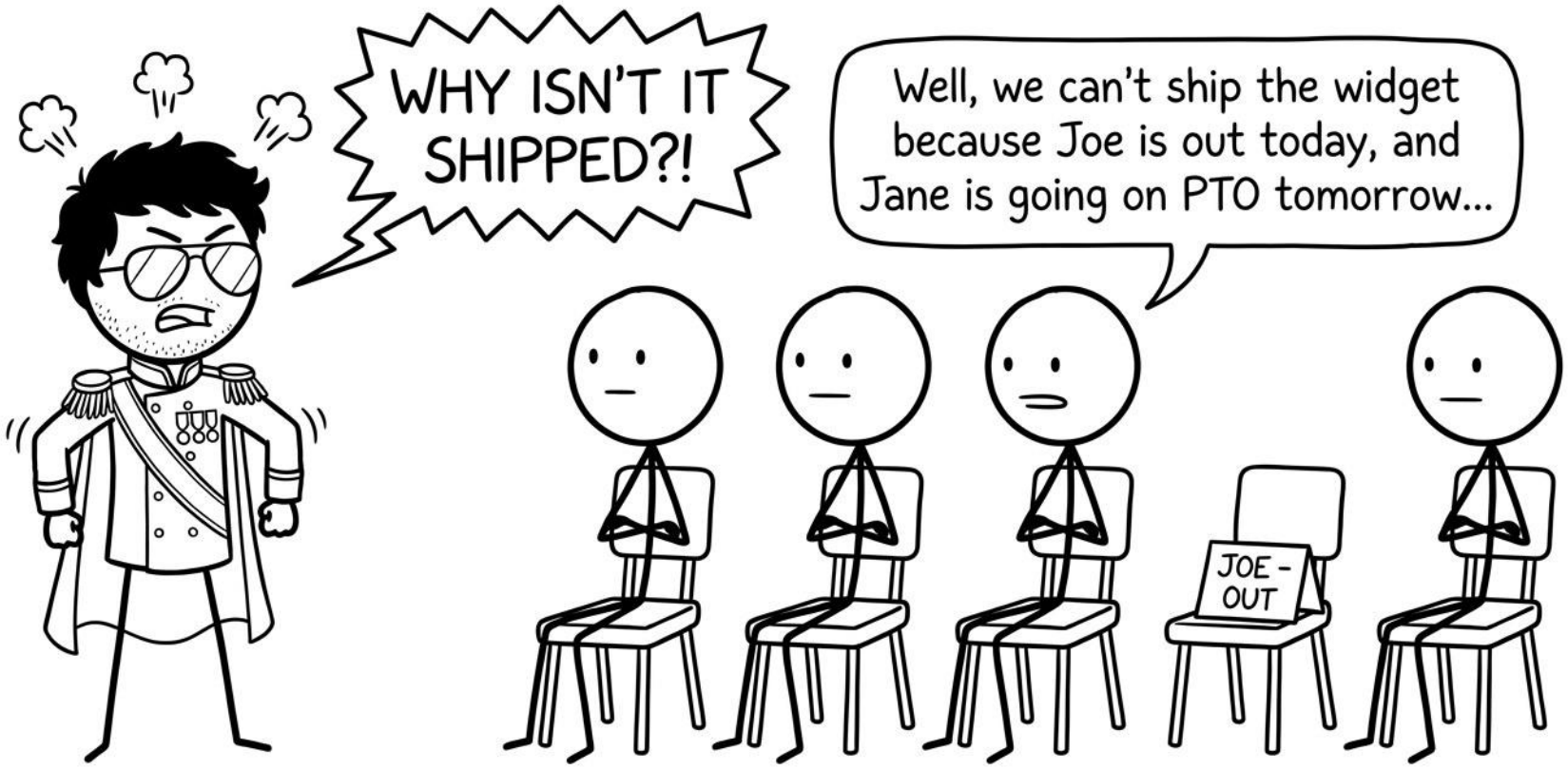


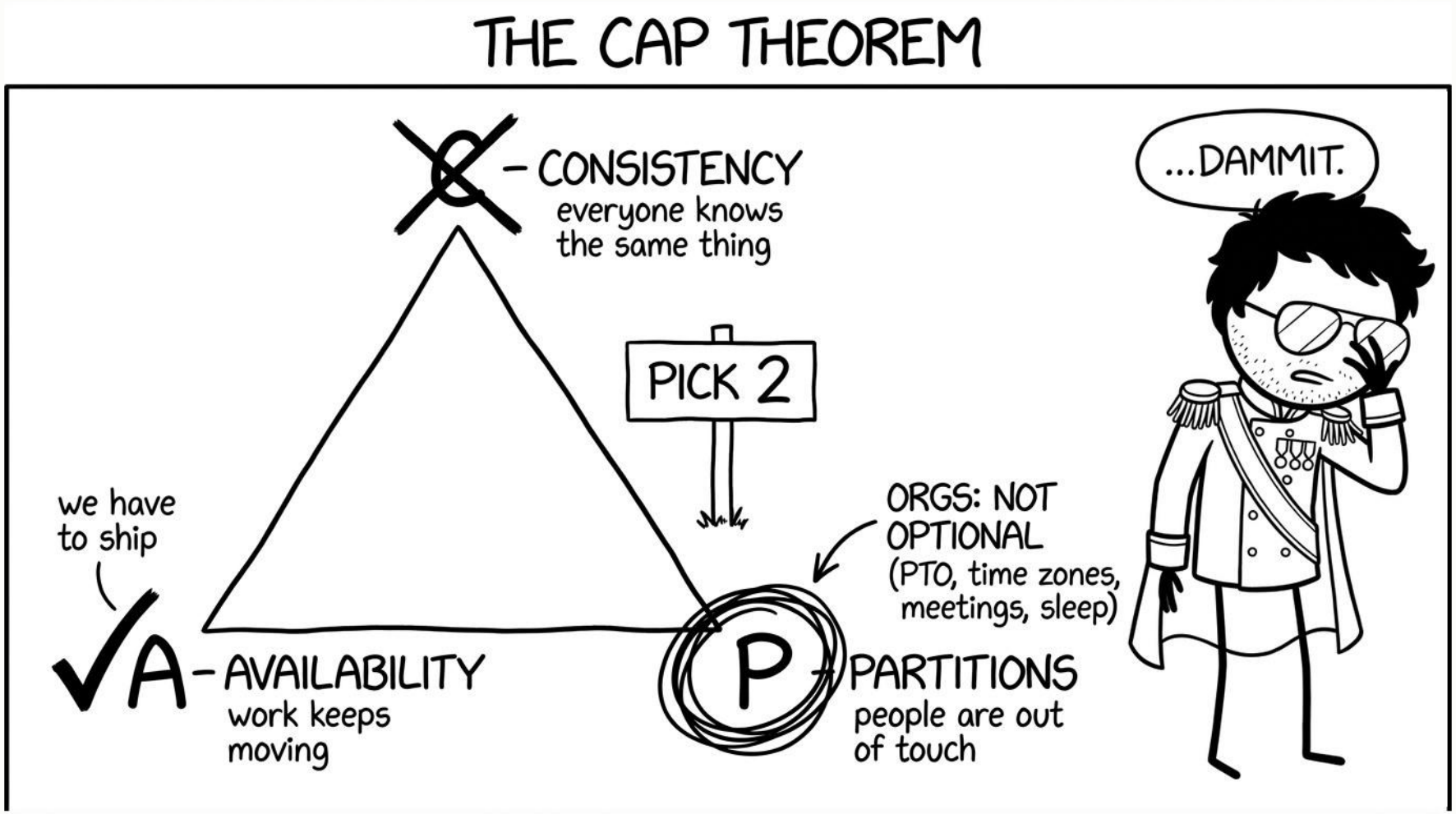
PARTITION: GONE FOR GOOD



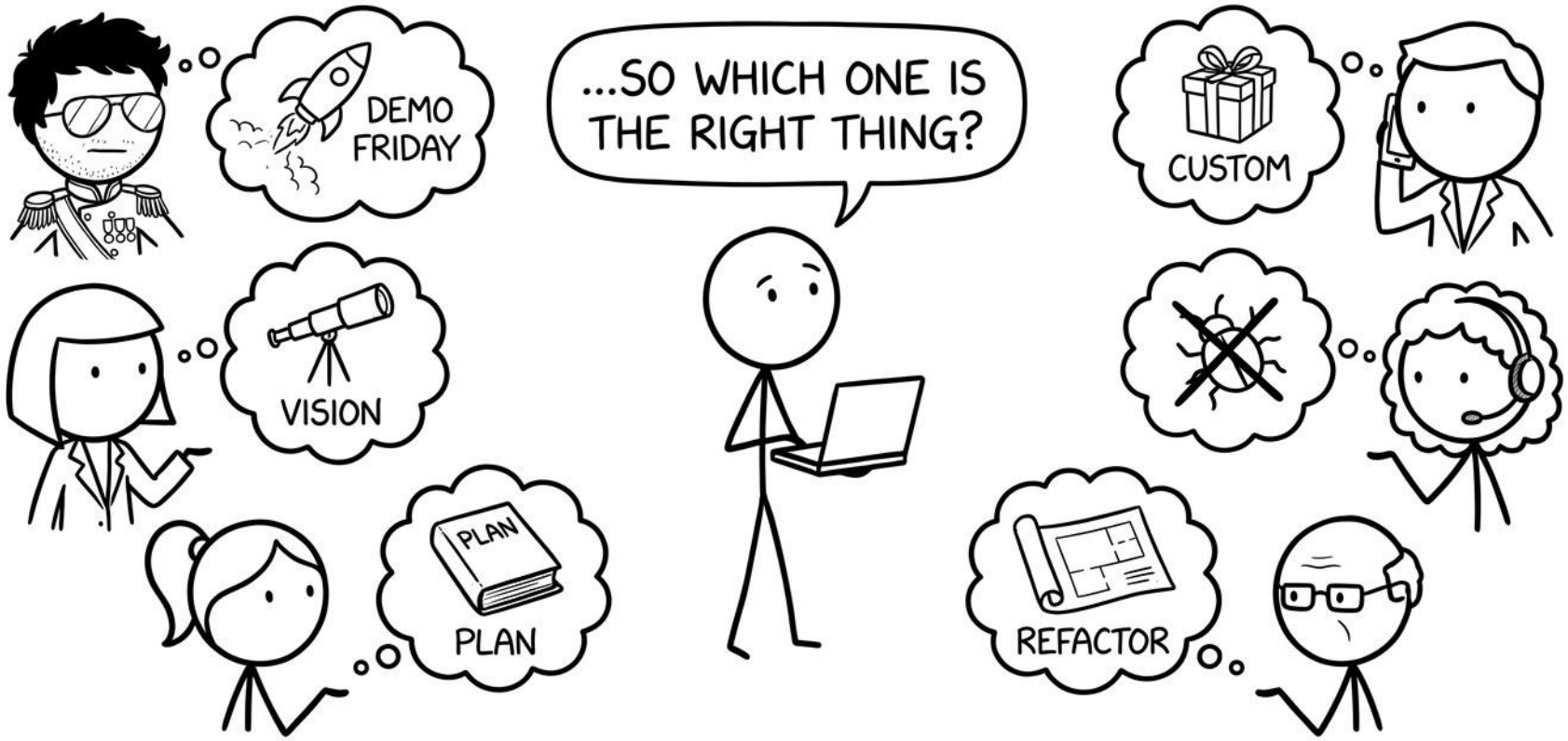

PARTITION: JUST BUSY
Pick up, pick up...
bzzt
bzzt
A or B?

AVAILABILITY: WORK NEVER STOPS... RIGHT?

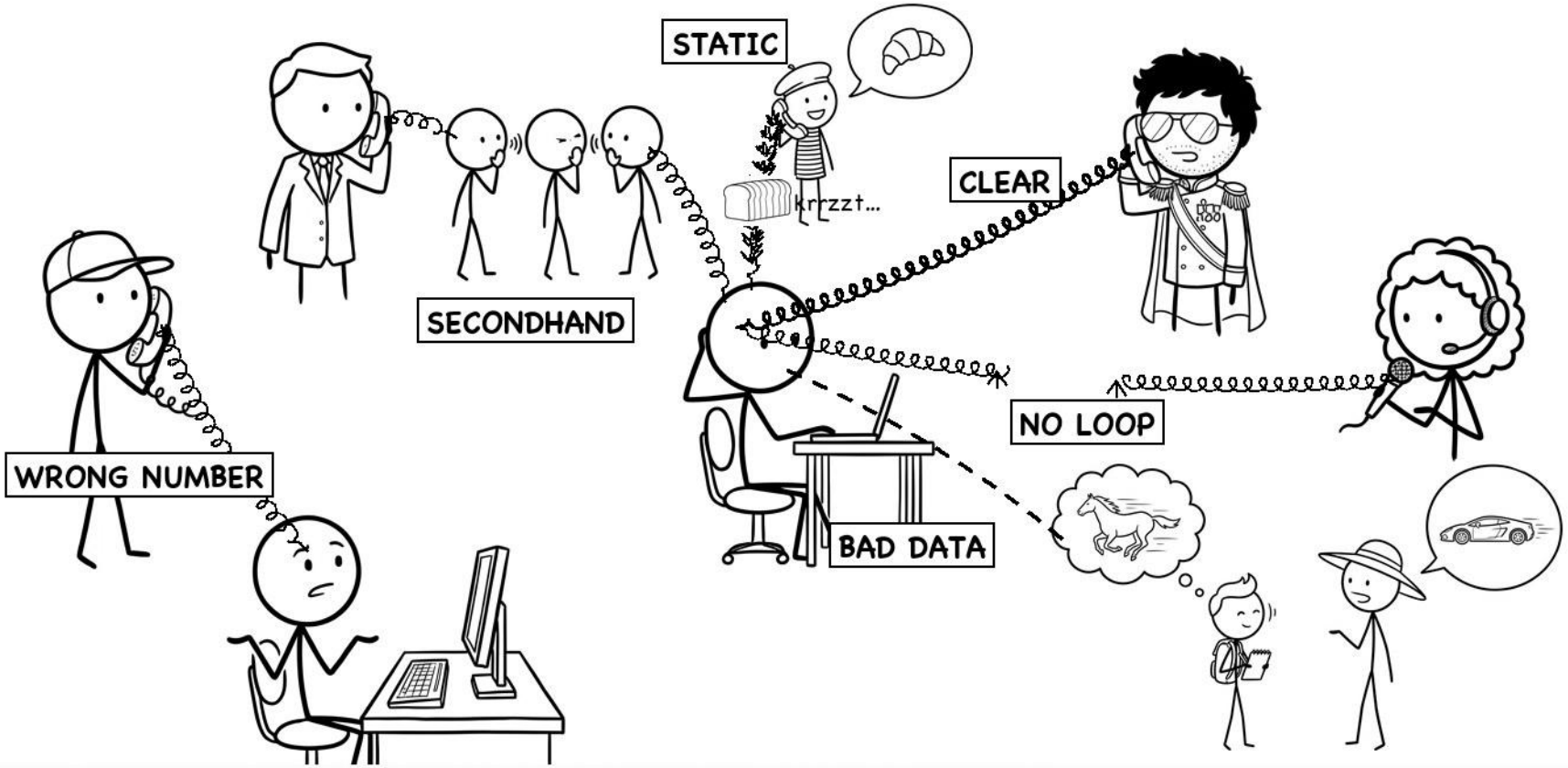


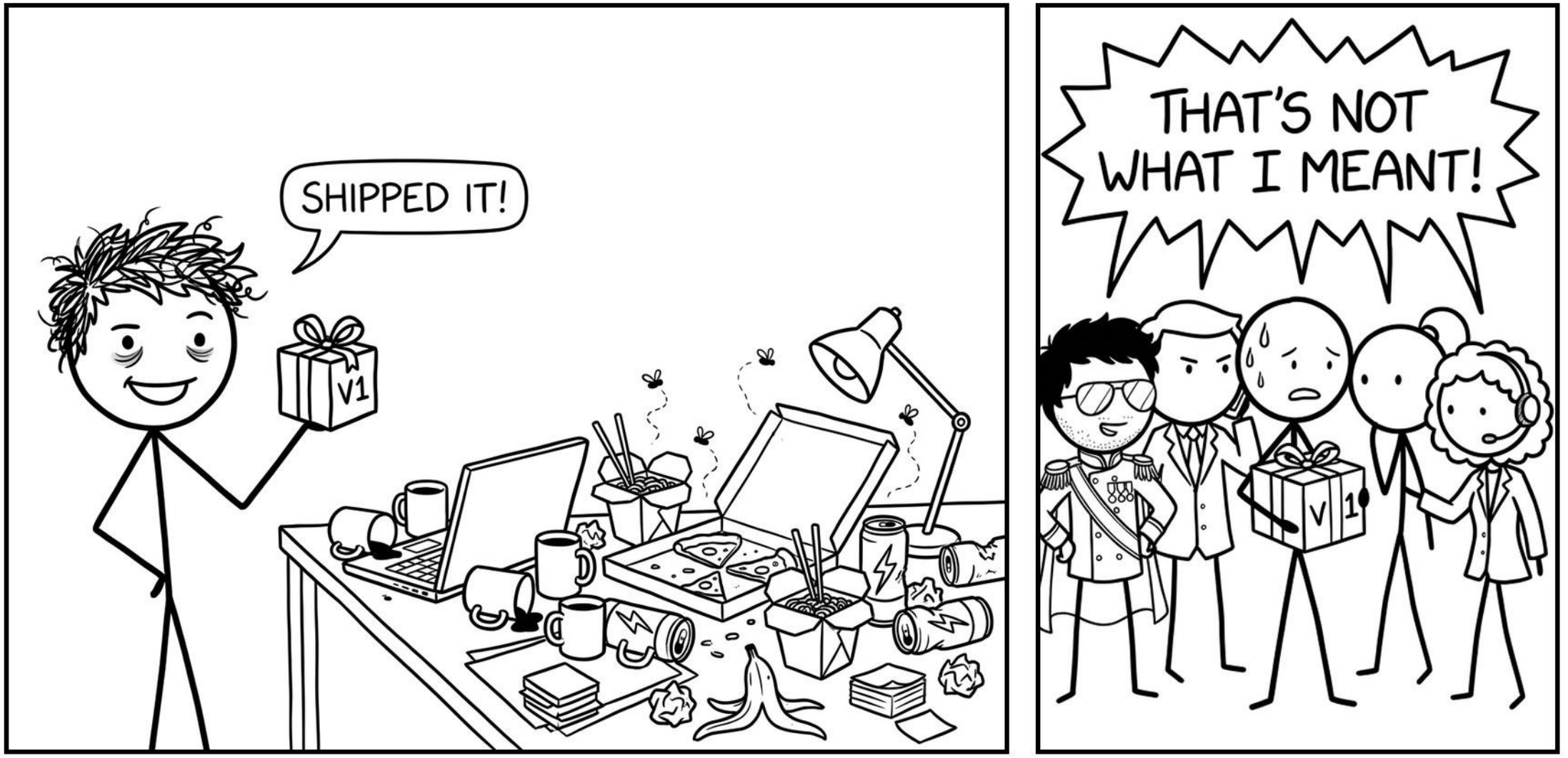

THE CAP THEOREM
C - CONSISTENCY
everyone knows the same thing
PICK 2
we have to ship
A - AVAILABILITY
work keeps moving
P - PARTITIONS
people are out of touch
ORGS: NOT OPTIONAL
(PTO, time zones, meetings, sleep)
...DAMMIT.

Everyone has a different "right thing".



...and every signal arrives differently.





And that's why we never build "the right thing".

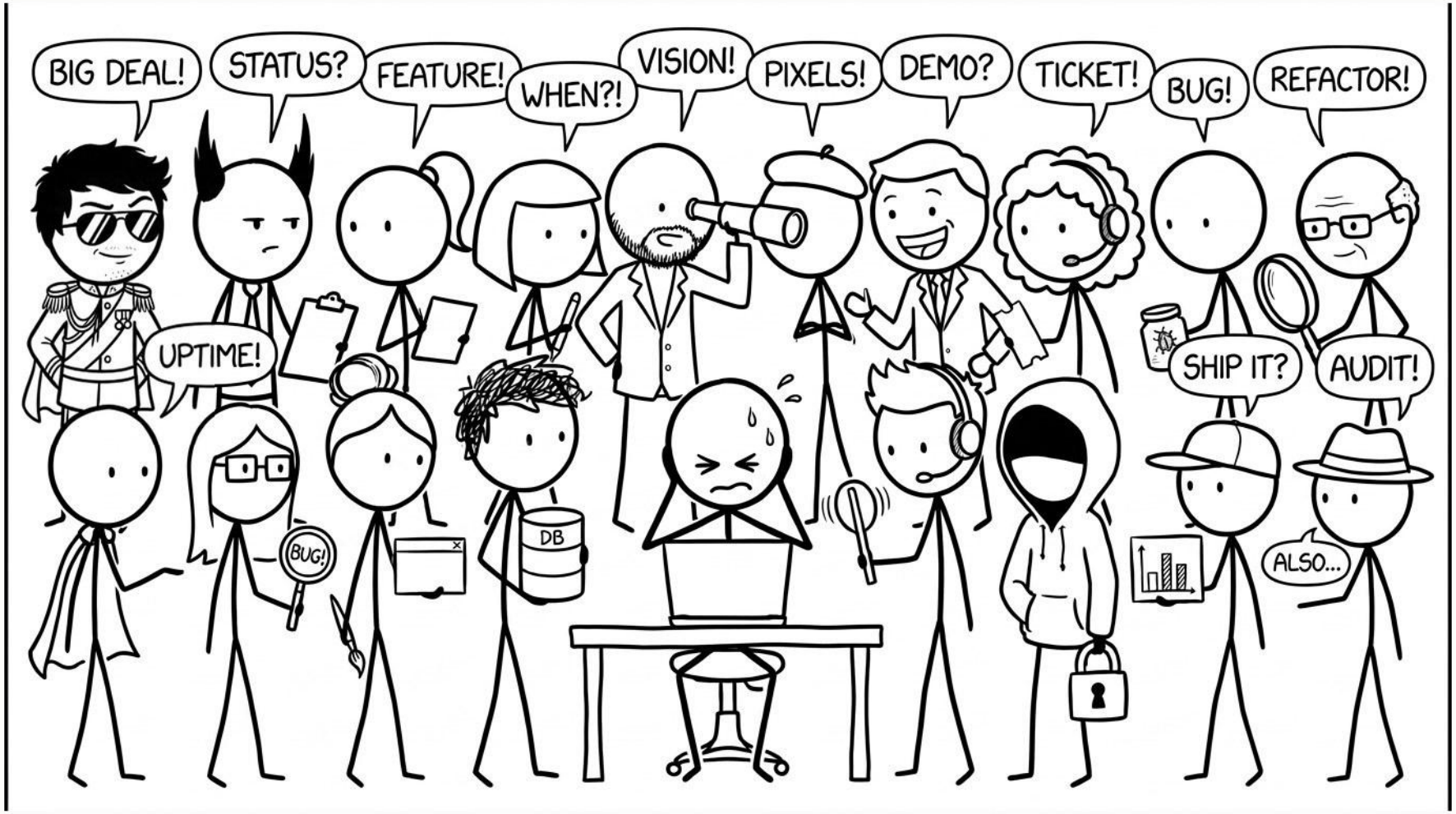
PART 3

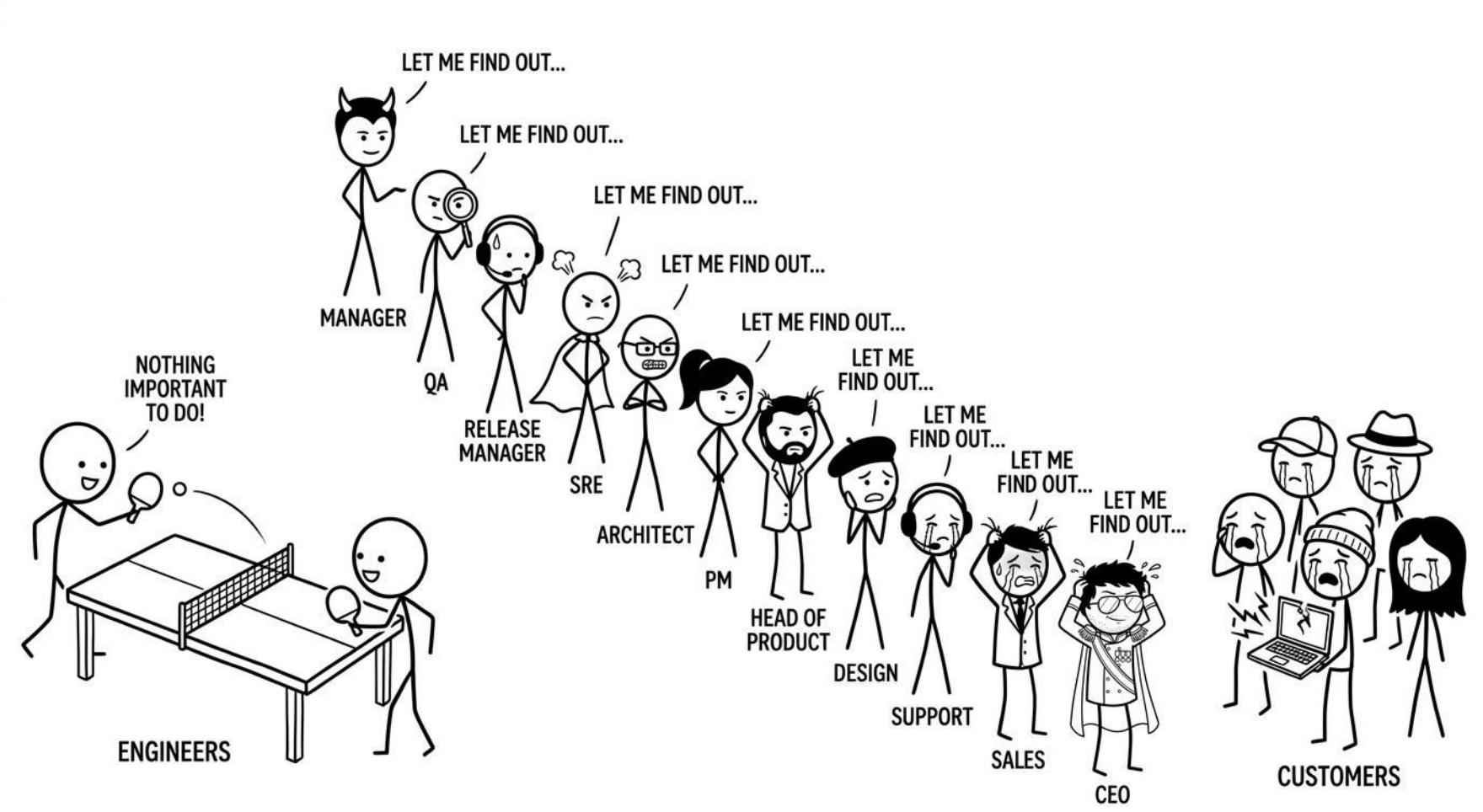
Common Org Failure Modes

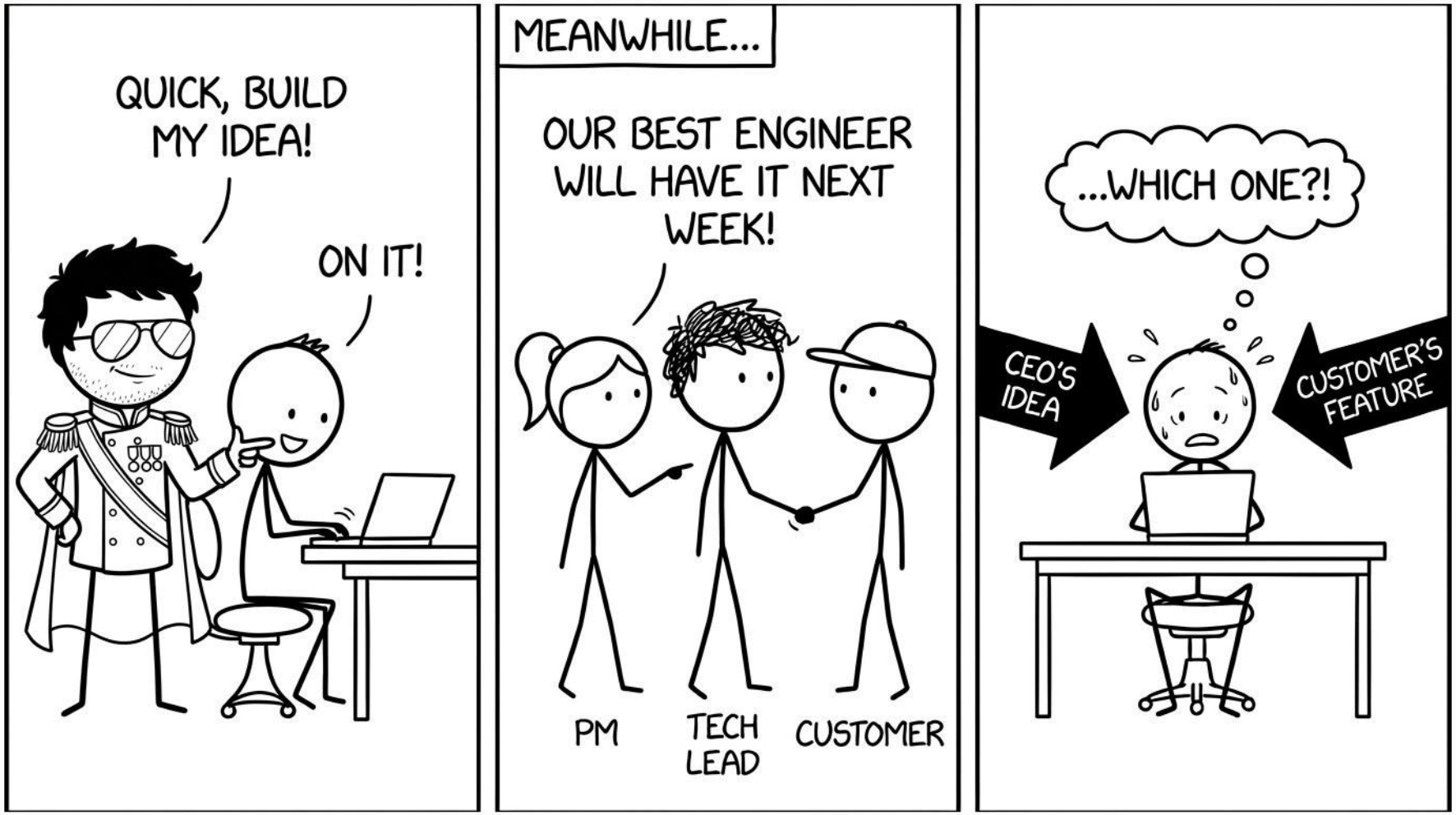
What it looks like when the org around engineering goes wrong

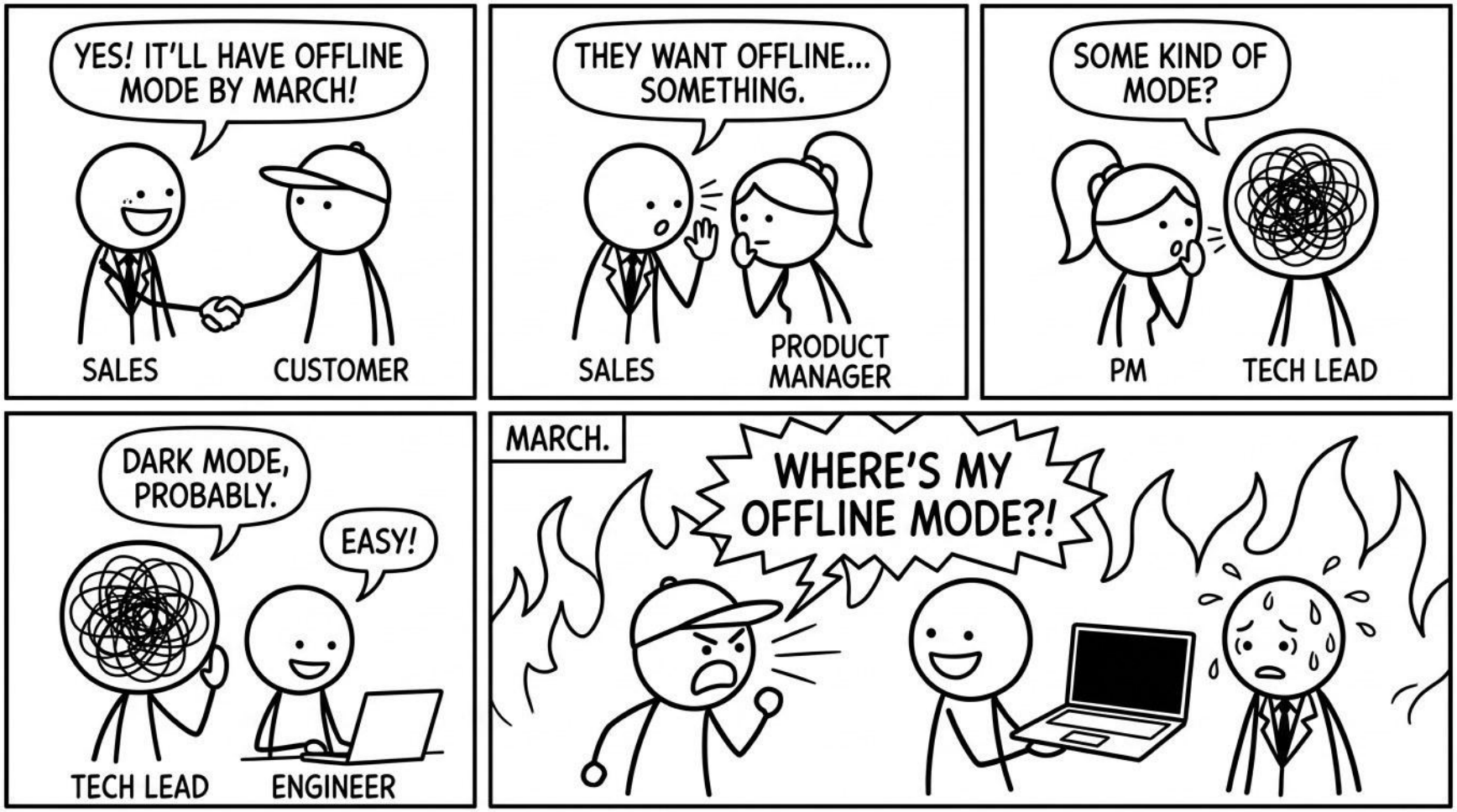


Guess there's nothing to do...


BIG DEAL!
STATUS?
FEATURE!
WHEN?!
VISION!
PIXELS!
DEMO?
TICKET!
BUG!
REFACTOR!
UPTIME!
BUG!
DB
SHIP IT?
AUDIT!
ALSO...


NOTHING IMPORTANT TO DO!
ENGINEERS
LET ME FIND OUT...
MANAGER
LET ME FIND OUT...
QA
LET ME FIND OUT...
RELEASE MANAGER
LET ME FIND OUT...
SRE
LET ME FIND OUT...
ARCHITECT
LET ME FIND OUT...
PM
LET ME FIND OUT...
HEAD OF PRODUCT
LET ME FIND OUT...
DESIGN
LET ME FIND OUT...
SUPPORT
LET ME FIND OUT...
SALES
LET ME FIND OUT...
CEO
CUSTOMERS


QUICK, BUILD MY IDEA!
ON IT!
MEANWHILE...
OUR BEST ENGINEER WILL HAVE IT NEXT WEEK!
PM
TECH LEAD
CUSTOMER
...WHICH ONE?!
CEO'S IDEA
CUSTOMER'S FEATURE


YES! IT'LL HAVE OFFLINE MODE BY MARCH!
SALES
CUSTOMER
THEY WANT OFFLINE... SOMETHING.
SALES
PRODUCT MANAGER
SOME KIND OF MODE?
PM
TECH LEAD
DARK MODE, PROBABLY.
EASY!
TECH LEAD
ENGINEER
MARCH.
WHERE'S MY OFFLINE MODE?!


REFACTOR FIRST!
FEATURE X!
BUG #42!
THE BIG CLIENT!
TOP PRIORITY
#4021
#4021
MUNCH MUNCH
POPCORN

Zoom Call: SEV-0 WAR ROOM

48 participants

3:07:42



Mute | Stop Video | Participants (48) | Share Screen | Chat (1,204) | Reactions | Leave

2 WEEKS BEFORE
THE BIG CUSTOMER LAUNCH



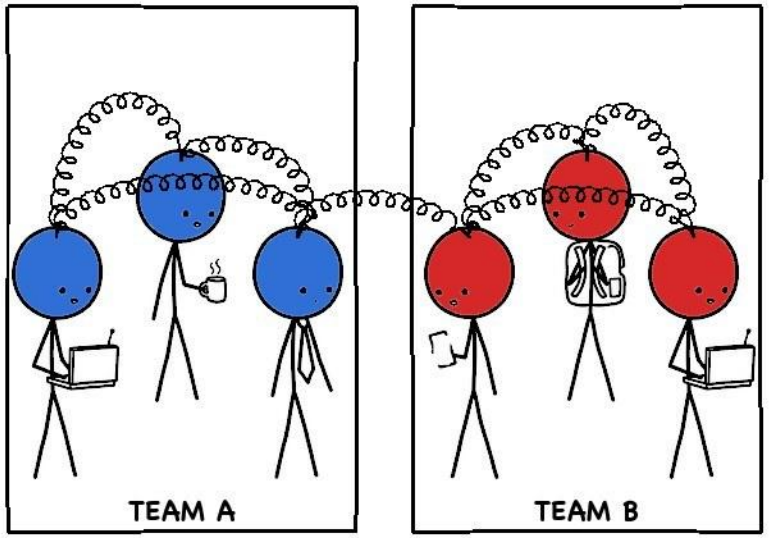
THE DAY BEFORE LAUNCH



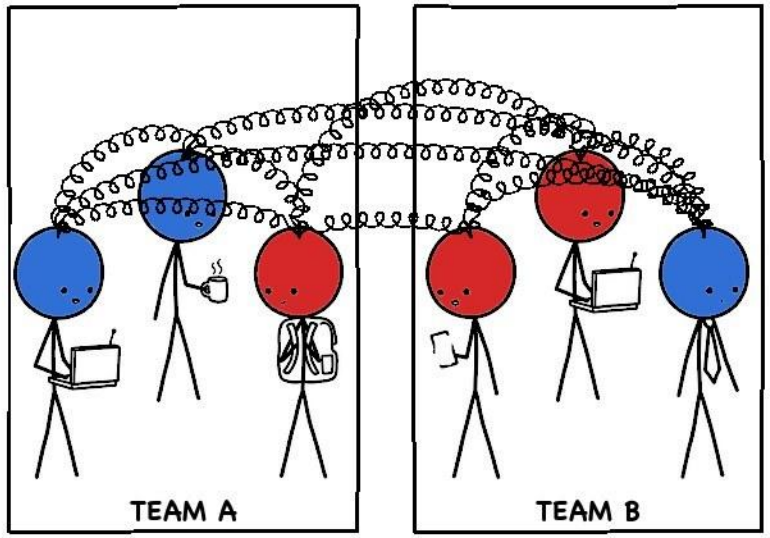
LAUNCH DAY



JANUARY



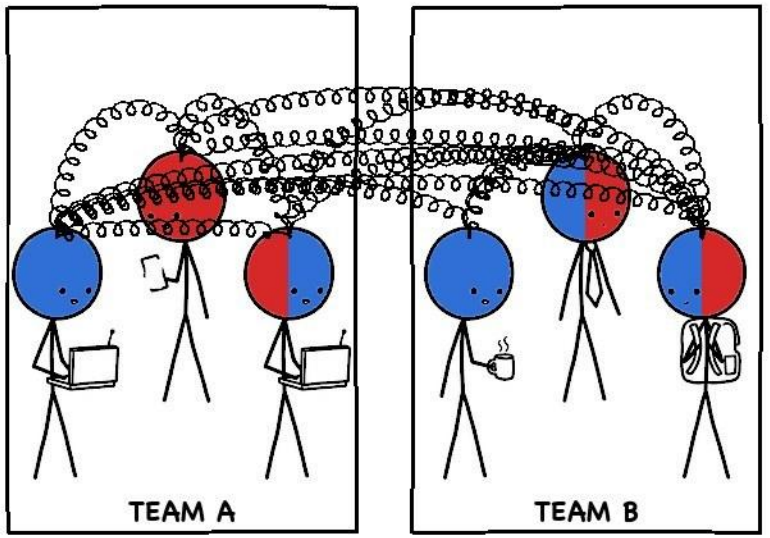
MARCH: REORG!



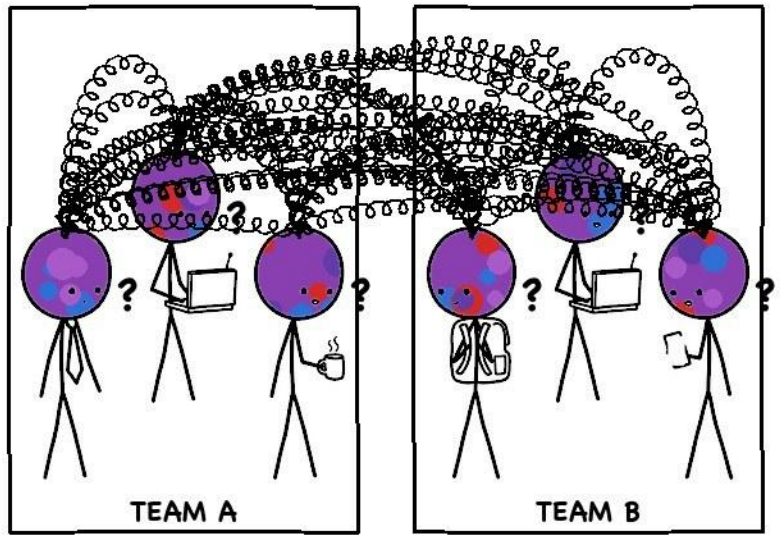
MEANWHILE, IN PRODUCT...

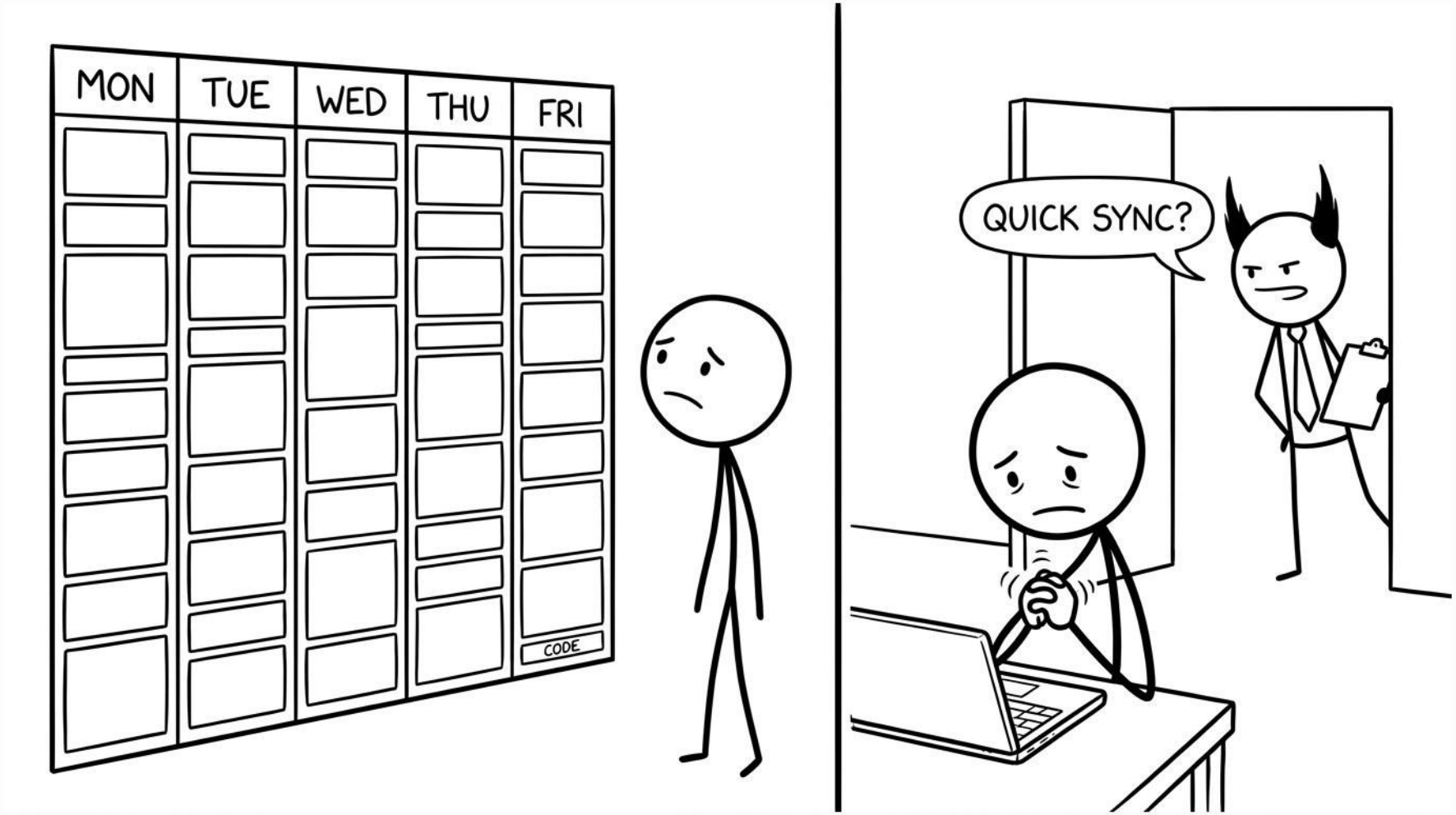


MAY: REORG!



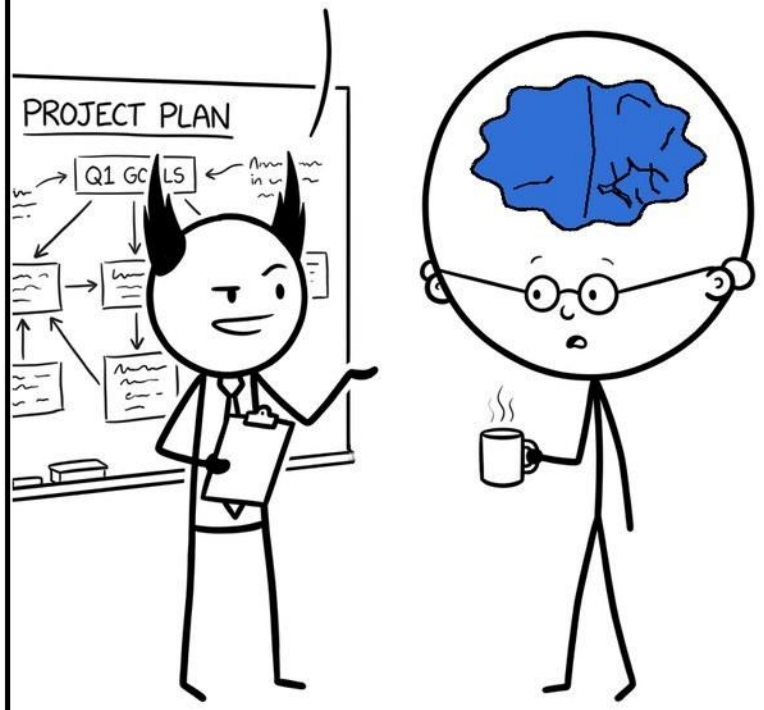
JULY: REORG!



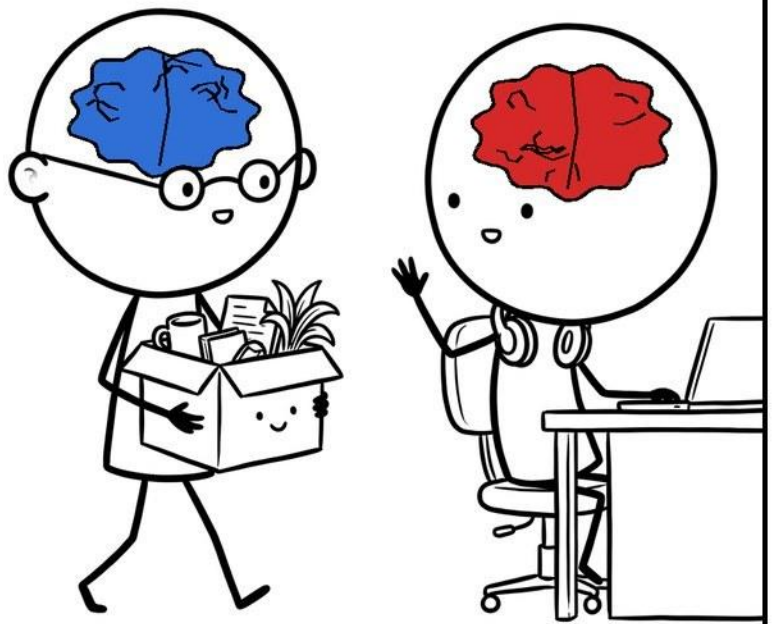

MON
TUE
WED
THU
FRI
CODE
QUICK SYNC?

MONDAY

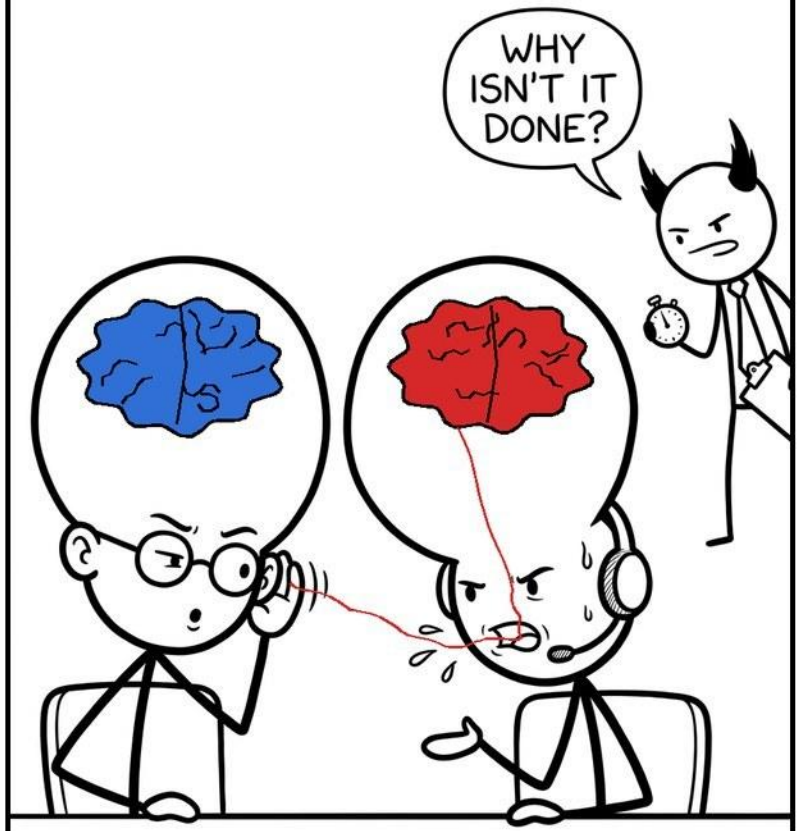
BORIS, I WANT YOU TO HELP OUT...



BORIS JOINS THE NEW TEAM



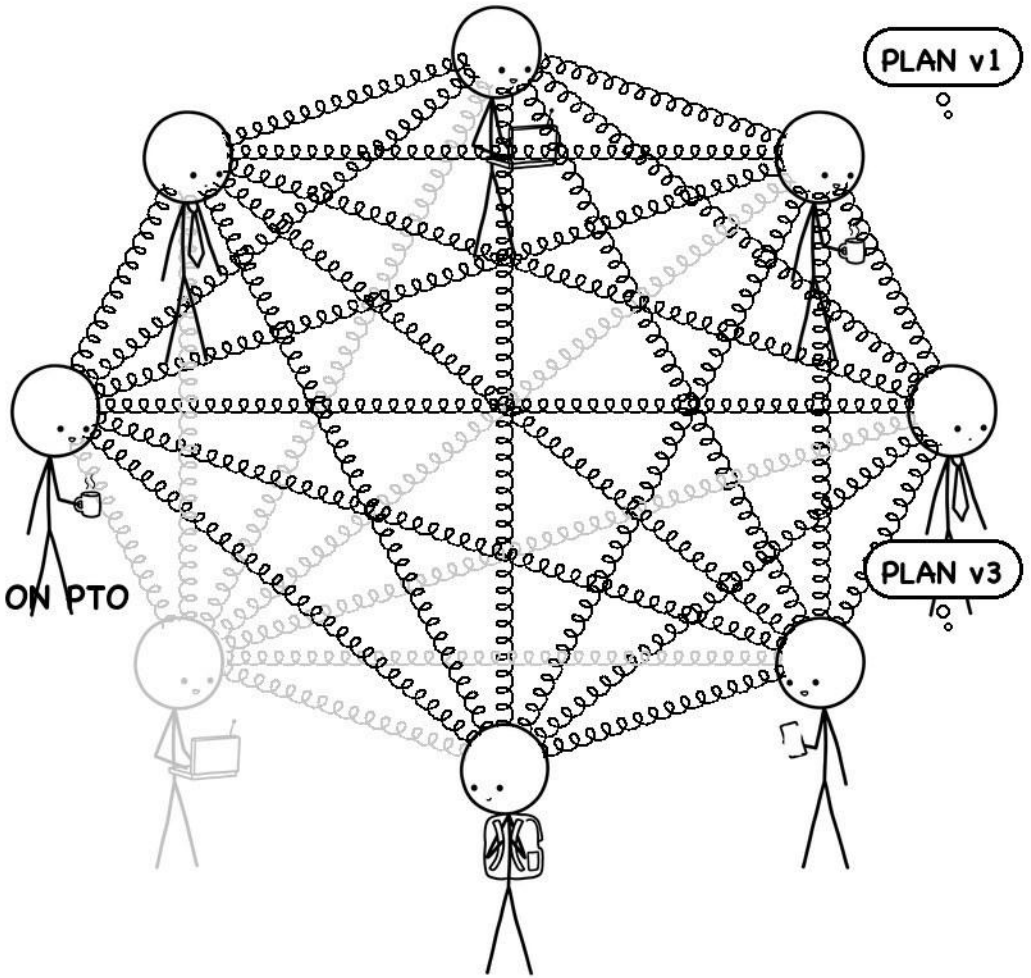
TWO WEEKS LATER



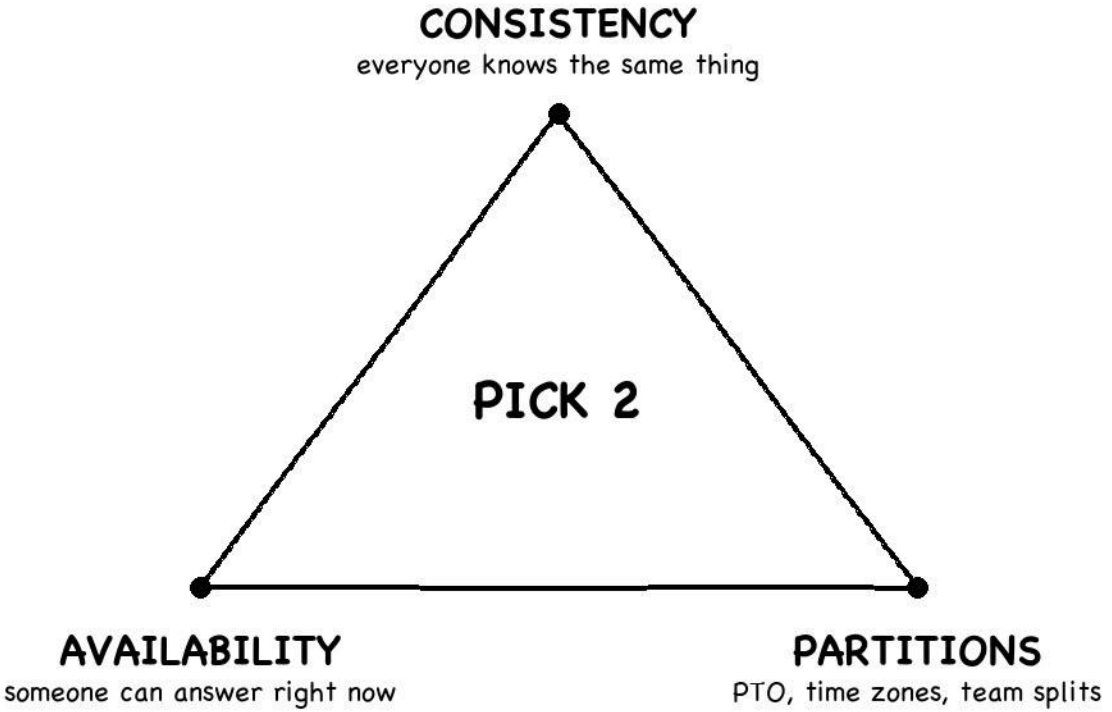
PART 4

Techniques for Solving This

YOUR ORG IS A DISTRIBUTED SYSTEM



8 people = 28 phone lines (N people ≈ N^2 lines)



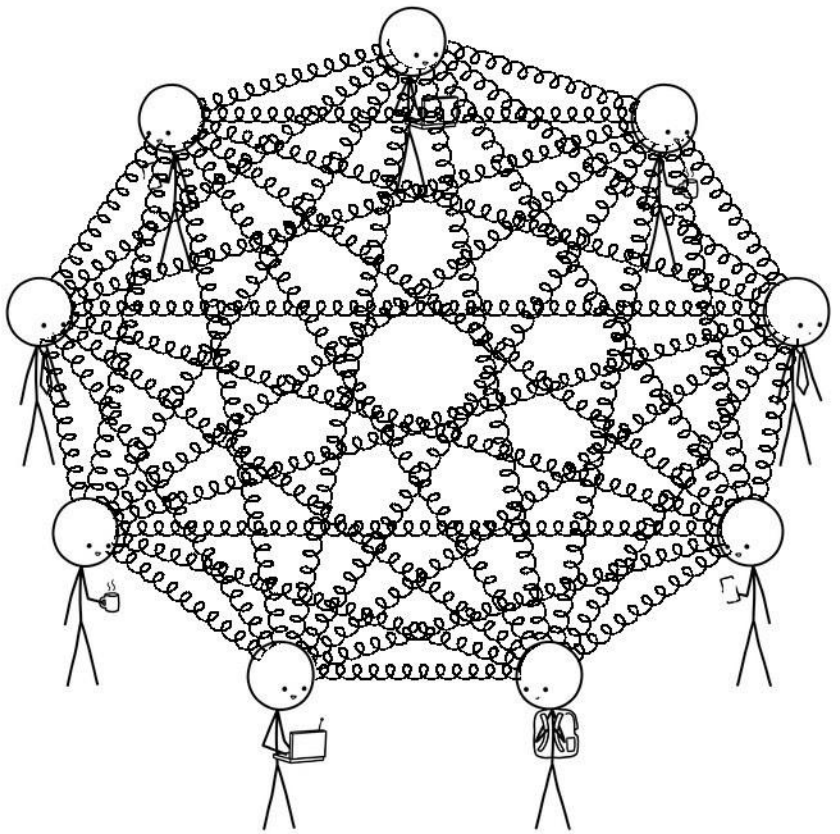
WHERE THE BRAINPOWER GOES



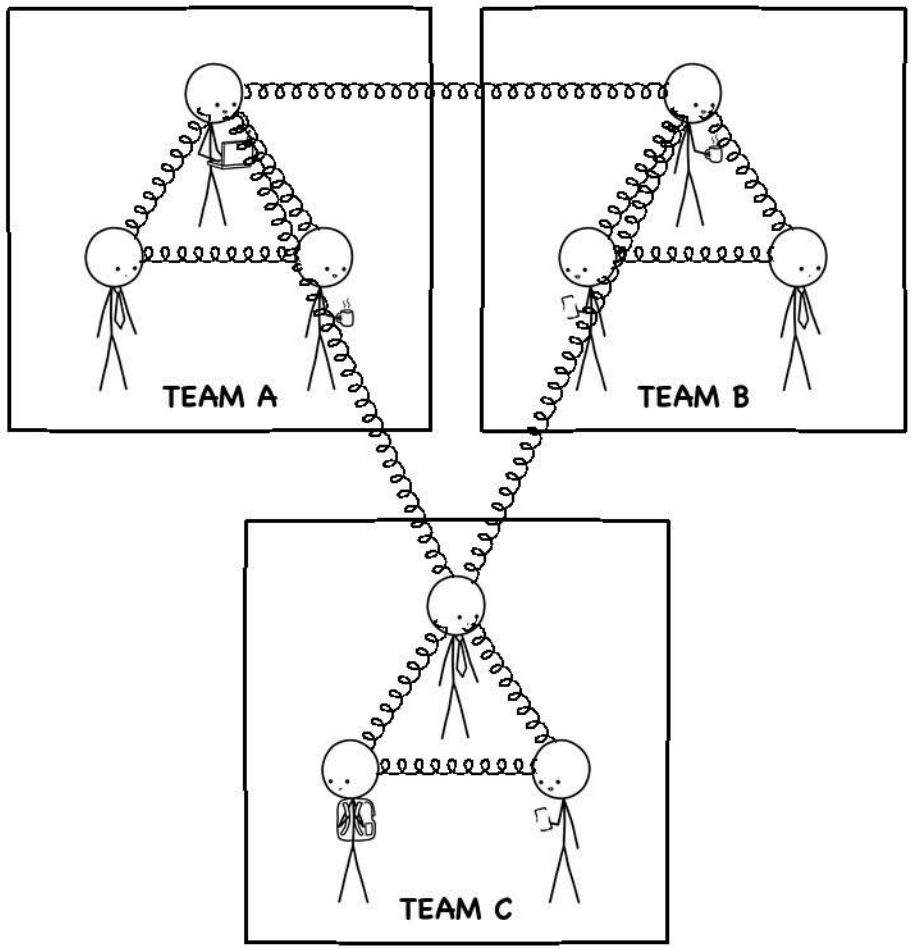
The job: just enough consistency,
without burning all the CPU on it.

SHARD IT: SMALL TEAMS, CLEAR OWNERS

Each team owns its own dragon, and teams talk through one interface.



1 team of 9: 36 phone lines

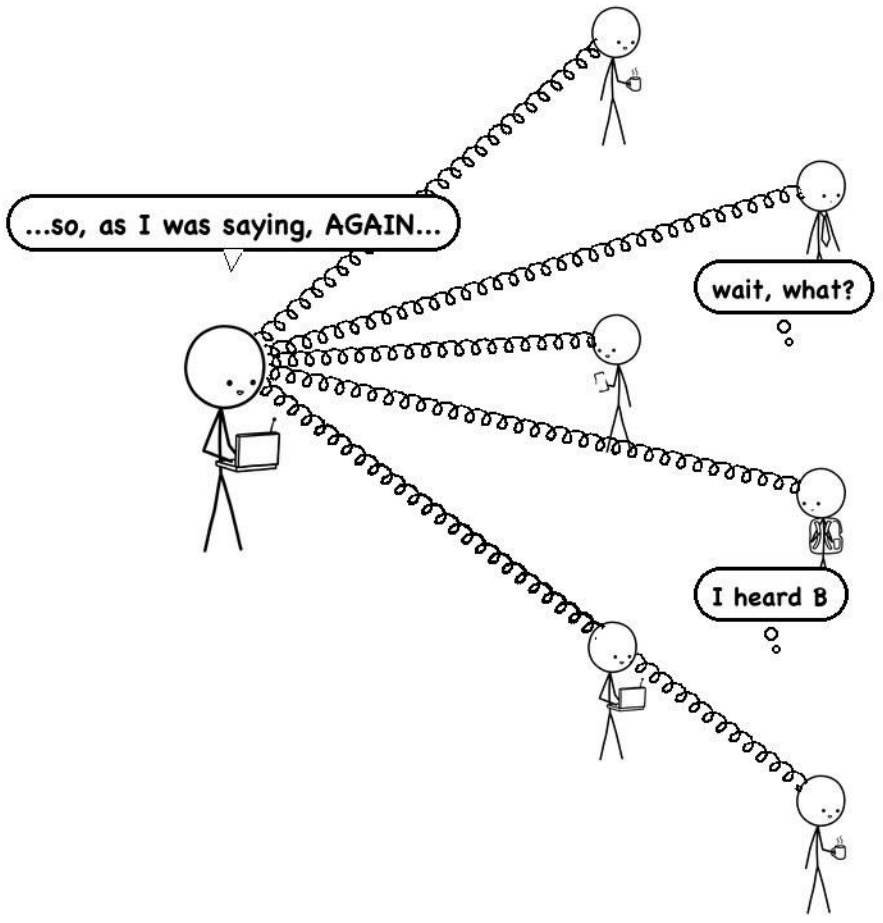


3 teams of 3: 12 phone lines

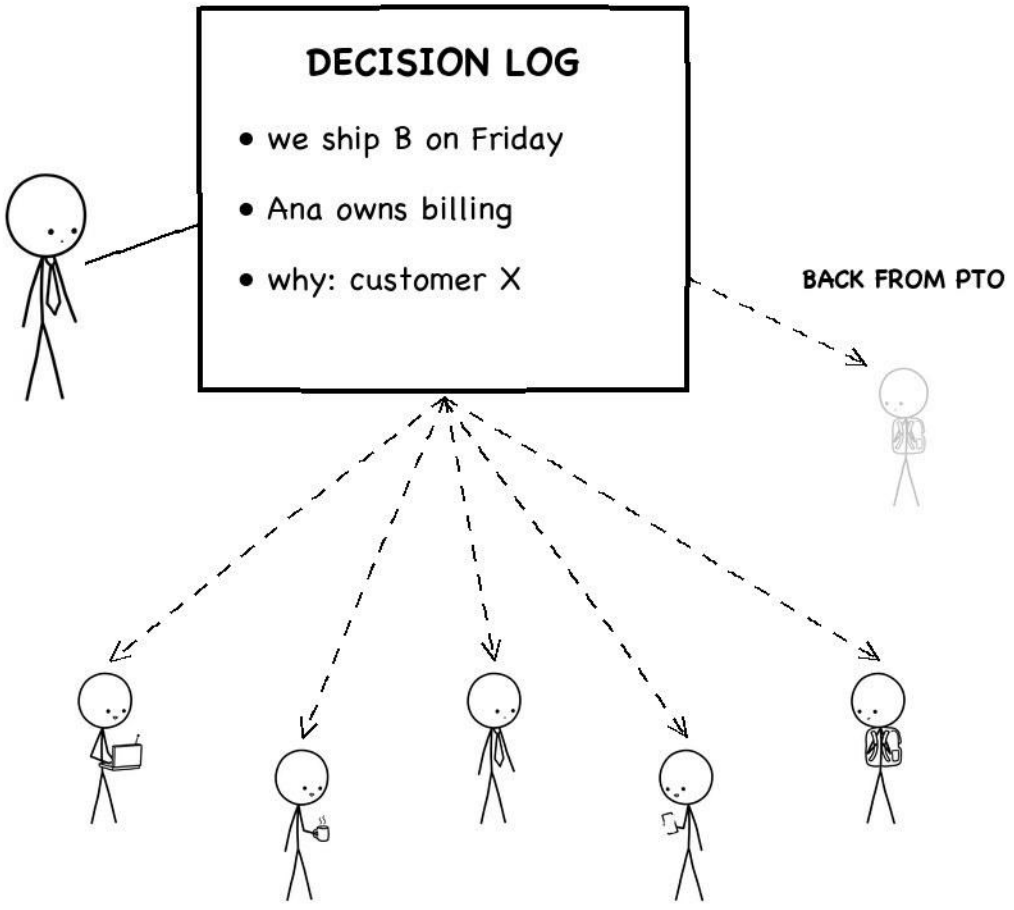
WRITE IT DOWN: BROADCAST, DON'T RPC

One write, many reads. Async. Still there when someone is on PTO.

TELLING EVERYONE, ONE BY ONE

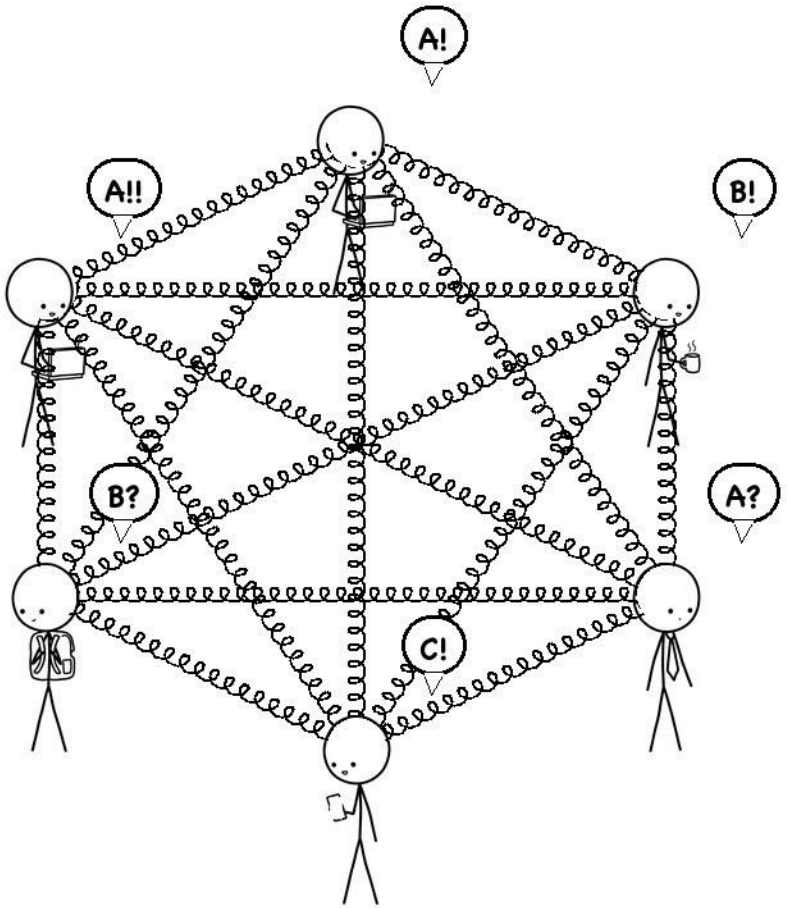


WRITE IT DOWN ONCE

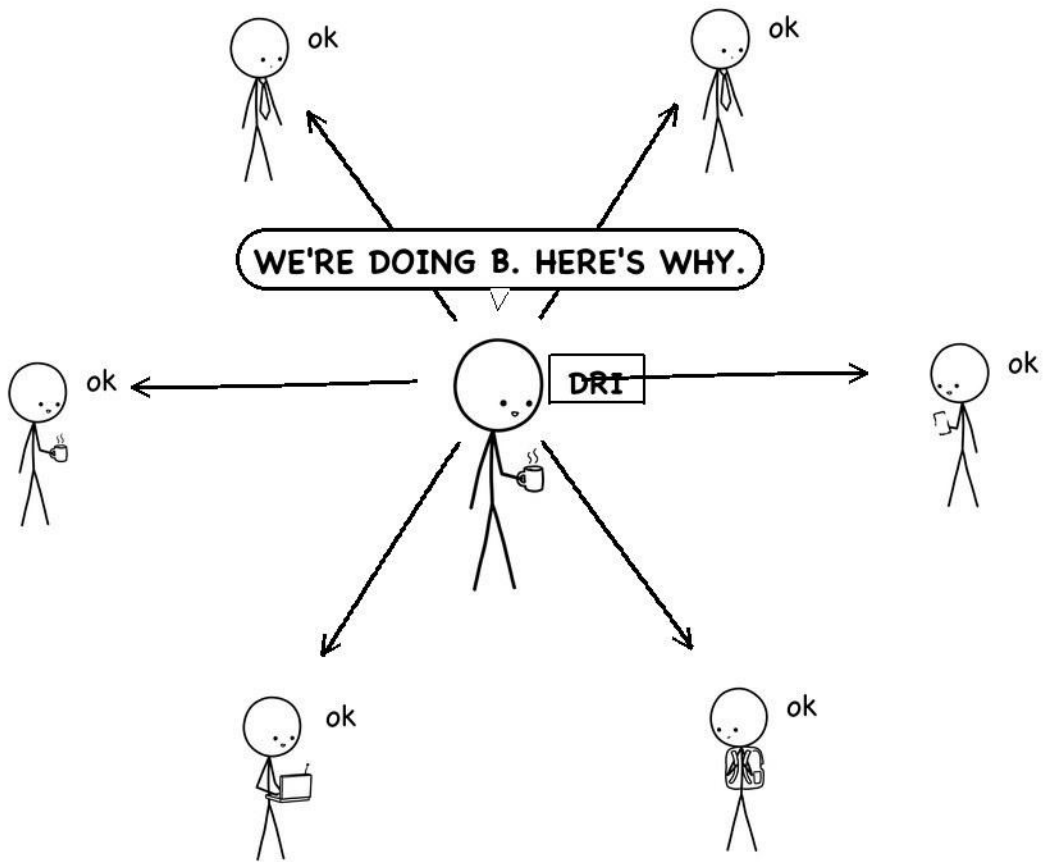


ONE OWNER PER DECISION

Consensus is expensive. Name a leader (the DRI); everyone else gets told.



EVERYONE DECIDES (FOREVER)

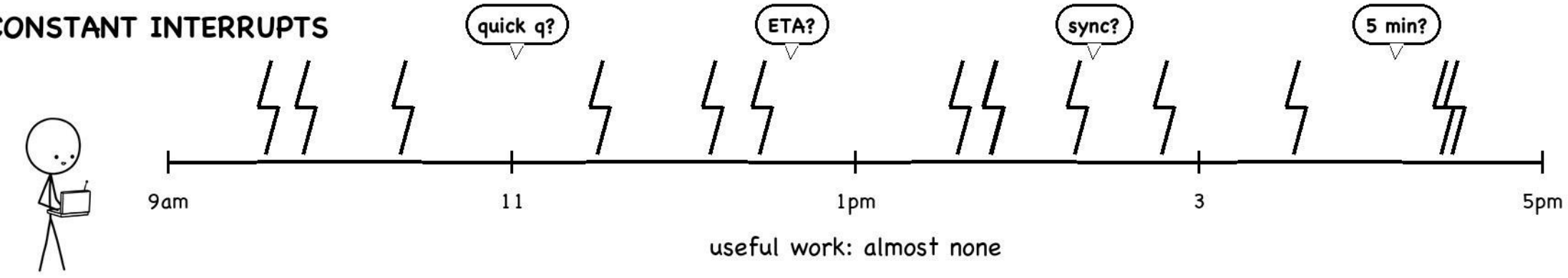


ONE DECIDES, EVERYONE HEARS WHY

HEARTBEATS & BATCHING

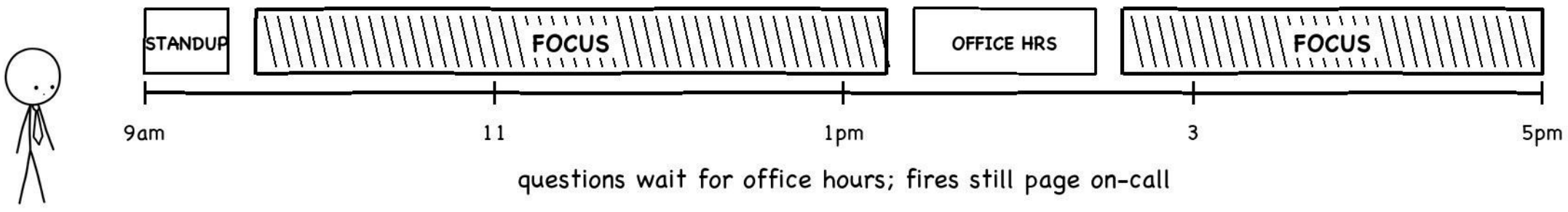
Predictable rituals beat constant interrupts.

CONSTANT INTERRUPTS



useful work: almost none

HEARTBEATS + BATCHING

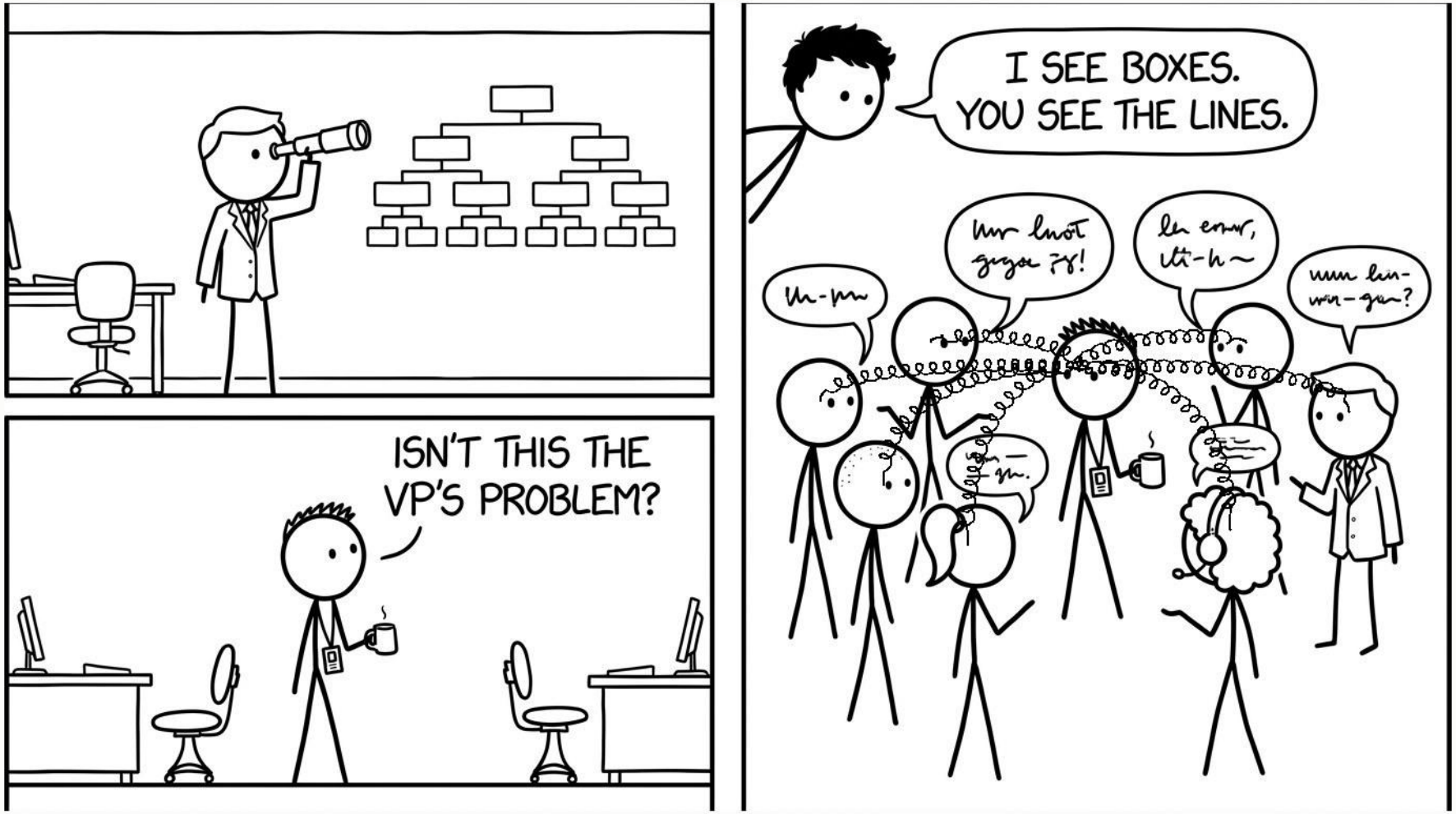


questions wait for office hours; fires still page on-call

PART 5

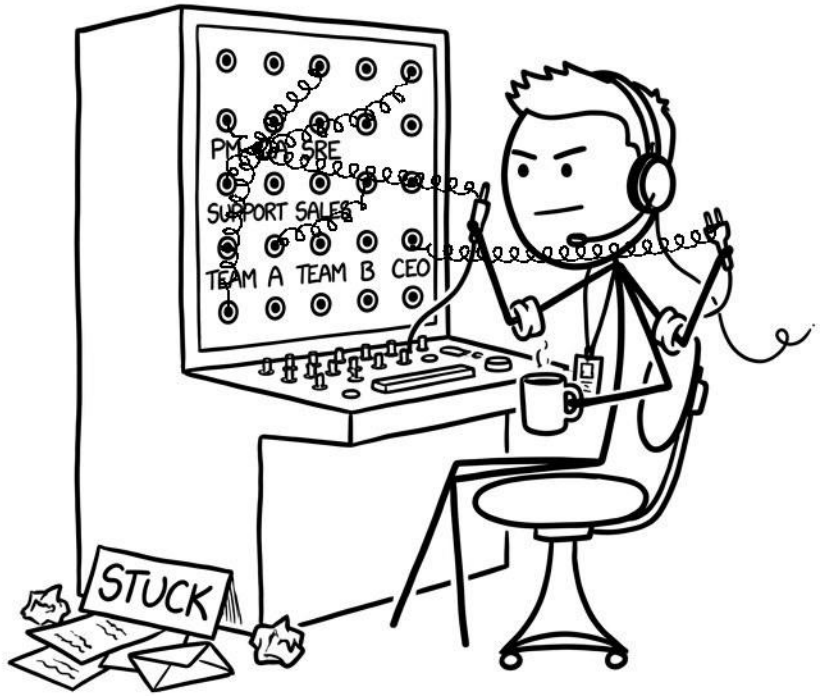
The EM's Responsibility

(Yes, you.)


ISN'T THIS THE VP'S PROBLEM?
I SEE BOXES. YOU SEE THE LINES.

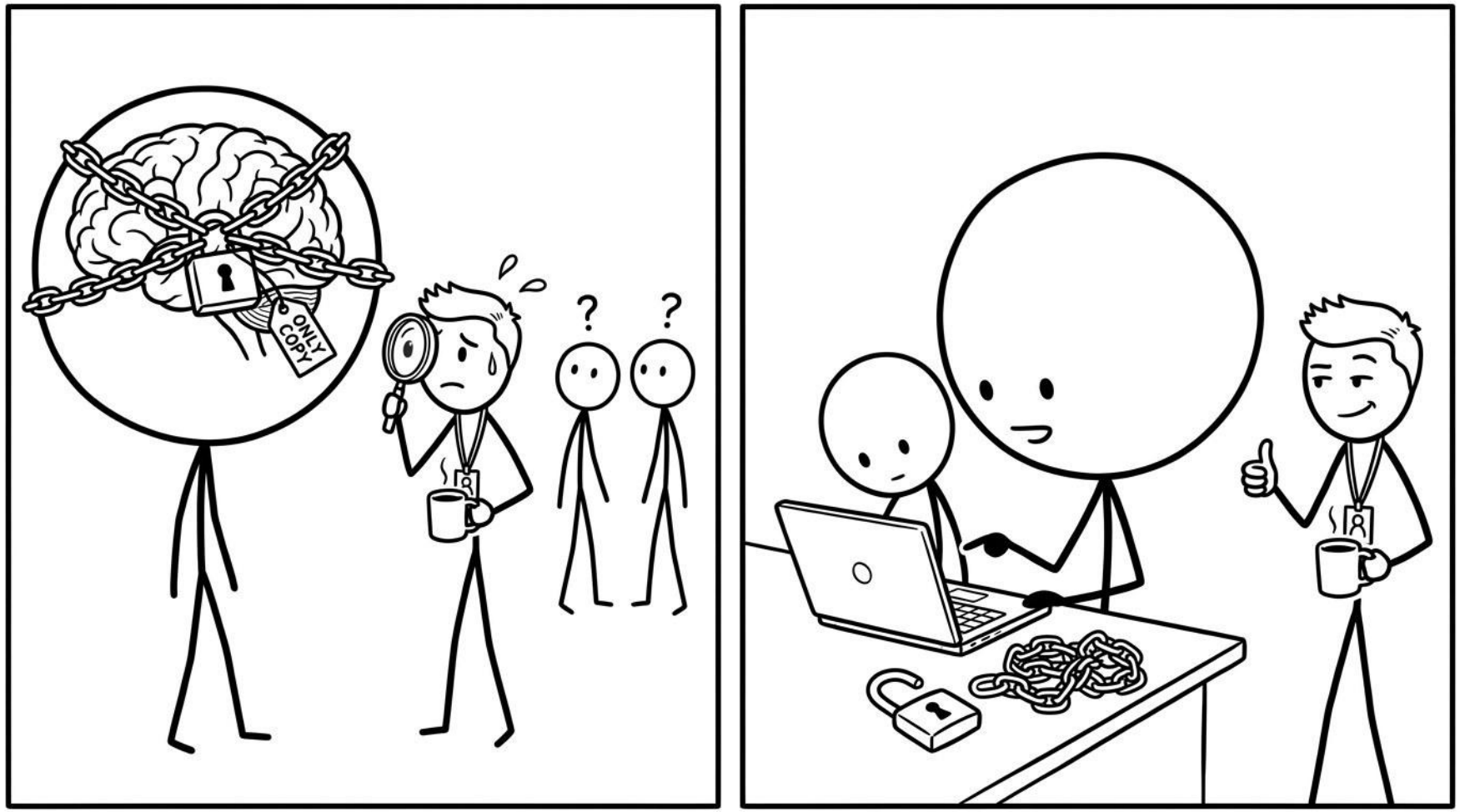
WHO needs to know WHAT, WHEN — and WHY?

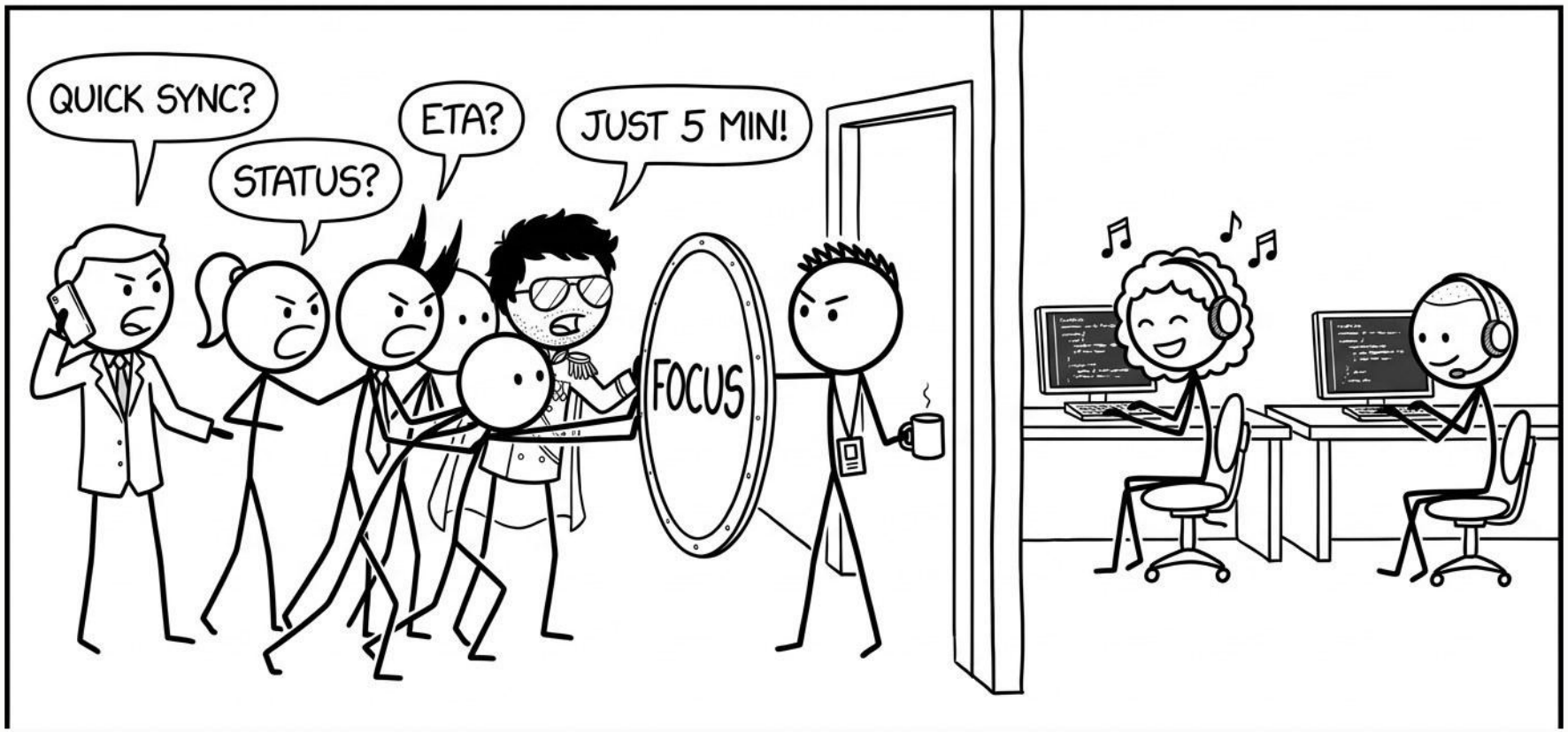
WHERE are they going to learn it?



Yes, you're just
a fancy router.

(Sorry.)


ONLY COPY
? ?


QUICK SYNC?
STATUS?
ETA?
JUST 5 MIN!
FOCUS

PICK THE CONSISTENCY LEVEL

Not everything needs everyone to know right now.

STRONG

everyone, right now

- incidents
- priority changes
- who owns what

HOW:

say it live AND write it down

cost: $$$

EVENTUAL

the right people, this week

- roadmap changes
- decisions + why
- new hires, departures

HOW:

weekly notes, decision log

cost: $$

NEED-TO-KNOW

whoever touches it

- implementation details
- how the cache works
- edge cases

HOW:

in the code, PR and docs

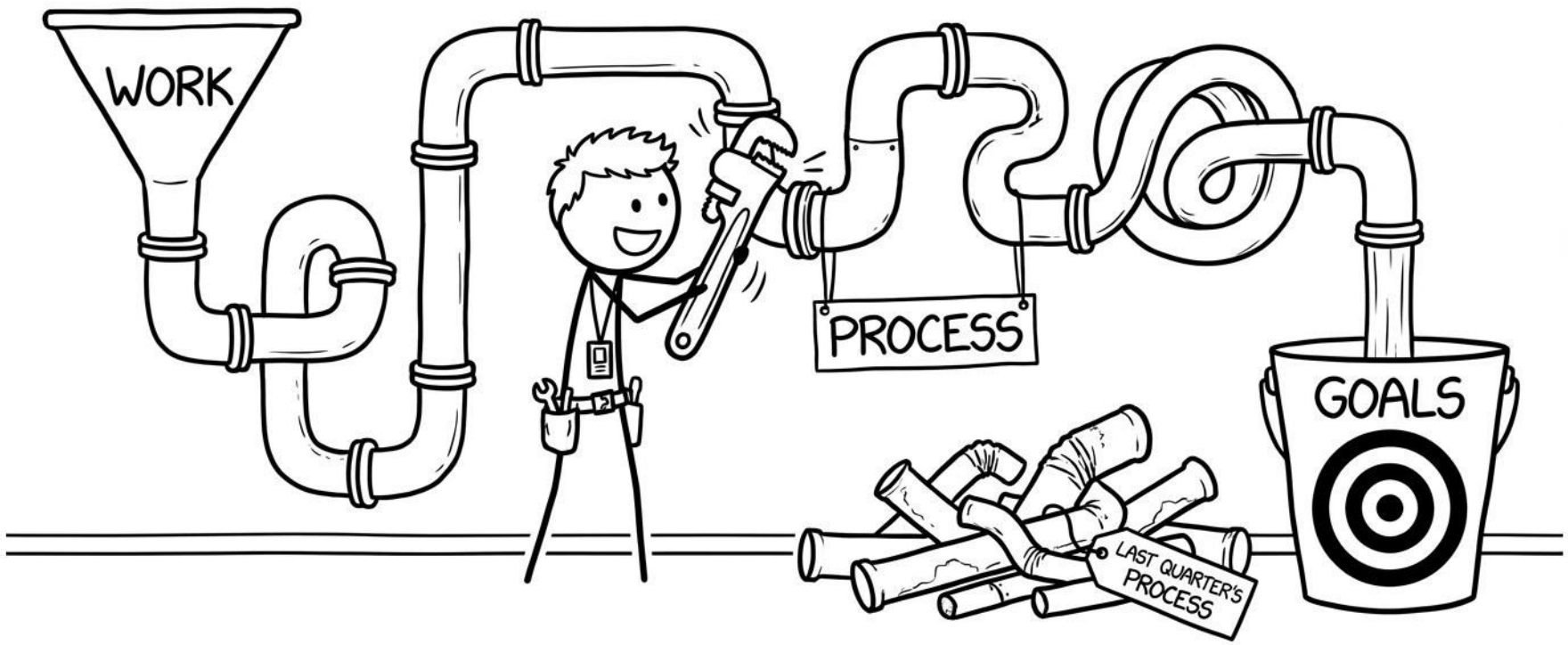
cost: $

...and that includes YOU. You don't need to know everything.

Don't burn all your cycles on info acquisition.

SHAPE PROCESS TO FIT GOALS

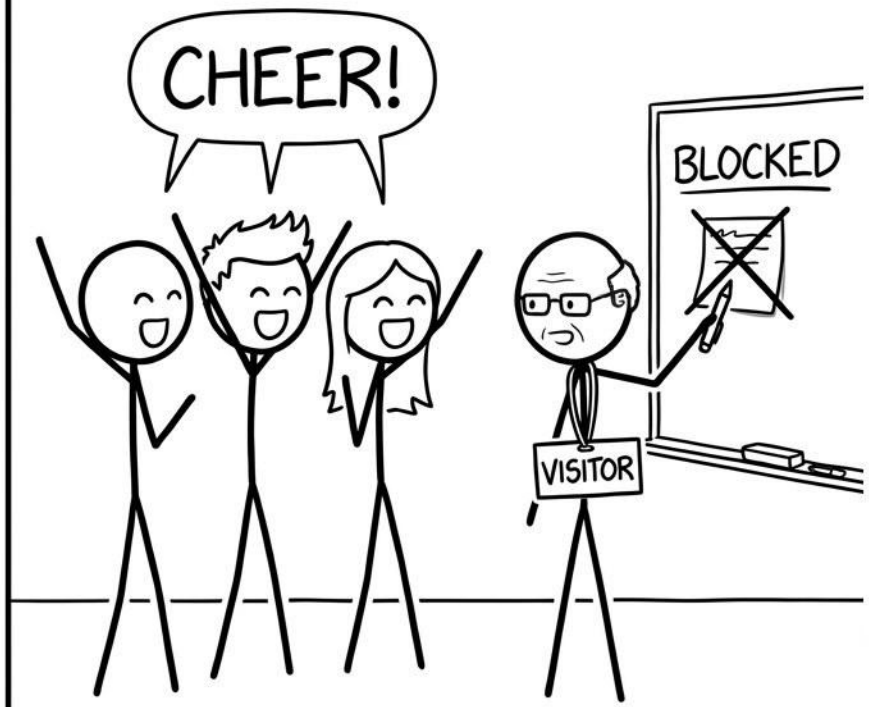
...and be comfortable changing it.



STANDUPS: SUPER HIGH BANDWIDTH, SUPER EXPENSIVE. USE JUDICIOUSLY.



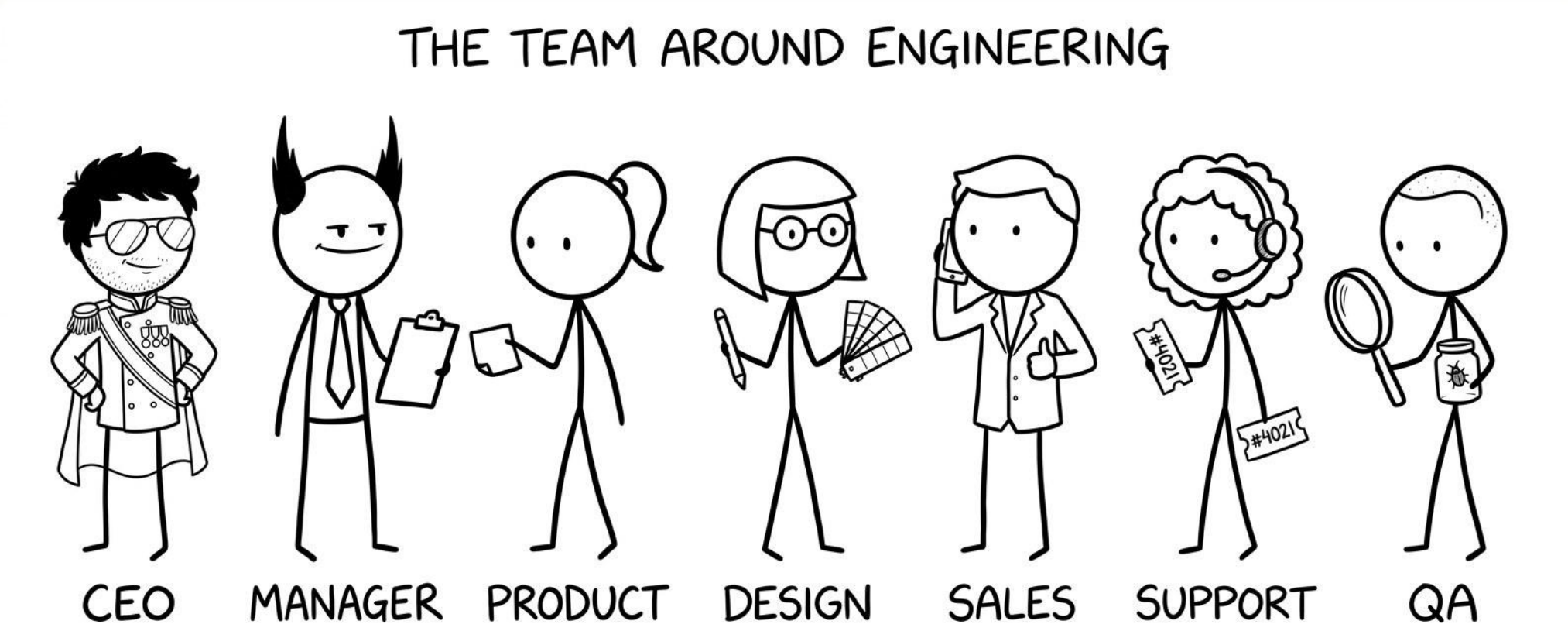
VIRTUAL TEAMS: BORROW (FOR A WHILE) WHOEVER UNBLOCKS YOU.

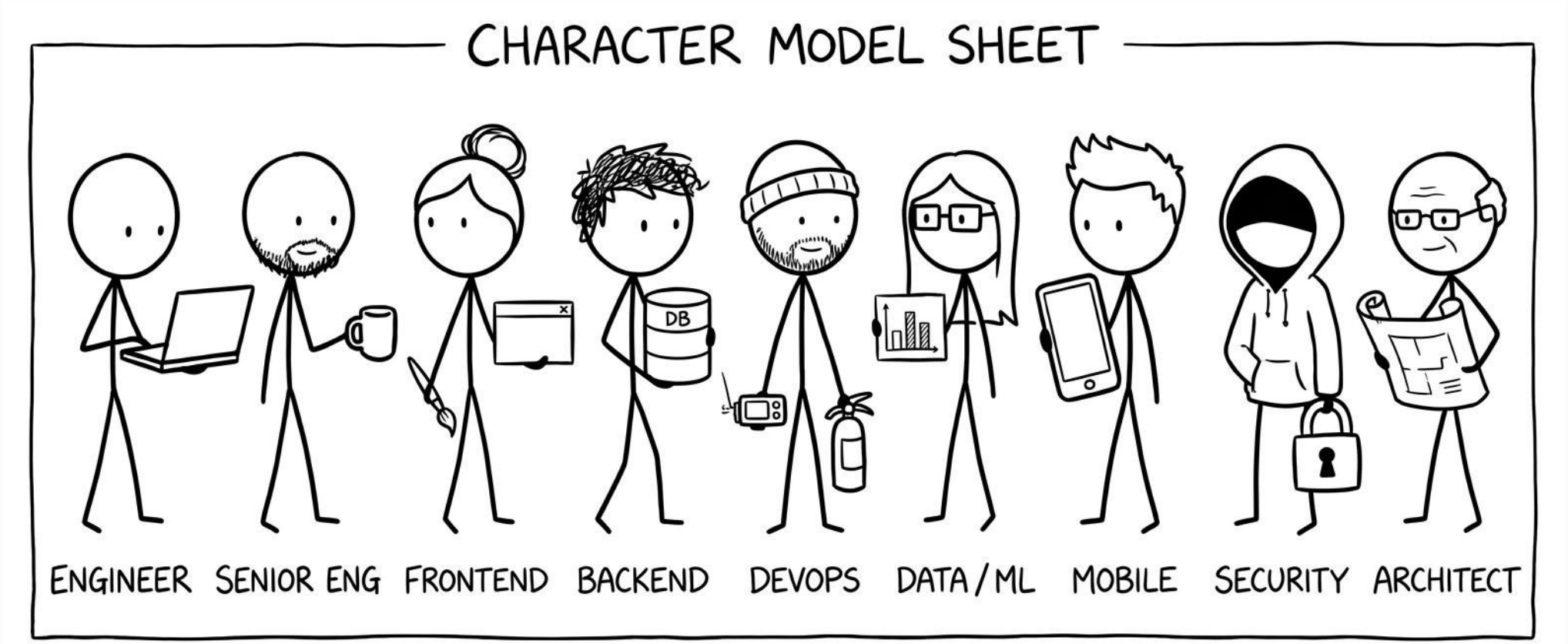


CUSTOMER WEEKLY · STORE HEALTH

Everyone who can affect a major goal. Weekly. No excuses for the BUSINESS not to be aligned.




THE TEAM AROUND ENGINEERING
#4021
#4021
CEO
MANAGER
PRODUCT
DESIGN
SALES
SUPPORT
QA


CHARACTER MODEL SHEET
DB
ENGINEER
SENIOR ENG
FRONTEND
BACKEND
DEVOPS
DATA / ML
MOBILE
SECURITY
ARCHITECT